

MAKINDE HENRY SANMI. Ph.D. Automatic Item Generation Using Multi-Agent Large Language Model (2026)

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The purpose of this study was to develop and evaluate a novel agentic framework for automatic item generation (AIG) using large language models (LLMs) to produce multiple-choice items for Physician Assistant Education Association (PAEA) End of Rotation™ examinations in cardiovascular disease across five clinical specialties: Emergency Medicine, Family Medicine, Internal Medicine, Pediatrics, and Surgery. The framework employed specialized AI agents: Cognitive Planning Agent, five track-specific Writer Agents, an automated Reviewer Agent, and an SME-Driven Rewrite Agent orchestrated through LangGraph using GPT-5 Nano for generation and GPT-5 Mini for automated review via PAEA's enterprise OpenAI subscription. Four research questions were addressed ranging from framework integration, alignment and validation, efficiency, and psychometric properties.

A two-phase approach generated 333 items, of which 210 (63.1%) passed automated review using an eight-dimensional quality rubric and entered multi-stage expert evaluation. Ten subject matter experts conducted small peer review (two per specialty), followed by six large peer review sessions with 6–8 Subject Matter Experts (SMEs) per session. During Phase 1, 50 enemy items were identified and eliminated for content overlap, and 35 items were processed through the Rewrite Agent with an 82.9% success rate. Across both phases, 76 items (36.2% of those entering human review) advanced to PAEA-administered pilot testing. Phase 2 demonstrated a 272% improvement over Phase 1 in pretesting advancement rates following inter-phase prompt refinement, with no enemy item concerns emerging.

Total computational cost was \$24.20 (7,741,445 tokens; 1,044 API calls), yielding \$0.16 per pretesting item in direct LLM costs and an 80–83% reduction in SME time compared to

traditional item development. Psychometric analysis of field-tested items employed classical test theory across all five examinations and one-parameter logistic (1PL) modeling for the Emergency Medicine examination, with item-level statistics compared to traditionally developed PAEA items to evaluate psychometric parity. Psychometric analysis of 33 field-tested AIG items revealed difficulty and discrimination values comparable to non-AIG field test items and operational items, with all but one item meeting Rasch model-data fit criteria. However, AIG items exhibited lower discrimination than historically calibrated items, consistent with their status as unrevised first-generation items. Computational costs were substantially lower than traditional development estimates, though direct comparison is limited because traditional estimates encompass the full development lifecycle including field testing and bank maintenance. The study contributes evidence that multi-agent LLM frameworks can produce assessment items that meet expert review standards and demonstrate adequate initial psychometric properties.

AUTOMATIC ITEM GENERATION USING MULTI-AGENT LARGE LANGUAGE
MODEL

by

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DEDICATION

To my father, Late Chief Ezekiel Akinlolu Makinde

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CHAPTER I: INTRODUCTION

Introduction

Educational assessment is undergoing a paradigm shift in the face of rapid technological advancements and data-driven methodologies. von Davier (2015) introduced the concept of computational psychometrics, which was defined as “a blend of stochastic processes theory, computer science-based methods, and theory-based psychometric approaches that may aid the analyses of complex data from performance assessments, including collaborative assessments.” This foundational development laid the groundwork for a new generation of testing research and development in which computational tools including natural language processing (NLP) and machine learning are applied directly to core measurement tasks such as item generation, scoring, and test assembly. The 2021 publication of *Computational Psychometrics: New Methodologies for a New Generation of Digital Learning and Assessment* (von Davier, Mislevy, & Hao, 2021) marked a significant advancement in the field. The volume brought together cutting-edge approaches that redefine how assessment is conceptualized and implemented in digital learning environments. The book highlighted methodologies such as supervised learning (Hao, 2021), unsupervised learning (Wong, 2021), large-scale Bayesian inference (Maris et al., 2021), virtual and performance-based assessments (Andrews-Todd et al., 2021), time series analysis (Halpin et al., 2021), stochastic processes (Halpin et al., 2021), social network analysis (Zhu, 2021), text mining (Flor & Hao, 2021), and automated scoring (Flor & Hao, 2021). These techniques reflect the evolving landscape of psychometric research and its intersection with data science. This study situates itself within this trajectory by applying LLM to the specific measurement challenge of automatic item generation.

Within the broad arsenal of computational psychometrics, NLP has emerged as a particularly transformative technology for educational assessment as evidenced by the publication of “Advancing Natural Language Processing in Educational Assessment” in 2023, which was a first of its kind book that provides evidence-based practices for the use of NLP-based approaches to automated text and speech scoring, language proficiency assessment, technology-assisted item generation, gamification, learner feedback, and beyond. Spanning historical context, validity and fairness issues, emerging technologies, and implications for feedback and personalization, these chapters provide a substantial treatment of NLP for educational measurement researchers, psychometricians, testing professionals, and policymakers (Yaneva & von Davier, 2023).

Assessment programs in education and credentialing rely on a steady supply of high-quality test items, yet producing these items through traditional manual writing is costly and labor-intensive. Automatic Item Generation (AIG) emerged as a “dream come true” for test developers by using computer algorithms to rapidly create new items, potentially yielding hundreds or thousands of items from a single design template (Gierl et al., 2021). Over the past two decades, research in AIG has advanced significantly (e.g., Drasgow et al. 2006); Gierl & Haladyna (2012, 2013); Gierl & Lai (2012, 2013, 2016, 2018); Gierl et al. (2008, 2012, 2016, 2017, 2018, 2021, 2022); Lai et al. (2016); Leighton & Gierl (2007); Sayin & Gierl (2024); Shin et al. (2022)), developing both theoretical foundations and practical tools needed to generate items at scale. However, much of this progress has centered on template-based AIG, where human experts design item models or templates that algorithms then use to instantiate new items. While template-based methods have proven effective over the years for certain applications, they depend on expert-crafted frameworks and may struggle to capture complex content domains (Shin, 2021).

To improve the efficiency of item development and to broaden item coverage in domains that template-based AIG cannot effectively address, the concept of non-template-based AIG was developed. This approach enables Subject Matter Experts (SMEs) to allocate their expertise toward the refinement, evaluation, and creative aspects of item writing rather than initial item construction. The existing methods that leveraged NLP methods to generate items (Aist, 2001; Brown, Frishkoff, & Eskenazi, 2005; Chen, Liou, & Chang, 2006; Devlin, Chang, Lee, & Toutanova, 2018; Mazidi, 2017; Narayan, Simoes, Ma, Craighead, & McDonald, 2020, Shin 2021) demonstrated the feasibility of reducing human involvement in item authoring but were limited by high rejection rates, reliance on hand-crafted features, and insufficient quality for operational use without extensive post-generation editing. Some other studies have combined both template-based AIG and non-template-based AIG (Shin 2021; Kiyak & Kononowicz 2024). As the field of NLP grows, large language models (LLMs) such as OpenAI's GPT variants, Anthropic Claude variants, Gemini variants and other foundational models, offers a new paradigm for non-template-based AIG. LLMs, unlike the earlier NLP methods, can generate plausible, natural-language assessment items without relying on hand-crafted item shells, by leveraging their vast trained knowledge (Lee et al. 2023; Bulut et al. 2024). This raises exciting possibilities for item generation, especially in content areas or item formats that elude simple templating.

Educational researchers have increasingly explored different approaches to implement LLMs in test development. Many studies have focused primarily on prompt engineering, with a few extending their scope to include fine-tuning of these models (Laverghetta Jr. & Licato, 2023; Amyeen, R. 2023; Kiyak et al, 2024). However, a common shortcoming across much of this emerging literature is the limited consideration of the validity of the generated items and their associated psychometric properties (Gierl & Lai, 2016; Lane et al., 2016; Shin, 2021). This gap

underscores the central motivation of this dissertation, determining how to effectively harness the capabilities of LLMs for item generation while ensuring the quality, style, validity, reliability and psychometric integrity of the resulting test items.

To address this, the present research adopts a framework for LLM item generation using an agentic approach; a cutting-edge approach in contemporary LLM applications. Within this agentic system, the LLM operates with human-in-the-loop "Agentic System" that can generate an item, review an item, provide evidence, has sophisticated reasoning, and iteratively refine its outputs through feedback and the use of supplementary tools or resources (Yao et al., 2023). By exploring this advanced methodology, the study aims to evaluate whether LLM-driven agents can produce educational assessment items that adhere to rigorous standards of educational measurement.

The subsequent sections of this chapter provide the broader context and rationale underpinning this research. Specifically, the evolution of AIG methodologies, template-based and non-template-based approaches. Additionally, LLM based AIG methods and the rationale behind the study are presented. This chapter concludes by articulating the specific research questions that guide the current inquiry and how to carry it out.

Background, Context, and Theoretical Framework

Evolution of Automatic Item Generation: From Traditional to AI-Driven Approaches

AIG began in the 1960s onwards with each phase contributing essential principles that influence contemporary practices (Haladyna, 2013). The foundational milestones that have shaped AIG evolution include prose-based AIG (Bormuth, 1970), facet theory (Guttman, 1953, 1959), item forms (Hively, 1974), concept formation (Markle & Tiemann, 1970), assessment engineering

(Luecht, 2006, 2007, 2013), and the influential 1998 ETS Invitational Seminar that culminated in "Item Generation for Test Development" (Irvine & Kyllonen, 2002).

Early AIG development was driven by practical needs in modern assessment. As computer-based testing expanded in the late 20th century, testing programs transitioning to computer-based testing (CBT) required large item banks to enable frequent administration, adaptive testing, and enhanced security (van der Linden & Glas, 2010). The traditional method of item development, solely dependent on SMEs developing one item at a time, proved extremely challenging, expensive, and time-consuming. Rudner (2010) estimated that developing a single item for a high-stakes exam costs between \$1,500-\$2,500, meaning that a company needing 1,500 items in their item bank would face costs of \$2,250,000 to \$3,750,000, with additional expenses for review, field testing, and potential revisions. The need for AIG became increasingly urgent as CBT required large pools of test items that can be used to construct parallel test forms, necessitating items of the same level of complexity and target construct (Kyllonen, 2009).

Throughout this dissertation, It is important to clarify the use of the term Automatic Item Generation (AIG). Gierl and colleagues (Gierl & Haladyna, 2013; Gierl & Lai, 2013) popularized this term in the context of their template-based cognitive modeling approach, establishing AIG as a specific methodology within educational assessment. However, the field has evolved to incorporate diverse computational techniques. This dissertation adopts a broader interpretation of AIG to refer to any technology-assisted item development process, including template-based approaches, NLP-driven methods, and LLM-based generation. Distinctions among these approaches are made explicit throughout to avoid confusion.

Template-Based AIG Foundations and Limitations

Template-based AIG represents the most established approach to AIG, characterized by systematic use of predefined structures that specify how item components can be manipulated to generate new instances (Gierl & Lai, 2013). The foundational three-step framework established by Gierl et al. (2012) involves: (1) developing cognitive models to identify and organize content domains; (2) creating item models that translate cognitive specifications into operational templates; and (3) systematically generating test items by combining feature values according to predefined constraints and rules.

Firstly, cognitive model development serves as the foundational theoretical framework guiding the systematic creation of item generation systems. Cognitive models provide explicit representations of task requirements and problem-solving processes that examinees must employ when responding to assessment items (Leighton & Gierl, 2007). Unlike traditional item development where cognitive representations remain implicit and unique to individual item writers, cognitive models provide formal, structured specifications that can be replicated and systematically manipulated for item generation. The advantages of cognitive models include their ability to identify and organize parameters, constraints, and instructions required to control the item generation process (Gierl et al., 2021). They facilitate inferences about examinees' thinking and problem-solving processes while serving as explicit representations that can be evaluated to ensure generated items address their intended assessment purposes (Leighton & Gierl, 2011).

The second component of template-based AIG is item modeling provides systematic frameworks for creating structured templates. The degree to which item templates constrain psychometric properties is governed by two theoretically distinct approaches, originally proposed by Drasgow et al. (2006): weak theory and strong theory. Under the weak theory approaches,

templates aim to generate isomorphic instances (i.e., items sharing identical cognitive demands but varying in surface features, yielding statistically equivalent item parameters) that maintain psychometric consistency, while strong theory approaches produce variants with predictable psychometric variation based on cognitive models (Gierl & Lai, 2013). Item models or templates serve as the "primary skeleton of test items" containing essential components to define and model test items from a parent item (Gierl & Lai, 2020). These models consist of three primary components: stems (main questions or prompts), options (response choices), and auxiliary information (supporting materials) (Gierl et al., 2015).

Despite their advantages, template-based frameworks exhibit inherent constraints in content novelty and domain coverage. Because templates are manually authored, they can become overly constrained, leading to repetitive item contexts and limited creativity in generated scenarios (Gierl et al., 2021). This limitation has prompted exploration of alternative approaches that minimize human intervention while maintaining quality standards.

Non-Template-Based AIG: Evolution and Recent Advances

Non-template-based frameworks have emerged as promising alternatives that minimize human intervention in the item generation process (Shin, 2021). These approaches leverage advances in NLP, machine learning, and artificial intelligence to generate items directly from source materials without requiring predefined templates. Lai and Gierl (2021) reviewed non-template-based AIG along three methodological stages defined by their underlying computational approach: syntax-based, semantic-based, and sequence-based systems. The development of LLMs has since changed the trajectory entirely, but the evolution of these three stages provides essential context for understanding the current landscape.

The first approach, syntax-based or rule-based AIG systems employ linguistically motivated algorithms to identify and extract potential item content from source texts. Early systems like the Mathematics Test Creation Assistant (TCA) allowed item writers to take existing problems and "variabilize" them by identifying surface features that could be changed without altering problem structure (Singley & Bennett, 2002). These systems utilize syntactic parsing, semantic analysis, and discourse processing to identify text segments suitable for item generation (Aist, 2001; Brown et al., 2005).

The second semantic-based approach utilizes corpus-based methods and NLP techniques to extract item content based on meaning rather than surface structure. These methods train statistical models on large text corpora to identify patterns and relationships exploitable for item generation (Mitkov & Ha, 2003; Heilman & Smith, 2009). Topic modeling approaches, particularly Latent Dirichlet Allocation (LDA) which is a probabilistic model that identifies latent thematic structures across large document collections by inferring the distribution of topics within individual texts (Blei et al., 2003), have shown promise for identifying thematic content within large document collections that can serve as the basis for item generation (Shin, 2021).

The third approach is the sequence-based approach. It uses neural networks or deep learning approaches. This marks a pivotal shift from manually defined rules to models that learn how to generate items from data. Du et al. (2017) developed one of the earliest neural systems for reading comprehension item generation using recurrent neural networks (RNNs), demonstrating the ability to generate various types of questions from reading passages. More recent approaches have leveraged transformer architectures, particularly Bidirectional Encoder Representations from Transformers (BERT), for item generation tasks (Narayan et al., 2020). von Davier's (2018, 2019) two-stage exploration illustrates the rapid advancement in neural architecture. Initially applying

RNNs to generate multiple-choice items, von Davier later shifted to transformer-based models by fine-tuning GPT-2 on medical corpora, enabling generation of more coherent, contextually appropriate item stems and distractors with improved applicability to domains such as medical certification exams.

A New Paradigm for Item Generation: Large Language Models

The emergence of LLMs represents a potentially transformative development in AIG, offering unprecedented capabilities for generating diverse, contextually appropriate assessment content across multiple domains (Song et al., 2025). Recent extensive analyses reveal that LLM applications in educational assessment have grown substantially, with most generated items being multiple-choice questions in computer and medical sciences (Song et al., 2025).

LLM based approaches used so far in item generation are three: prompt engineering, fine-tuning, and retrieval-augmented generation (RAG). Prompt engineering technique involves crafting specific input prompts to guide a pre-trained LLM in generating desired content without modifying the model's underlying parameters. This approach leverages zero-shot and few-shot in-context learning to steer item generation. In zero-shot learning, the model generates items based solely on instructions without seeing examples, while few-shot learning provides the model with a small number (typically 1-5) of exemplar items within the prompt to guide generation. Sayin and Gierl (2024) demonstrated this approach for reading comprehension, constructing "text analysis cognitive models" encoded as prompts that guide GPT-3.5 to produce inferential questions while maintaining high linguistic quality and variability. This method enables flexible and rapid item creation without the computational overhead of model retraining.

Fine-tuning adapts a pre-trained LLM to a specific domain or task by updating its parameters using domain-relevant data. This approach customizes the model to generate items that

better align with particular content areas or assessment standards, often improving relevance and coherence compared to out-of-the-box models. Bulut and Yildirim-Erbasli (2022) fine-tuned open-source transformers on reading comprehension texts, finding that domain-adapted models produced richer stories and more assessment-aligned items than off-the-shelf counterparts, albeit with continued reliance on human review for quality assurance.

RAG enhances LLM output by integrating external knowledge sources during generation. By retrieving relevant information from curated databases or curricular repositories, RAG grounds generated content in accurate, up-to-date information, thereby improving factual correctness and alignment with test specifications. Prior research has shown that LLM-generated assessment items can perform comparably to human-written questions when evaluated under controlled conditions. Separately, studies applying RAG in medical question-answering contexts have reported improvements in topical relevance and factual fidelity relative to non-RAG approaches. In addition, Guinet et al. (2024) used RAG to produce task-specific exam questions and evaluated model performance across datasets, demonstrating the feasibility of RAG workflows for structured question synthesis. Together, these findings suggest strong potential for RAG-enhanced item generation in medical assessment, while acknowledging ongoing challenges related to complex clinical reasoning, novel item formats, and domain-specific content constraints.

However, several challenges persist in LLM-driven AIG. Model hallucination poses risks for assessment validity, requiring robust detection and correction mechanisms (Gorgun & Bulut, 2024). Bias in training data can propagate into unfair item pools, necessitating systematic bias audits and inclusive fine-tuning strategies (Belzak et al., 2023). Moreover, psychometric calibration beyond difficulty estimation remains largely manual, pointing to the need for tighter

integration of LLM outputs with item response theory frameworks. This dissertation employs a different approach called Agentic AI.

Agentic systems represent the most advanced LLM methodology that address limitations associated with single-model approaches to complex assessment tasks. Rather than relying on a single LLM, an agentic approach employs multiple specialized AI agents that collaborate through structured workflows. Liu et al. (2024) developed a modular framework for designing and evaluating Generative Intelligent Tutoring Systems (ITS), leveraging GPT-4 for Socratic-style interactions and personalized feedback with specialized AI agents including Instructional Advisors, Data Analyzers, and Feedback Assessors, demonstrating significant improvements in English vocabulary (mean scores increased from 26.4 to 30.7, $p < 0.05$), grammar (18.2 to 23.1, $p < 0.05$), and sentence construction (19.3 to 23.2, $p < 0.05$) among 30 undergraduate students. Similarly, Outay et al., 2024 developed the Assess-Predict-Oversee-Transit (APOT) model that demonstrates how specialized agents can enable intelligent management of competency building, with different agents handling learning style assessment, performance prediction, progress monitoring, and transition functions between learning stages.

In the context of AIG, an agentic framework enables distribution of responsibilities across specialized agents: content analysis agents understand domain requirements (Cognitive model); generation agents produce initial item drafts; reviewer agents evaluate content for accuracy and appropriateness; and quality assurance agents ensure items meet specified standards. This division of labor allows optimization of each agent for specific functions while enabling sophisticated coordination and feedback mechanisms.

Addressing these challenges requires a validation framework that integrates classical validity concerns with the governance demands specific to AI-generated content. The Duolingo

English Test (DET) Responsible AI framework (Burstein et al., 2022) provides such a structure. Originally developed for a digital-first language assessment that relies on AI for item generation, scoring, and security, the DET framework organizes validation evidence around four standards: Validity and Reliability, Fairness, Privacy and Security, and Accountability and Transparency. These standards align with established validity theory preserving the integrated evidence-based argumentation of Messick (1989) and Kane (2006) while adding AI-specific dimensions that classical frameworks were not designed to address, such as documentation of algorithmic processes, bias mitigation in training data, and transparency of automated decision-making. This dissertation adopts the DET framework as the organizing structure for validation, mapping each stage of the agentic generation and review pipeline to specific evidentiary claims across all four standards.

Purpose of the Study

The purpose of this study is to develop, implement, and evaluate a novel agentic system for non-template-based AIG that utilizes LLMs to produce cardiovascular disease assessment items aligned with Physician Assistants Education Association (PAEA) End of Rotation™ examination standards. PAEA is the sole U.S. national organization representing Physician Assistant (PA) educational programs within higher education.

This research aims to address the critical gap between the theoretical potential of LLM-based item generation and its practical implementation in demanding medical education contexts. This research aims to advance both the theoretical understanding and practical implementation of non-template-based automatic item generation by addressing three fundamental challenges: (1) how to structure LLM-based generation processes to maintain content related validity and construct alignment without explicit human-designed templates; (2) how to validate AI-generated

assessment content using the DET Responsible AI framework that addresses both traditional validity concerns and AI-specific issues; and (3) whether AI-generated items can achieve psychometric quality comparable to traditionally developed items. The study uses cardiovascular disease assessment in the context of PAEA End of Rotation™ examination standards as a demanding test case, but the primary contribution is methodological rather than content domain. The cardiovascular context provides rigorous evaluation conditions due to its complex content requirements, high professional standards, and well-established assessment frameworks, making it an ideal proving ground for advanced AIG methodologies.

The overarching purpose is to advance AIG from a promising but limited research area to a practical methodology that can produce assessment items meeting professional standards for validity, reliability, and fairness across diverse educational contexts. This involves demonstrating not merely that LLMs can generate plausible-looking items, but that they can produce items that function appropriately in assessment contexts and support valid inferences about examinees' competencies. Validation should encompass evaluation of raw LLM outputs, human-edited drafts, and comparison with traditionally authored items to establish the relative quality and effectiveness of each approach.

Rationale for the Present Study

Despite significant advances in AIG research and the emergence of powerful LLMs, current approaches continue to depend heavily on expert-designed templates and substantial human oversight to ensure quality, particularly in specialized domains (Kiyak & Kononowicz, 2025; Agarwal et al., 2026). These limitations are especially evident in medical education assessment, where the complexity of domain knowledge, strict psychometric and validity

standards, and the demand for scalable item production present persistent challenges for existing AIG methodologies.

The fundamental problem addressed by this research is the lack of a proven, systematic approach to leveraging LLMs for AIG that can produce assessment items meeting the rigorous standards required for high-stakes medical education evaluation. Current LLM-based item generation approaches which rely primarily on prompt engineering techniques, face several interconnected challenges that limit their practical applicability and adoption.

First, content-related validity presents persistent challenges in LLM-based item generation. While LLMs can generate plausible-sounding assessment items (von Davier, 2019), ensuring that these items accurately reflect intended learning objectives and measure target competencies remain problematic. Without the explicit cognitive models that guide template-based approaches, LLM-generated items may drift from intended content specifications or inadvertently measure unintended skills. In cardiovascular disease assessment, this challenge is particularly acute given the complex pathophysiological relationships and the critical importance of clinical accuracy in item content.

Second, quality control and consistency issues plague non-template approaches. Traditional template-based methods achieve consistency through systematic variation of predefined elements, but LLM-based generation lacks these built-in constraints. Each generated item is potentially unique in structure and approach, making it difficult to ensure consistent difficulty levels, appropriate distractors, and reliable psychometric performance. Recent studies have documented significant variability in the quality of LLM-generated assessment items, with acceptance rates varying widely depending on domain, prompt design, and review criteria (Kıyak & Emekli, 2024).

Third, while traditional validation frameworks including expert committee review and pretesting remain essential for ensuring all items meet high standards of clinical accuracy, structural quality, and psychometric soundness, the initial review process for AI-generated content could be made more efficient by addressing AI-specific issues earlier in the development pipeline. AI-generated items may exhibit distinctive patterns such as factual hallucinations, systematic biases from training data, or unexpected content generation behaviors (Ji et al., 2023; Huang et al., 2025) that, if detected and addressed during the generation phase rather than post-hoc review, could reduce the burden on human reviewers. The goal is not to replace established validation standards, but to augment the development process with AI-specific quality checks that allow expert reviewers to focus their expertise on higher-order concerns such as clinical relevance and pedagogical effectiveness rather than correcting basic factual or structural errors.

Fourth, scalability and efficiency concerns limit the practical deployment of existing approaches. While LLMs offer the potential for rapid item generation, extensive human review and revision by subject matter experts remains essential to ensure items meet operational standards. The challenge is that current approaches often produce initial outputs requiring substantial revision, which can diminish the efficiency gains that motivate LLM-based AIG adoption. The objective is not to eliminate human expert review which remains indispensable for maintaining quality and validity but rather to improve initial generation quality and develop systematic approaches that allow expert reviewers to work more efficiently. This includes reducing time spent on correcting basic errors and enabling reviewers to focus their expertise on higher-level validation tasks such as ensuring clinical accuracy, pedagogical appropriateness, and alignment with assessment frameworks.

Fifth, domain-specific requirements in medical education create additional complexity. Cardiovascular disease assessment requires items that not only test factual knowledge but also clinical reasoning, diagnostic skills, and therapeutic decision-making. Items must reflect current clinical practice standards, incorporate appropriate patient diversity, and avoid content that could promote misconceptions or unsafe practices. Ensuring that AI-generated items meet these domain-specific requirements while maintaining pedagogical effectiveness presents significant challenges.

The convergence of these challenges creates a situation where the theoretical potential of LLM-based AIG remains largely unrealized in practical applications. Testing organizations may either avoid using AI-generated content due to quality concerns or employ it with such extensive human oversight that efficiency gains are minimal. This situation represents a critical barrier to addressing the growing demand for high-quality assessment content in medical education and other specialized domains. The core problem this dissertation intends to address is the absence of a systematic, validated approach to AIG using LLMs that can consistently produce assessment items meeting standards required for operational use in medical education assessment. Specifically, there is no established methodology for structuring LLM-based generation processes to ensure content related validity, maintain construct alignment, achieve consistent quality, and enable efficient validation in complex domains such as cardiovascular disease assessment. This problem is particularly urgent given the rapid advancement of LLM capabilities and the growing interest in their educational applications. Without rigorous research establishing how these technologies can be effectively and responsibly integrated into assessment development processes, there is risk of either premature adoption that compromises assessment quality or excessive caution that prevents realization of potentially transformative benefits.

This dissertation addresses these challenges by developing an agentic framework that integrates LLMs with an electronic review system to produce cardiovascular assessment items aligned with PAEA test specifications and content standards. The framework is evaluated through content-related validity and fairness evidence, including multi-stage SME review with explicit validation criteria and checkpoints, and through psychometric analysis of item difficulty, discrimination, model fit statistics for the field-tested items compared to traditionally developed items from previous and current test administrations. Together, these components provide a systematic methodology for determining whether LLM-based generated items can meet the standards required for operational use in high-stakes medical education assessment.

Research Questions

This study was guided by four interconnected research questions that collectively address the advancement of AIG methodology and the evaluation of AI-generated assessment items through content related validation, psychometric analysis, and Duolingo English Test (DET) responsible AI principles.

- 1) How can large language models be integrated into a designed agentic framework to develop assessment items?
- 2) To what extent do we have alignment between the agentic framework items and SME reviewed items based on established test specifications and content standards?
 - a. How effective is the electronic review system developed to help SME review AI generated items for medical education assessment?
 - b. What validation criteria and checkpoints are necessary to ensure content related validity, fairness, and psychometric properties of AI-generated test items, and how do these compare to those used for human-authored items?

- 3) To what extent does the agentic LLM framework improve efficiency in terms of item development time, review cycles, and cost per item compared to conventional item development methods?
- 4) What are the psychometric properties (difficulty, discrimination, reliability, dimensionality) of test items generated by the agentic LLM approach, and how do these properties compare to those of traditionally developed items from previous and current test administrations?

Significance of Study

This study contributes to the field of educational measurement in five ways. First, it introduces the first empirically evaluated multi-agent LLM framework specifically designed for high-stakes medical education assessment, extending AIG beyond template-based and single-prompt approaches to a structured agentic architecture with specialized generation, review, and revision agents. Second, it documents the complete end-to-end AIG pipeline from cognitive model specification through generation, automated review, multi-stage SME validation, and field testing, providing a transparent, reproducible workflow adaptable to other content domains and assessment contexts. Third, it demonstrates how responsible AI principles can be systematically mapped to validity evidence within the assessment development process, bridging the gap between AI governance frameworks and classical validation theory. Fourth, it introduces a replicable multi-agent framework that operationalizes the full item development lifecycle, offering testing organizations a documented methodology for integrating LLMs into established item development workflows. Fifth, it provides empirical evidence on the psychometric viability of LLM-generated assessment items by field-testing AI-generated items alongside operational items on live high-

stakes examinations an evaluation context largely absent from the existing AIG literature, which has relied primarily on expert ratings or small-scale pilot administrations. Together, these contributions address the practical question facing testing organizations: how AI-generated items can meet the content, psychometric, and ethical standards required for operational use in high-stakes medical education assessment.

The study employs mixed-methods design combining technological innovation with empirical validation. The agentic framework distributes responsibilities across specialized AI agents handling content analysis, generation, review, and quality assurance, advancing AIG to a practical methodology meeting professional standards. The remainder includes four chapters: Chapter 2 reviews AIG methodologies and theoretical foundations; Chapter 3 details methodology and system architecture; Chapter 4 presents evaluation findings; Chapter 5 synthesizes results, discusses implications, and provides recommendations.

CHAPTER II: LITERATURE REVIEW

In chapter one, we established the critical need for advancing AIG methodologies in educational assessment, particularly in complex domains like medical education. The chapter introduced the practical limitations of traditional item development including prohibitive costs, scalability constraints, and reliance on subject matter experts. It articulated the rationale for a new framework that integrates LLM within an agentic, human-in-the-loop system. Four research questions were postulated:

- 1) How can large language models be integrated into a designed agentic framework to develop assessment items?
- 2) To what extent do we have alignment between the agentic framework items and SME reviewed items based on established test specifications and content standards?
 - a. How effective is the electronic review system developed to help SME review AI generated items for medical education assessment?
 - b. What validation criteria and checkpoints are necessary to ensure content related validity, fairness, and psychometric properties of AI-generated test items, and how do these compare to those used for human-authored items?
- 3) To what extent does the agentic LLM framework improve efficiency in terms of item development time, review cycles, and cost per item compared to conventional item development methods?
- 4) What are the psychometric properties (difficulty, discrimination, reliability, dimensionality) of test items generated by the agentic LLM approach, and how do

these properties compare to those of traditionally developed items from previous and current test administrations?

This literature review provides the theoretical and empirical foundations necessary to address these questions. The chapter begins by tracing the historical evolution of AIG from the 1960s to the present, then examines template-based frameworks including cognitive design systems and item modeling that provide the baseline against which agentic approaches will be compared. It next turns to non-template-based methods, progressing from rule-based systems through neural approaches, LLMs, and multi-agent systems, before addressing validity frameworks spanning Messick's unified theory, Kane's argument-based validation, and the Duolingo English Test Responsible AI standards. The chapter concludes by synthesizing key findings and identifying research gaps that this dissertation addresses.

Evolution of Automatic Item Generation

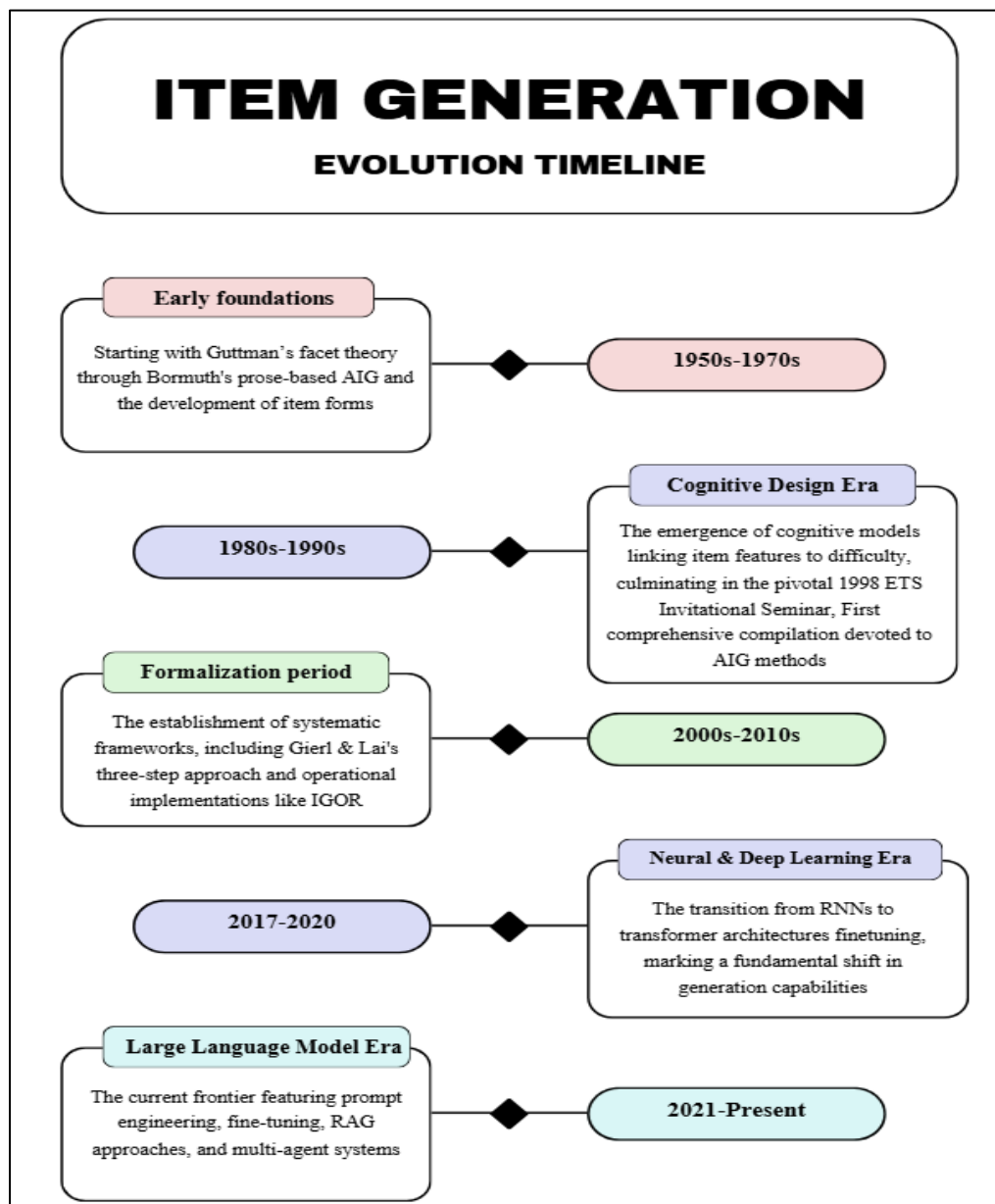
The concept of AIG begins historically in the 1960s and continues till date (See Figure 1 and Appendix A for the full visual breakdown of the evolution of AIG), each phase contributing essential principles that continue to influence today's contemporary practices. Haladyna (2013) identified five pivotal milestones that have shaped the evolution of AIG: 1) prose-based AIG (Bormuth, 1970); 2) facet theory (Guttman, 1953, 1959); 3) item forms (Hively, 1974); 4) concept formation (Markle & Tiemann, 1970), and 5) the 1998 ETS Invitational Seminar culminating in the publication of "Item Generation for Test Development" (Irvine & Kyllonen, 2002).

The prose-based AIG approach involves transforming key sentences from well-written prose into test questions, particularly WH-questions that seek content information. Guttman's facet framework emphasized the importance of operationally defining test content domains and

systematically varying specific facets to generate item families with predictable psychometric properties (Haladyna, 2013). The development of item forms by Hively (1974) advanced the field by introducing fixed syntactic structures that could accommodate systematic content variation. Markle and Tiemann's (1970) framework provided systematic methods for analyzing concepts and rapidly producing test items that measured different levels of conceptual mastery, from basic recognition to application in novel contexts (Haladyna, 2013). Lastly, the 1998 ETS Invitational Seminar represented a watershed moment in AIG development, bringing together diverse perspectives from psychology, measurement theory, cognitive science, and computer technology. This interdisciplinary collaboration laid the groundwork for subsequent theoretical and methodological advances that characterize contemporary AIG research (Irvine & Kyllonen, 2002).

The motivation for AIG arose from practical needs in modern assessment. As computer-based testing expanded in the late 20th century, testing programs required ever-growing item banks to enable frequent administration, adaptive testing, and enhanced security. The need for large number of items for sustaining the item banks became a necessity as computers allow continuous testing and testing on-demand (van der Linden & Glas, 2010), requiring a steady supply of new items to maintain adequate volume and prevent item exposure. However, traditional item development depends on SMEs to develop one item at a time a process that is both time-consuming and costly. Rudner (2010) estimated the cost of developing a single high-stakes exam item at \$1,500–\$2,500. For a testing organization needing approximately 1,500 items annually, this translates to \$2,250,000–\$3,750,000, and these items must still be reviewed, field tested, and potentially revised if psychometric characteristics prove inadequate.

Figure 1. Evolution of Automatic Item Generation Timeline



Automating item creation dates back several decades. Embretson (1983, 1998) was among the first to outline a cognitive design system approach, proposing that item features and difficulty could be systematically linked via cognitive theory. This approach treated test items as outcomes of underlying cognitive processes and sought to identify how specific item features (e.g.,

stimulus complexity, response format) influence item difficulty. Around the same time, Bejar (1985–1986) and colleagues at ETS explored the algorithmic generation of varying item stimuli and observed how those variations affected difficulty. Item models in these early studies were essentially templates with certain elements that could be systematically manipulated (numbers, words, figures, etc.) to produce new but structurally similar items. A classic example was the generation of isomorphic math word problems: an initial problem schema would be “variablized” by substituting values or contexts to create many new problems. These rule-based item generators assumed that the new items would inherit the psychometric properties of the original model item. Indeed, early research provided evidence for this inheritance in an experimental Graduate Record Examination (GRE) quantitative test; items generated on-the-fly from item models showed difficulty and discrimination statistics similar to their source items, yielding high correlations (around 0.80–0.87) between examinee scores on generated vs. original test forms (Someswar, 2024). Such findings demonstrated the promise of AIG for producing *equivalent* items in bulk without extensive trial-and-error.

The formalization of AIG as a field accelerated in the early 2000s. A landmark was the volume *“Item Generation for Test Development”* (Irvine & Kyllonen, 2002), one of the first compilations devoted to AIG methods, which also influenced the newer compilations *“Automatic Item Generation: Theory and Practice”* (Gierl & Haladyna, 2013) and *“Advanced Methods in Automatic Item Generation”* (Gierl et al., 2021). Irvine outlined two essential pillars for AIG: (1) creating a computerized generative structure (templates and algorithms) for assembling items, and (2) understanding the item feature difficulty relationships so that generated items have predictable psychometric properties. In the ensuing years, interdisciplinary advances in cognitive science, psychometrics, and computing converged to enrich AIG methodologies. Several test design

frameworks emerged that, while distinct, all facilitated more systematic item development: Embretson's cognitive design system, Mislevy's evidence-centered design, Luecht's assessment engineering, and Gierl template-based AIG were prominent examples. These frameworks guided AIG research and practice throughout the 2000s and 2010s. Figure 2 illustrates the workflow and data transformation required to generate items under Gierl et al.'s (2021) template-based AIG approach.

Figure 2. Workflow and data transformation required to generate Items (Gierl et al., 2021)



By the 2010s, AIG had matured from conceptual prototypes to operational implementations in various domains. Gierl and colleagues have published extensively on AIG (Gierl & Haladyna, 2012, 2013; Gierl & Lai, 2013, 2016; Gierl et al., 2008, 2012, 2018) processes, including a three-step strong and weak theory AIG approach that became a de facto standard for template-based AIG. Figure 2 illustrates the workflow and data transformation requirement in generating items. Entire item banks have been generated for high-stakes exams; for example, the Medical Council of Canada used AIG software Item Generator (IGOR) to create thousands of bilingual medical exam items from a set of modeled templates. Practical evaluations have also been conducted; a cost-benefit analysis by Kosh et al. (2018) found that AIG becomes more cost-effective than manual writing when roughly 200 or more items are needed within a single content area. They estimated that developing 40-item models with an initial cognitive model and software deployment incurs a high upfront cost, but once in place, hundreds of items can be generated

quickly, making up for the investment when large volumes are required. In K-12 math, they projected the break-even point at approximately 173–247 generated items per content domain.

As AIG matured, methodological diversity expanded to include what we now call non-template-based AIG approach consisting of rule-based, statistical/ NLP, neural learning approaches, and now LLM, each addressing distinct limitations of template-based systems. Rule-based frameworks codify expert-defined heuristics into generation rules, enabling explicit control over item features and supporting combinatorial test assembly via optimization methods (Geerlings et al., 2013). Statistical/NLP pipelines exploit corpus-based analyses of lexical and syntactic patterns to inform the generation of plausible stems and distractors, mitigating rigidity in rule-based systems while maintaining interpretability (Lai et al., 2016; Shin & Lee, 2024). Neural architectures, notably RNNs and LLMs, have introduced open-ended content synthesis capabilities, allowing algorithmic extrapolation beyond predefined templates and enhancing linguistic naturalness (Davier, 2018, 2019; Laverghetta & Licato, 2023). Falcão et al.'s review in medical education underscores the rising prominence of both rule-driven and neural approaches, highlighting the methodological diversification of contemporary AIG research (Falcão et al., 2022). In summary, over roughly four decades, AIG has evolved from a novel idea to a feasible solution for the test development bottleneck.

Today, two major paradigms coexist: template-based AIG, which employs systematic cognitive and item model frameworks, and non-template-based AIG, which leverages computational approaches ranging from rule-based systems to neural networks and transformer models (von Davier, 2018, 2019; Narayan et al., 2020). More recent explorations have incorporated LLM approaches such as prompt engineering, fine-tuning, and RAG (Laverghetta & Licato, 2023; Gorgun, 2024; Shin, 2021; Lee et al., 2023). This dissertation proposes a new

approach involving the use of an AI agentic system in assessment item generation, capitalizing on multi-agent workflow architectures that have demonstrated effectiveness in other domains such as software engineering, scientific discovery, and automated reasoning (Zhang et al., 2025). The following sections examine each of these approaches in detail, beginning with template-based AIG frameworks before turning to non-template-based methods.

Template based AIG

Template-based AIG represents the most established and widely implemented approach to automatic item generation, characterized by the systematic use of predefined structures or templates that specify how item components can be manipulated to generate new instances, as shown in Fig. 2.1. Gierl and Lai (2013) established a foundational three-step framework for template-based AIG that has become the standard approach in the field: developing cognitive models to identify and organize content domains, creating item models that translate cognitive specifications into operational templates, and systematically generating test items by combining feature values according to predefined constraints and rules.

Cognitive Model Development

Cognitive model development represents the foundational theoretical framework that guides the systematic creation of item generation systems in template-based AIG. Cognitive models serve as explicit representations of the task requirements and problem-solving processes that examinees must employ when responding to assessment items (Leighton & Gierl, 2007). Unlike traditional item development, where cognitive representations remain implicit and unique to individual item writers, cognitive models provide formal, structured specifications that can be replicated and systematically manipulated for item generation see Figure 3. Cognitive models in

educational testing can be defined as descriptions of human problem-solving on standardized educational tasks that help characterize the knowledge and skills examinees at different levels of learning have acquired (Gierl & Lai, 2017). These models organize cognitive and content-specific information to provide SMEs with formal, structured representations of how to create and solve tasks on tests. The fundamental purpose of cognitive models in AIG extends beyond simple item creation to encompass the specification of content, parameters, constraints, and instructions that control the behavior of the generative system.

A critical advantage of cognitive models lies in their explicit nature, which enables replication and evaluation. While traditional item development relies on implicit mental representations that vary across item writers, cognitive models provide detailed documentation of the content, logic, and reasoning skills required to generate and solve assessment tasks. This explicit specification allows for systematic quality control and ensures that generated items maintain consistency with intended learning outcomes and assessment objectives.

Advantages of Cognitive Models for AIG

Gierl et. al, 2021 stated three major advantages of Cognitive Models for AIG. The first is that the cognitive model is used to identify and organize the parameters, constraints, and instructions required to control the item generation process, with variables described using cognitive and content-specific information within a formal, structured representation (Embretson & Gorin, 2001). This organizational function enables SMEs to specify the cognitive and content features that must be manipulated to produce new items systematically.

Second, cognitive models facilitate inferences about examinees' thinking and problem-solving processes by providing structured representations of the content, parameters, constraints,

and instructions used to create assessment tasks. By identifying the knowledge and skills required to generate new items, cognitive models enable predictions about how examinees are expected to select correct and incorrect options from generated items, supporting both item development and score interpretation (Leighton & Gierl, 2011).

Third, cognitive models serve as explicit and formal representations of task-specific problem solving that can be evaluated to ensure generated items address their intended assessment purposes. This evaluation capability is particularly important in medical education contexts where items must accurately represent clinical reasoning processes and maintain content validity across multiple generations.

Stages of Cognitive Model Development

Cognitive model development occurs through a systematic four-stage process designed to ensure detailed specification and validation. In the initial stage, SMEs clearly define and articulate the requisite knowledge domains, content specifications, and reasoning processes necessary for constructing various assessment tasks, thereby establishing clear boundaries for the testing domain within which item generation will take place. This stage typically involves analysis of test specifications organized in two-way matrices representing content areas and cognitive skills (Lane et al., 2016), often structured hierarchically from simple to complex topics.

The second stage focuses on creating working cognitive models using parent items that represent the domain of interest. Parent items provide prototypical structures and exemplars for the types of items to be generated, though they are not strictly required for model development. Cognitive models are created inductively, with SMEs reviewing parent items to identify and describe information that could be used to generate new items based on parent characteristics. This

stage requires SMEs to be highly specific about content and parameters that will be manipulated, as these specifications directly affect the knowledge and skills examinees use to solve generated problems.

Stage three emphasizes model evaluation using formal procedures to assess accuracy and completeness. Given the direct relationship between cognitive model accuracy and the quality of generated items, evaluation procedures typically involve two-member SME teams where one SME develops the model and a second provides independent feedback. The evaluation focuses on three critical components: the content, which is the model's capabilities for generating test items, the logic that deals with the accuracy of knowledge and skills required to produce correct options, and the representation, that is, the adequacy of the model to reflect and cover the testing domain of interest (Gierl et al., 2021).

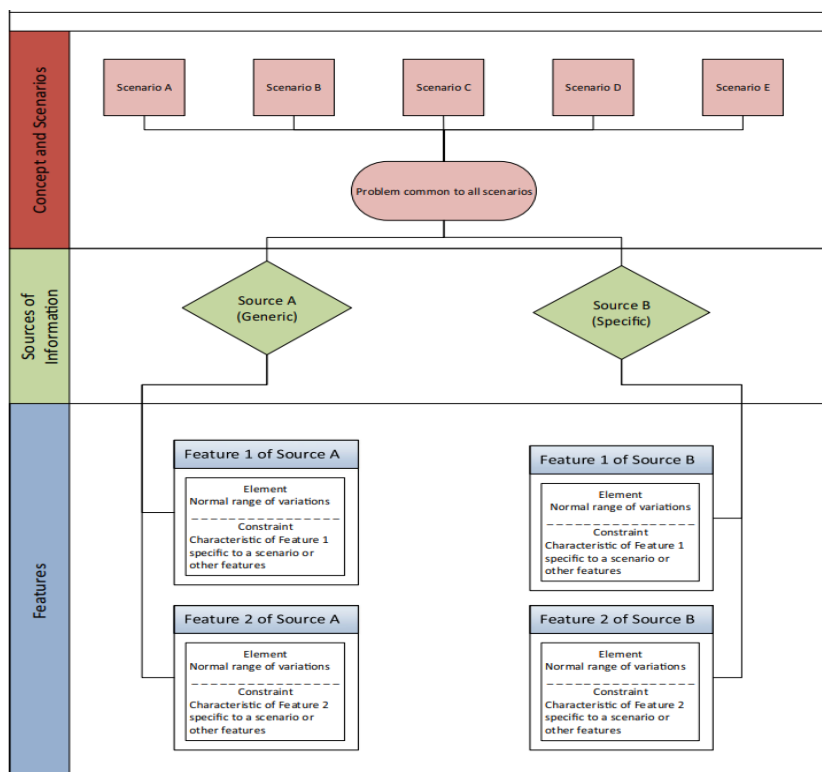
The fourth stage involves a thorough evaluation by independent SME groups focused on operational implementation and use. This stage employs standardized rating procedures, often using rubrics that evaluate different components of cognitive model quality. The goal is to identify and document cognitive models that are considered error-free and ready for operational use in item generation systems.

Types of Cognitive Models

Two primary types of cognitive models are commonly employed in AIG applications: the logical structures and key features models (Gierl & Lai, 2017). Logical structures and cognitive models are utilized when a specific concept or set of closely related concepts is operationalized as part of the generative process. These models are most suitable for measuring examinees' ability to use concepts across various content representations, often implementing formulas, algorithms, or

logical outcomes. The defining characteristic of logical structure models is that while content items can vary, the underlying concept remains fixed across generated items.

Figure 3. A general cognitive model structure for AIG (Gierl,Lai, & Tanygin, 2021)



Key features: Cognitive models are employed when task attributes or features are systematically combined to produce meaningful outcomes across item feature sets. These models focus on measuring examinees' ability to assemble and apply key features within domains, emphasizing problem-solving using systematic combinations of permissible features. The defining characteristic of key features models is that both content and key concepts can vary across generated items due to meaningful combinations of features, with systematic feature combination defined by constraints specified in the cognitive model.

Medical Education Applications

In the context of medical education, cognitive models have proven particularly valuable for generating items that assess clinical reasoning and diagnostic skills. Medical cognitive models typically incorporate multiple sources of information, including patient history, physical examination findings, laboratory results, and imaging studies. These models can specify how different clinical features should be combined to create realistic patient scenarios while maintaining diagnostic complexity and clinical authenticity. Gierl has worked extensively in these content areas, like abdominal injury (Gierl, Lai, & Zhang, 2018b), infection and pregnancy (Gierl & Lai, 2016b), chest trauma (Gierl & Lai, 2018), hernia (Gierl & Lai, 2013), post-operative fever (Gierl, Lai, & Turner, 2012), jaundice (Gierl & Lai, 2017), and dentistry (Lai et al., 2016), and many more.

Another example from (Gierl & Lai, 2017), where he defines a cognitive model for respiratory illness diagnosis that specify scenarios involving common cold versus seasonal flu, with features including patient demographics, symptom presentation, physical examination findings, and onset characteristics. The model includes constraints ensuring that specific symptom combinations align with clinically appropriate diagnoses, while allowing systematic variation in presentation details to generate diverse but realistic clinical scenarios. A plausible example of a medical cognitive model is shown in Figure 3. To illustrate this model, a medical example from surgery is presented to describe how physicians could manage a patient who is experiencing postoperative fever. Figure 4 contains the logic specified in the cognitive model required to coordinate the content presented in the item model in Table A1. The item model, as defined by Gierl et al. (2015), is a template that highlights how the features in an assessment task can be

manipulated to produce new items and translates the cognitive processes identified in Figure 3 into a structured framework for systematic item generation.

The explicit nature of medical cognitive models enables validation by clinical experts who can assess whether generated scenarios accurately represent real-world diagnostic challenges and maintain clinical plausibility. This validation capability is essential in medical education, where assessment items must not only measure knowledge and reasoning skills but also reflect current medical practice standards and patient safety considerations. The framework's emphasis on construct representation ensures that generated items maintain fidelity to the underlying cognitive processes that characterize expert medical practice (Someshwar, 2024).

Figure 4. A cognitive model for generating post-operative fever assessment tasks

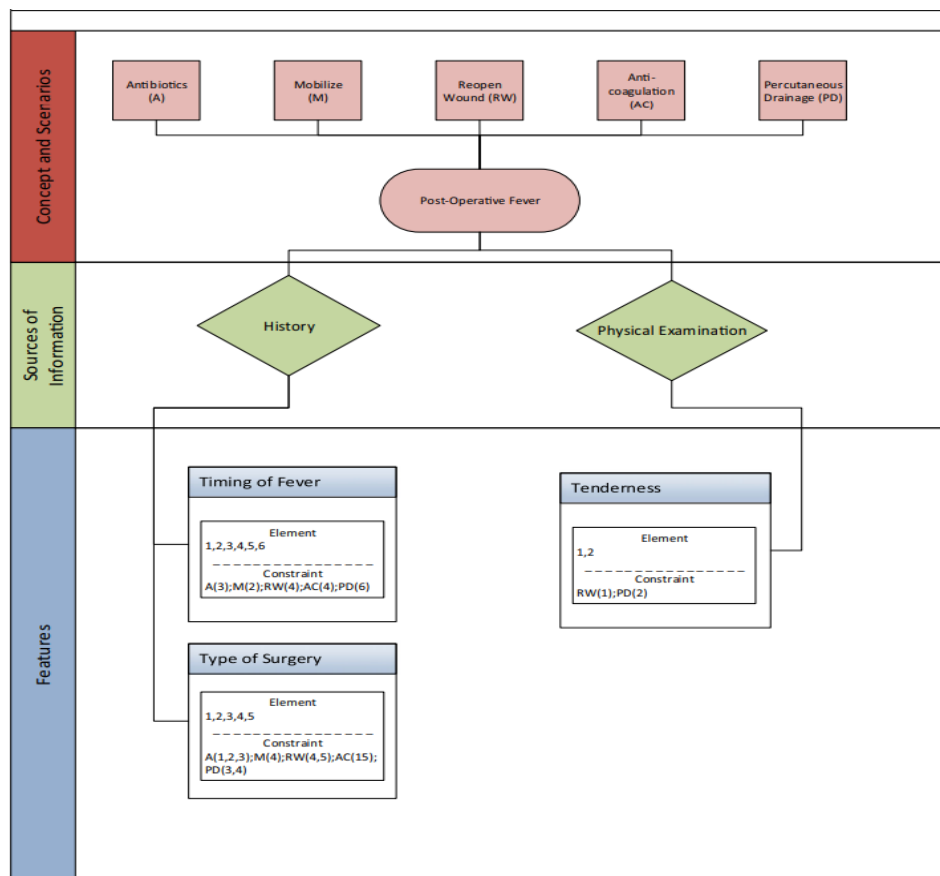


Table 1. An item Model for Generating Post-Operative Fever Assessment Tasks

Stem	A 40-year-old man has a [Type of Surgery]. On postoperative day [Time of Fever] he has a temperature of 35.5 C. [Tenderness]. Which one of the following is the best next step?
Elements	Timing of Fever: 1:1 2:2 3:3 4:4 5:5 6:6 Type of Surgery: 1. Gastrectomy 2. Right hemicolectomy 3. left hemicolectomy 4: appendectomy 5: laparoscopic cholecystectomy Tenderness: 1: Physical examination reveals a red and tender wound at the opening. 2: Physical examination reveals tenderness in the abdominal region with guarding and rebound. 3: Empty
Options	1: Antibiotics (A) 2: Mobilize (M) 3: Reopen wound (RW) 4: Anticoagulation (AC) 5: Percutaneous drainage (PD)
Key	Variable

Item Modeling

Having established that cognitive models provide the theoretical foundation for identifying content structures and difficulty-driving features, the next step in Gierl and Lai's (2013) three-step template-based AIG framework is item modeling. While cognitive models define *what* to assess and *why* certain features influence difficulty, item models operationalize this knowledge into structured templates that specify *how* items are systematically generated. Item modeling represents

the operational component of template-based AIG, providing the systematic framework for creating structured templates that can generate multiple item instances while maintaining psychometric consistency or predictable psychometric properties. Item models, also referred to as item templates, schemas, item forms, or item shells in the literature, serve as the primary skeleton of test items that contain essential components to define and model test items from a parent item (Gierl & Lai, 2020). It can also be defined as an item model (LaDuca et al., 1986; Bejar, 1996, 2002; Bejar et al., 2003; Gierl et al., 2008) that serves as an explicit representation of the variables in an assessment task, which includes the stem, the options, and oftentimes auxiliary information.

Components of Item Models

Item models consist of three primary components that work together to create a complete assessment task (Gierl et al., 2015), stem, options, and auxiliary information. The stem contains the main question or prompt to which the examinee is required to respond, incorporating variables that can be systematically manipulated to create different versions. For example, in medical assessments, the stem might include variables for patient age, symptoms, or clinical presentation that can be modified while maintaining the underlying diagnostic challenge. The options component specifies the structure of response choices in multiple-choice items, ensuring that the correct answer is consistently present while distractors are systematically varied. In medical contexts, distractors often represent common diagnostic errors or alternative conditions with similar presentations, providing educational value beyond mere assessment. Auxiliary information encompasses any supporting materials necessary for item completion, including images, tables, graphs, or contextual details. In medical item models, this might include laboratory results, imaging studies, or patient history data that can be systematically varied to create realistic clinical

scenarios while maintaining the core assessment objective. A third sub-component, elements, refers to the specific components within the stem, options, and auxiliary information that can be varied or manipulated to create new items. These are the specific components within the stem, options, and auxiliary information that can be varied or manipulated to create new items. The elements within each component represent the specific variables that can be manipulated during item generation. These elements can be strings (text-based variables) or integers (numerical values), and it define how content will vary across generated items Bejar et al., 2003. In medical assessments, elements might include patient demographics, symptom descriptions, laboratory values, or treatment options.

Item Model Taxonomy and Structure

Gierl, Zhou, and Alves (2008) developed a taxonomy for classifying item model types based on stem and option characteristics. The stem taxonomy categorizes models as independent, which are defined as elements unrelated to each other, dependent are elements directly related, mixed is defined as a combination of independent and dependent elements, and fixed elements, which are a constant format with no variation. Option classification includes randomly selected, basically distractors chosen randomly from content pools, constrained options are generated according to specific formulas or relationships, and fixed options are invariant options across all generated items. This taxonomy provides essential guidance for item model design and helps ensure systematic coverage of different item structures.

Theoretical Approaches to Item Model Development

Item models can be developed using two primary theoretical approaches, each with distinct advantages and limitations. The weak theory approach focuses on generating calibrated items

automatically from design guidelines using a theory of invariance (Drasgow et al., 2006). This approach begins with a parent item whose psychometric properties are known and systematically manipulates incidental elements' surface features that create apparent differences without altering underlying psychometric characteristics. For example, in a medical item assessing diagnosis of a respiratory condition, incidental elements might include the patient's name, age, or presenting location (e.g., emergency department vs. outpatient clinic), which change the surface appearance of the item without altering the clinical reasoning required to answer correctly. Incidental elements are elements that serve as surface features of the item models that cause generated items to appear differently without changing the psychometric properties of the parent item (Gierl & Lai, 2013). The strong theory approach represents a more ambitious but complex methodology that aims to generate calibrated items automatically from design principles using cognitive theories of difficulty (Drasgow et al., 2006). This approach emphasizes the cognitive processes required for item solutions and attempts to predict psychometric properties based on theoretical models of task complexity. According to Gierl and Lai, while strong theory approaches offer the potential to eliminate field testing by predicting item characteristics a priori, their practical implementation remains limited due to the scarcity of well-developed cognitive theories in most assessment domains.

Layered Item Models

Item model sophistication can be enhanced through layered approaches that increase generative capacity and content diversity. These approaches are divided into two 1-layer item models and N-layer item models. The 1-layer item model manipulates a relatively small number of elements at a single level, providing straightforward implementation but limited variation.

These models work well for domains with clear structural relationships, such as mathematical problem-solving or basic medical scenarios. N-layer item models allow manipulation of larger numbers of elements across multiple levels, enabling nonlinear manipulations that dramatically increase generative capacity (Gierl & Lai, 2012). In medical education, n-layer models can simultaneously vary patient demographics, clinical presentation, diagnostic workup, and treatment options, creating complex scenarios that more authentically represent clinical practice. An n-layer medical item model might include a demographic layer (age, gender, medical history), a presentation layer (symptoms, physical findings, vital signs), a diagnostic layer (laboratory results, imaging findings), and a management layer (treatment options, monitoring parameters). This layered approach enables the generation of diverse clinical scenarios while maintaining clinical authenticity and assessment validity.

Gierl and Lai's (2013) three-stage framework provides the most widely adopted methodology, beginning with cognitive model development to identify mental representations required for item solutions. The framework proceeds through content identification for item models, leveraging cognitive theories to specify elements and constraints, and concludes with computer-assisted item generation using systematic algorithms. This structured approach has been successfully implemented in various systems including IGOR, which demonstrates the practical feasibility of large-scale item generation from well-designed models Gierl et al. 2012.

Advantages of template-based AIG

Template-based Automatic Item Generation (AIG) offers a host of advantages across cost, psychometric quality, security, scalability, and content control. It dramatically improves cost efficiency and scalability by enabling rapid generation of large volumes of items from a single

template, hundreds or thousands of items can be produced from a well-designed model, making it a far more economical alternative to traditional manual item writing (Gierl & Haladyna, 2013; Kosh, 2018). This mass production capability not only reduces the per-item development cost but also addresses item scarcity issues in large-scale assessments (Someswar, 2024). In her dissertation, she examined the impact of variation within AIG item families on examinee scores, demonstrating that well-designed item models can maintain score precision even when generating large numbers of items from a single template. Her findings underscore the practical viability of template-based AIG for sustaining item banks at scale, provided that quality control mechanisms are in place to monitor within-family variation. In terms of psychometric quality, template-based AIG produces isomorphic items that are structurally and psychometrically equivalent by manipulating content variables within fixed cognitive and item models, thus maintaining construct alignment and difficulty balance (Gierl, Lai, & Zhang, 2016; Shin, 2021). Moreover, strong-theory AIG systems allow developers to predict item difficulty a priori by systematically associating specific item features with psychometric properties, enabling potential pre-calibration without field testing (Fu, 2022; Gierl & Lai, 2013). AIG also enhances test security by facilitating the creation of extensive item banks with interchangeable variants, reducing the likelihood of item exposure and cheating (Gierl & Haladyna, 2013; Handbook of Test Development, 2015). Because each template can yield a nearly infinite number of content variants, it becomes harder for examinees to memorize or share specific test items, especially in high-stakes computer-based testing environments (Someswar, 2024). Finally, the template-based approach promotes standardization and content control since all items originate from predefined item models, they adhere to structural consistency, minimize item-writing flaws, and reduce the influence of

individual writer biases (Gierl & Lai, 2016; Shin, 2021). This ensures high-quality item pools are systematically developed and easier to manage across operational cycles.

Despite their advantages, template-based frameworks exhibit important constraints in content coverage and flexibility. Because they rely on subject-matter experts to construct detailed cognitive and item models, the process can be time-intensive and difficult to scale, especially in domains where content cannot be easily deconstructed into discrete features and values. As Shin (2021) explains, template-based AIG is particularly challenging to implement for assessments like reading comprehension, where evaluating inferential knowledge, sentiment, and main ideas require a level of abstraction not easily captured through fixed templates. This leaves room for subjectivity and error in item creation (Rush et al., 2016). For instance, Masters et al. (2001) indicated that more than 70% of the textbook questions in a nursing item bank were erroneous due to item-writing flaws. These limitations have prompted the exploration of non-SME's based item generation called non-template-based AIG.

Non-Template-based AIG

In the past decade, template-based approaches have dominated AIG research and practice. Non-template-based frameworks have emerged as promising alternatives that minimize human intervention in the item generation process. According to Shin (2021), non-template-based AIG creates the essential components to define an item, the stem, options, and auxiliary information directly without any predefined item templates. These leverage advances in natural language processing, machine learning, and artificial intelligence to generate items directly from source materials without requiring predefined templates. (Aist, 2001; Brown et al., 2005; Chen et al., 2006; Devlin et al., 2018; Mazidi, 2017; von Davier, 2018, 2019; Shin, 2021; Ackerman & Balyan, 2023; Gorgun, 2024; Laverghetta & Licato, 2023; Lee et al., 2023; Narayan et al., 2020). Table 2

summarizes the key distinctions between template-based and non-template-based approaches. This review will build on the three approaches for non-template-based AIG from Shin's (2021) dissertation work to more modern approaches, organized into four broad categories: syntax-based approaches, NLP methods, neural/deep learning, and LLM-based approaches.

Table 2. Comparison of Template-Based and Non-Template-Based AIG Approaches

	Template-Based AIG	Non-Template-Based AIG
Cost	Low marginal cost after setup. Ideal for	Low authoring cost, but
Efficiency	large-scale use.	review/editing raises concern.
Scalability	One template can generate hundreds of consistent items.	Can generate many items fast.
Psychometric Quality	Consistent quality and difficulty control with theory-driven design.	Item difficulty and quality are unpredictable. Needs testing.
Content Flexibility	Effective for structured content like math or science.	Good for diverse or unstructured content like reading.
Operational Feasibility	Used in exams with high SME-reviewed acceptance rates.	Experimental use, not yet common in high-stakes exams.
Test Security	Supports security via item variants and rapid regeneration.	Promising for individualized items

Rule-Based Approaches

Rule-based AIG systems employ linguistically motivated algorithms to identify and extract potential item content from source texts. Early item generation systems often fell in this category.

For example, the Mathematics Test Creation Assistant (TCA) described by Singley & Bennett (2002) allowed item writers to take an existing math problem and “*variabilize*” it. The user would identify surface features that could be changed without altering the problem structure, such as numeric values, object names, or contexts, and then specify lists of permissible substitutions. The software would then generate new items by combining these variations. These systems typically utilize syntactic parsing, semantic analysis, and discourse processing to identify text segments suitable for item generation (Aist, 2001; Brown, Frishkoff, & Eskenazi, 2005). Chen, Liou, and Chang (2006) developed a semi-automated grammar question generation system that employed pattern-matching techniques to identify grammatical structures suitable for assessment. Their system used regular expression matching to identify question candidates and generate plausible but incorrect distractors based on common error patterns.

Similarly, Brown et al. (2005) created a vocabulary assessment system that utilized WordNet, a large lexical database, to generate six types of vocabulary questions, including definitions, synonyms, antonyms, hypernyms, hyponyms, and cloze items. This system demonstrated the potential for leveraging existing linguistic resources to automate item generation in specific domains. Another example of rule-based generation is figural or puzzle items. Researchers like Freund et al. (2008) showed that by implementing well-defined algorithms, one can automatically generate *matrix reasoning items* (à la Raven’s Progressive Matrices) with known psychometric properties. In their approach, rules for combining shapes, colors, and patterns were encoded such that each generated matrix puzzle had a predictable solution and difficulty level. However, the approach is limited by the completeness and quality of the rules. Writing a set of rules that covers all the nuances of language and content is challenging. Inevitably, rule-based systems might create some awkward or nonsensical items if the rules don’t account for an

exception. For instance, a naive cloze question rule might remove a word that makes the sentence ambiguous or unanswerable (Mostow & Jang, 2012).

Statistical/NLP Methods

As computing power and data availability grew, researchers explored statistical and natural language processing (NLP) methods for item generation. This AIG method utilizes corpus-based methods and natural language processing techniques to extract item content and generate assessment tasks. These methods typically involve training statistical models on large text corpora to identify patterns and relationships that can be exploited for item generation (Mitkov & Ha, 2003; Heilman & Smith, 2009). Semantic-based approaches focus on analyzing the meaning structure of texts to generate items that assess deeper levels of comprehension. Aquino, Chua, Kabiling, Pingco, and Sagum (2011) developed a semantic-based system that analyzed text meaning to generate reading comprehension items, while Gütl et al. (2011) employed combined statistical, structural, and semantic methods for automatic question generation.

Topic modeling approaches, particularly Latent Dirichlet Allocation (LDA), have shown promise for identifying thematic content within large document collections that can serve as the basis for item generation. These methods can automatically discover latent topics and generate items that assess understanding of key concepts and relationships (Shin, 2021). Shin applied a hybrid Automatic Item Generation (AIG) approach that integrated template-based and non-template-based methods to generate reading inference test items. A key methodological component was the use of *Latent Dirichlet Allocation* (LDA) this is a probabilistic topic modeling algorithm that identifies hidden thematic structures within large collections of text by representing each document as a mixture of topics, particularly a sentiment-weighted LDA, to uncover subtopic

structures within narrative texts (e.g., the *Harry Potter* series). The weighted LDA incorporated sentiment scores to enhance topic coherence and differentiation. After identifying subtopics through this enhanced topic modeling, Shin used rule-based parsing to categorize subtexts and then employed item models to generate inference-based reading comprehension items. The LDA-supported process allowed for logical arrangement and meaningful variation in content, improving the system's ability to generate complex items suited for reading comprehension tasks. NLP methods have significantly broadened the scope of AIG. They enabled item generation in domains where creating an explicit cognitive model was impractical. For example, generating reading comprehension questions from arbitrary passages can be attempted by NLP techniques that identify important sentences or infer possible questions, something that a rigid template from an expert might not handle across all topics. These methods have been the steppingstone to even more advanced AI-driven generation, as they introduced less human templating and leveraged large text datasets and language technologies to create items.

Neural/Deep Learning Approaches

The past decade has seen the rise of neural network approaches to automatic item generation, leveraging the advances in deep learning and natural language generation. These approaches mark a shift from manually defined rules or simple statistics to models that learn how to generate items from data. Neural AIG systems typically use sequence-to-sequence models or other neural architectures to generate questions, prompts, or even entire items that include the correct answer and distractors based on input data such as a text passage or a knowledge base. Du, Shao, and Cardie (2017) developed one of the earliest neural systems for reading comprehension item generation using recurrent neural networks. Their system demonstrated the ability to generate

various types of WH-questions from reading passages, with human evaluation indicating that the generated items were linguistically fluent and appropriately calibrated in difficulty. More recent approaches have leveraged transformer architectures, particularly BERT, for item generation tasks. Narayan, Simoes, Ma, Craighead, and McDonald (2020) developed a transformer-based sequence-to-sequence model that could generate factual questions directly from reading passages, demonstrating improved performance over earlier neural approaches. Gao et al (2019) introduced difficulty-controlled item generation using sequence-to-sequence models that could incorporate specified difficulty levels into the generation process.

In a two-stage exploration of deep learning for AIG, von Davier (2018, 2019) first applied RNNs to generate multiple-choice items from open-access datasets, showing that RNNs could replicate the structural features of human-authored items, though the generated content occasionally exhibited grammatical inconsistencies and lacked the clinical specificity required for operational use. In a subsequent 2019 study, he shifted to Transformer-based models by fine-tuning OpenAI's GPT-2 on a large medical corpus from PubMed. This transition enabled the generation of more coherent, contextually appropriate item stems and distractors, demonstrating improved language modeling and applicability to domains such as medical certification exams. The progression illustrated the rapid advancements in neural architecture and their potential to support scalable, high-quality item development in educational assessment.

The evolution of neural-based approaches from RNNs to transformer-based architectures marked a pivotal shift in automated item generation, significantly enhancing the coherence, contextual relevance, and scalability of generated content. However, these early deep learning applications, while powerful, still rely on models trained on narrow domains or limited datasets,

often requiring extensive fine-tuning and computational resources. The emergence of large language models (LLMs), such as GPT-3, Claude, Gemini, and beyond, represents a new paradigm in which general-purpose models pretrained on massive, diverse corpora can be adapted for educational assessment with minimal additional training or no training at all. This shift opens up a new methodological landscape. Rather than relying solely on training data, LLMs enable novel approaches like *prompt engineering*, *retrieval-augmented generation (RAG)*, and *agentic systems* to guide item creation. These methods leverage pretrained knowledge and external data sources to generate high-quality, domain-aligned items efficiently, offering a more flexible, modular, and scalable framework for automatic item generation.

Large Language Models in AI

The emergence of Large Language Models (LLMs) represents a potentially transformative development in all fields, and in our case, automatic item generation offers unprecedented capabilities for generating diverse, contextually appropriate assessment content across multiple domains. This section examines the current state of LLM applications in AIG, their capabilities and limitations, and their potential integration with existing AIG frameworks. Specifically, we review three principal paradigms by which LLMs have been applied to AIG tasks or their potential. Prompt engineering, fine-tuning, and retrieval-augmented generation (RAG) drawing on recent empirical studies and theoretical analyses.

Prompt engineering leverages zero- and few-shot in-context learning of pre-trained LLMs to steer item generation without modifying model weights. Sayin and Gierl (Sayin & Gierl, 2024) used this approach for reading comprehension, constructing “text analysis cognitive models” encoded as prompts that guide GPT-3 to produce inferential questions across passages. Their work

demonstrates that carefully designed prompts can elicit items aligned with cognitive targets while maintaining high linguistic quality and variability (Sayin & Gierl, 2024; Login, 2024) used prompt templates to generate plausible multiple-choice distractors for Russian reading comprehension, showing that the strategic insertion of context and exemplar distractors can markedly improve distractor plausibility over naïve generation. Zelikman et al. demonstrated a two-stage approach in which GPT-4 was first prompted to generate initial test items according to expert-curated rules and then to simulate student responses for automatic difficulty and ambiguity estimation, thereby automating both item production and preliminary calibration (Zelikman et al., 2023). These works illustrate that prompt design, ranging from zero-shot directives to chain-of-thought exemplars, serves as an accessible entry point for AIG practitioners, requiring no model weight updates while offering flexibility across domains.

When prompt-based methods reach their limits, fine-tuning provides a more tailored path to optimizing LLMs for AIG tasks by updating model parameters on domain-specific corpora or item-response data. Bulut and Yildirim-Erbasli fine-tuned open-source transformers on reading comprehension texts and associated questions, finding that domain-adapted models produced richer stories and more assessment-aligned items than their off-the-shelf counterparts albeit with continued reliance on human review to ensure quality (Bulut & Yildirim-Erbasli, 2022). Alshehri advanced this concept in the cybersecurity education context by constructing a multi-layer agent framework that integrates fine-tuned LLMs at distinct stages of question generation, reasserting the potential for enhanced item design across diverse domains (Alshehri, 2025). These studies confirm that fine-tuning on curated items or response datasets substantially enhances an LLM's capacity to generate assessment items with desired psychometric properties and domain relevance. Instruction tuning, a specialized fine-tuning variant, further refines LLM behavior by training large

collections of (instruction, output) pairs, thereby aligning model outputs with user intents across diverse tasks. Gorgun and Bulut (2024) explored this strategy using an instruction-tuned Llama 3-8B to automate quality control checks including error classification, item improvement suggestions, and validation of criteria such as content relevance and answerability on automatically generated cloze items. Addressing limitations of existing evaluation methods that rely on reference items, resource-intensive field testing, or slow and inconsistently documented human review (Amidei et al., 2018; Kurdi et al., 2020), their findings indicated that the model could effectively classify item quality, suggest targeted revisions, and assess psychometric properties, offering a scalable alternative to evaluating all generated items rather than a selected subset. Their feasibility study highlighted that instruction-tuned LLMs can serve as scalable second readers in AIG pipelines, mitigating the bottleneck of human expert review without extensive additional annotation efforts. This approach suggests a promising route to embedding quality assurance within the LLM generation loop itself.

Retrieval-Augmented Generation (RAG) methods offer yet another dimension for enhancing LLM-based AIG by grounding item generation in external knowledge bases or curricular repositories. Although RAG has been extensively applied in open-domain question answering, its application to AIG remains nascent; integration could allow LLMs to retrieve domain-specific facts, formulas, or standards and then generate contextually accurate items that reflect the latest curricular content or empirical data. This approach was extended to medical item generation by Makinde (2025), who presented findings at the National Council on Measurement in Education (NCME) annual conference demonstrating that LLM-generated medical assessment items performed indistinguishably from human-written items in expert evaluation. By combining retrieval of canonical content with generative flexibility, RAG-powered AIG systems stand to

improve both the fidelity and topical relevance of generated items, especially in rapidly evolving fields such as computing and biomedicine.

Despite these advances, several challenges persist in LLM-driven AIG. Model hallucination, in which generated items include factually incorrect or nonsensical content (Ji et al., 2023; Huang et al., 2025), poses risks for assessment validity and requires robust detection and correction mechanisms (Gorgun & Bulut, 2024). Bias in training data can propagate into unfair or unbalanced item pools, necessitating systematic bias audits and inclusive fine-tuning strategies (Bulut & Yildirim-Erbasli, 2022). Moreover, psychometric calibration beyond difficulty estimation, such as discrimination indices and test-information functions, remains largely manual or simulation-based, pointing to the need for tighter integration of LLM outputs with item-response theory (IRT) frameworks and automated simulation environments. This dissertation extends these LLM-based AIG paradigms by developing an agentic AI system that integrates generation and reviewing agents within a structured framework. Drawing inspiration from multi-agent workflows in which individual agents function as nodes (discrete processing units that perform specific tasks) connected by edges (directional links that route data and instructions between agents) to form coordinated processing pipelines (Zhang et al., 2025), specialized agents were designed for the cognitive model, item models, item reviewer, and finally, human-in-the-loop correction via the SME Review system.

Agents and Multi-Agent Systems in Educational Assessment

In educational assessment, Mittal (2017) defined an agent as an autonomous software entity that can perform specialized tasks independently or in collaboration with other agents. Multi-agent systems (MAS) comprise multiple agents working in concert to handle complex problems that

would be difficult for a single system to solve (Jiang et al., 2024). Key characteristics of agents include autonomy (acting without continuous human oversight), proactivity (taking initiative based on goals and data), and sociability (communicating and coordinating with other agents). In an MAS, tasks can be decomposed into sub-tasks and distributed among different agents, which then solve them collaboratively to achieve better results (Jiang et al., 2024). This paradigm is well-suited to educational contexts, where complex, distributed activities such as ongoing assessment, scoring, feedback provision, and item generation benefit from being handled by multiple coordinated agents. Mittal's MASME system (Multiagent System for Management and Evaluation of Examinations) demonstrates this approach in testing administration. In MASME, an Examiner Agent is responsible for assembling question papers, evaluating student responses, and compiling results, while a Student Agent presents the questions to test-takers and collects their answers. These agents operate autonomously using an item bank and predefined scoring rules, cooperating to simulate the roles of a human examiner and proctor. The implementation of MASME with the JADE agent framework (Java Agent Development Environment) showed that such agents can perform exam tasks and reduce human effort in evaluation, with automated scores correlating well with human-assigned scores. This example underlines how agent-based systems can manage assessment workflows by delegating sub-tasks to dedicated agents, thereby increasing efficiency and consistency.

Agents for Item Generation and Test Development

A particularly promising application of agents in assessment is in AIG. Traditional AIG research has produced large numbers of items using cognitive models and templates (item models), but maintaining quality and alignment with test specifications remains challenging (Gierl & Lai, 2013). Recent advances with large language models (LLMs) enable more flexible item generation,

yet quality control is a persistent concern (Lee et al., 2025). Here, an agent-based approach can offer a systematic solution. The idea is to mirror the collaborative workflow of human test development teams by assigning different agents to different item development roles. For example, one agent assigned as an “Item Writer Agent” can focus on generating draft questions based on content requirements, while another agent acts as a “Content Reviewer” to check the alignment of each item to the test specifications and subject matter accuracy (Lee et. al. 2025). Additional agents might serve as a “Bias and Fairness Reviewer,” detecting insensitive or biased language, or as a “Language Editor,” ensuring clarity and correct grammar. Each agent operates with a degree of autonomy in its specialty, but they communicate their findings or suggestions to the group. This reflects the multi-agent principle where specialized knowledge is encapsulated in different agents that work together to produce a high-quality outcome.

Importantly, these agent roles in AIG are grounded in well-established test development practices. Test construction typically involves content experts, psychometricians, and editors who sequentially and iteratively refine items. An agentic system simply maps these expert functions to AI agents. For instance, a Content Reviewer agent would emulate the work of a subject-matter expert, ensuring content validity. By dividing responsibilities, the system can embed checks and balances. The generation agent proposes an item, the review agents critique it, and perhaps a Coordinator agent or workflow engine decides if the item needs revision or can advance to the next stage. This approach echoes the evidence-centered design principle of separating the task (item stimulus) from the evidence evaluation, but here each aspect is handled by an intelligent agent specialized for that task.

The use of multiple agents in AIG facilitates a modular, iterative workflow that directly addresses many challenges of LLM-driven item generation. Instead of relying on a single large model to “do it all” (and then catching errors after the fact), an agent-based system builds quality control into the generation process. Lee and colleagues (2025) introduced an LLM-based multi-agent AIG framework (LM-AIG) that exemplifies this approach. In their framework, different agents handle item writing, content review, linguistic checks, bias detection, and item revision, with a final integration of human feedback before an item is considered complete. Notably, the agents operate in a cooperative manner: the generation agent produces an initial draft, then passes it to the review agents, which may flag issues or suggest modifications, and an iterative loop ensues until the item meets all predefined quality criteria. This design embeds multi-stage quality assurance throughout the AIG pipeline rather than limiting review to a post hoc step. By the time an item is finalized, it has effectively gone through a multi-stage vetting process akin to a human committee’s review, but much faster and more consistently. Early studies report that such multi-agent AIG systems can indeed improve item quality and diversity. For example, a recent multi-agent item generation system for medical licensing exams (USMLE) produced questions that were judged to be more relevant and diverse compared to those generated by a single LLM prompt, while also adhering closely to content standards (Jiang & Feng, 2025). These findings suggest that breaking the task into agent-mediated sub-tasks can harness the strengths of LLMs and mitigate their weaknesses through targeted oversight at each step.

Conceiving an AIG system as a collection of interacting agents provides a powerful organizational framework. It leverages the strengths of distributed autonomous processes, anchoring them in sound test development practice through human guidance. This multi-agent perspective not only addresses the challenges of scale, efficiency, and complexity in modern

assessment development but also creates opportunities for enhanced adaptability and innovation. As the next chapter (Methodology) will detail, this dissertation builds on the multi-agent-based approach, where it implements a coordinated set of agent-like components to tackle the specific challenges of large language model-driven item generation in a principled way. By integrating cognitive model, item models, and reviewers into a cohesive workflow, the system aims to ensure that automated item generation is at once scalable, theoretically grounded, and quality-controlled, ultimately marrying the speed of AI with the wisdom of human experts in assessment design.

While AI methods offer promising advancements in automatic item generation, critical decisions with high stakes, particularly those involving life-dependent implications, should not yet be entrusted solely to autonomous systems. Any of the four approaches discussed can support such development, but until we can confidently ensure the validity, reliability, and fairness of these systems, maintaining human oversight remains essential and ethically responsible. This oversight may take the form of human-in-the-loop or human-on-the-loop frameworks. In the following section, we will examine validity frameworks for automatic item generation in greater detail.

Validity in Automatic Item Generation

The application of artificial intelligence to assessment development introduces fundamental challenges to traditional approaches of validation, requiring careful examination of how established validity frameworks apply to AI-generated content. While the core principles of validity theory remain relevant, the unique characteristics of AI generation processes necessitate expanded approaches to evidence collection and evaluation. This section synthesizes foundational validity frameworks, examines the centrality of content related validity for AI-generated items,

and presents the Duolingo Responsible AI framework as a practical approach to validating AI-powered assessments.

Foundational Theories of Validity in Assessment

Validity theory in assessment has evolved dramatically since the early 20th century, progressing from fragmented conceptualizations to sophisticated unified frameworks that emphasize evidence-based argumentation. Early work in the 1910s–1940s emphasized criterion validity, correlating test scores with external benchmarks and content validity by ensuring test content matched the domain. By the 1950s, construct validity emerged as a formal concept. Cronbach and Meehl’s landmark 1955 paper introduced construct validity as the overarching notion encompassing all sources of evidence. Through the 1960s and 1970s, educational testing often treated validity as a stable “property” of the test itself, employing a componential or "trinitarian" view with separate content, criterion, and construct validity. Textbook definitions of the era focused on content coverage (Cureton, 1951; Ebel, 1961) and criterion correlations as the "gold standard." By the late 1970s and 1980s, however, scholars like Cronbach began to question the fragmented approach. Cronbach (1980, 1988) and others noted that treating validity as disjoint pieces led to overlap and confusion. This paved the way for a unified conception.

A major shift occurred in the 1980s when Samuel Messick integrated all traditional validity types under a single construct-based theory. Messick (1989) argued that content, criterion, and construct-related evidence were not independent types of validity but rather complementary facets of a unified validity framework, and that the social consequences of test score interpretation and use should also be considered as part of validity assessment. Specifically, Messick demonstrated that content and criterion evidence are subsumed as part of construct validity, and he introduced

the idea that test consequences, social or ethical impacts, are part of validity itself. This unified validity view expanded the focus from merely technical correlations to broader inferences with multiple lines of evidence: content, substantive, structural, generalizability, external, and consequential. However, this synthesis was not without controversy. Many scholars expressed concern over its practicality and its ability to provide actionable guidance for test developers. Lissitz and Samuelsen (2007), for example, argued that the unified framework obscured the centrality of content validity in evaluating educational tests and proposed separating test evaluation into internal aspects including content representation, internal structure, and latent processes with external aspects such as criterion relationships and social consequences. The framework prioritized evaluating test content independent of construct theory. These debates paved the way for the next major development, which is the argument-based approach advanced by Michael Kane in the 1990s, which sought to provide a more practical structure for validation by laying out an interpretation/use argument supported by specific sources of evidence.

Kane (1992, 2006, 2013) articulated interpretation-use arguments (IUA) that explicitly lay out the chain of reasoning from observed scores to inferences and decisions. Kane's work helped operationalize validation by constructing an argument supported by multiple lines of evidence. At the same time, scholars like Shepard (1993, 2016) and Chalhoub-Deville (2009, 2016, 2020) stressed consequences and usage. Shepard, for instance, emphasized evaluating intended and unintended test consequences (washback) as part of validity.

By the 2010s, validity theory embraced an eclectic, evidence-centered stance. The 2014 AERA/APA/NCME *Standards for Educational and Psychological Testing* formally outlined five sources of validity evidence (content, response processes, internal structure, relations to other

variables, and consequences). As standards note, content evidence involves alignment of test content with the intended construct, while internal structure evidence pertains to the degree to which item relationships conform to the construct. At this point, validity was no longer viewed as an inherent property of a test but as a relationship between test scores and their intended interpretations supported by diverse evidence. New frameworks also emerged, one of which is evidence-centered design (Mislevy et al., 2003), it emphasized designing assessments around evidentiary claims, and Embretson's (2007, 2017) integrated framework that combined cognitive design with psychometric evidence. Collectively, these paradigms, Messick's unified theory, Kane's argument-based validation, Mislevy's ECD, Lissitz and Samuelsen (2007), Embretson's (2007, 2017), and Chalhoub-Deville (2020) contemporary synthesis recast validity as an integrated evidence-based argument rather than a checklist of disparate types, laying the groundwork for a new validity framework tailored to AI-generated assessments.

This progression from fragmented validity types to integrated evidence-based argumentation directly informs the present study's approach to validating AI-generated items. Rather than selecting one framework to the exclusion of others, this dissertation draws on the complementary strengths of each: Messick's insistence on unified evidence, Kane's practical argumentation structure, Lissitz and Samuelsen's prioritization of content representation, Mislevy's design-centered specification, and Chalhoub-Deville's emphasis on impact by design. The challenge is identifying an operational framework that can synthesize these contributions while addressing the AI-specific concerns that none of these foundational theories were designed to accommodate.

Content-related validity for AI-Generated Items

The evolution of content validity conceptualization in educational measurement provides crucial theoretical foundations for understanding its primacy in AI-generated assessment contexts. The 1974 Standards originally defined content validity as "required when the test user wishes to estimate how an individual performs in the universe of situations the test is intended to represent" (AERA, APA, & NCME, 1974, p. 28). This foundational definition established the central concern that remains critical for AI-generated items: ensuring that test content adequately represents the intended performance domain.

By the 1985 Standards, the terminology shifted to "content-related evidence," reflecting the field's movement toward a unified conception of validity while maintaining focus on the fundamental question of content representativeness. The current Standards define content-related validity evidence as "evidence based on test content that supports the intended interpretation of test scores for a given purpose" and emphasize that such evidence must address "fidelity of test content to performance in the domain in question and the degree to which test content representatively samples a domain" (AERA, APA, & NCME, 2014, p. 218).

Messick's (1989) influential conceptualization further clarified that content-related evidence "is evaluated by showing how well the content of the test samples the class of situations or subject matter about which conclusions are drawn" (p. 16). This definition proves particularly relevant for AI-generated items because it emphasizes the systematic relationship between test content and the broader domain of inference, a relationship that must be explicitly programmed into automated generation systems rather than relying on human expertise to maintain domain alignment.

Given the unique challenges posed by AI-generated assessment content, content-related evidence warrants particular emphasis within the integrated validity framework not as a separate type of validity, but as the evidentiary foundation upon which all other sources of validity evidence depend. This prioritization is consistent with Lissitz and Samuelson's (2007) argument that content representation deserves centrality in evaluating educational tests, and it is especially salient in AIG contexts where automated generation introduces risks of deviation from intended constructs, learning objectives, or content standards. Unlike evidence based on relations to other variables or consequences, which can be evaluated post hoc, content-related evidence requires systematic attention to the generation process itself, ensuring that AI-produced items faithfully represent the intended performance domain before other forms of validation can meaningfully proceed. The established methods for collecting content validity evidence are professional judgment, analyses documenting relevance and representativeness to targeted domains, and think-aloud protocols. This can be adapted for AI-generated contexts where the generation process itself becomes a critical component of content validation. Professional judgment remains essential but must now evaluate not only the appropriateness of generated items but also the adequacy of generation specifications and algorithms. Domain representativeness analyses become more complex when automated systems must maintain appropriate sampling across potentially thousands of generated items Gierl et al. 2016. The systematic evaluation procedures that characterize traditional content validation must be extended to encompass the algorithmic processes that determine how AI systems translate domain specifications into actual test content. This foundation in established content validity theory provides the conceptual framework for developing validation approaches that address the unique characteristics of AI generation while maintaining the rigor required for high-stakes assessment applications.

Mislevy's Evidence-Centered Design (ECD) framework provides essential methodological foundations for establishing content validity in AI-generated items. Mislevy et al. (2003: 1) describe assessment design as fundamentally "an exercise in evidentiary reasoning," where test developers must articulate explicit claims about examinees, specify the evidence needed to support those claims, and design tasks that elicit appropriate evidence. This approach offers advantages for AI-generated content because it requires explicit specification of the relationships between task characteristics and construct evidence before generation begins. In the ECD framework, task models specify how assessment tasks should elicit observable behaviors, while evidence models define how those observations should be interpreted as evidence about underlying constructs. For AI-generated items, these models could provide the theoretical foundation for ensuring that generation processes maintain appropriate construct alignment. The explicit nature of ECD models enables systematic evaluation of whether AI systems are producing items that adequately represent intended construct domains a central requirement for content validity evidence. Mislevy's emphasis on validity by design becomes particularly important for AI-generated items because it embeds validity considerations directly into the generation process rather than attempting to establish validity post hoc. This design-centered approach addresses one of the primary challenges in AI-generated assessment: ensuring that automated processes maintain systematic relationships to intended constructs across multiple generations. This study operationalizes ECD principles through the agentic framework's blueprint-driven cognitive planning, where clinical scenarios are specified before item generation begins ensuring that construct-relevant evidence is designed into the generation process rather than evaluated only after items are produced.

Lissitz and Samuelson's (2007) advocacy for a validity framework that prioritizes content representation and internal structure evidence provides additional theoretical support for

emphasizing content-related evidence as the foundational concern in AI-generated assessment contexts. They argue that Messick's unified framework, while theoretically sophisticated, may obscure the fundamental importance of establishing that tests measure what they purport to measure. Their narrower definition of validity emphasizes that content representativeness and internal structure coherence should be established before other forms of validity evidence can be meaningfully interpreted. For AI-generated items, the Lissitz and Samuels framework offers practical advantages by focusing attention on the most immediate validity concerns: whether automated generation processes produce items that adequately sample the intended content domain and whether those items exhibit appropriate psychometric relationships. This focused approach acknowledges that while consequences and external relationships are important, they depend fundamentally on establishing that the generated content appropriately represents the intended constructs.

Chalhoub-Deville and O'Sullivan's (2020) examination of validity in their seminal work *Validity: Theoretical Development and Integrated Arguments*, provides crucial theoretical grounding for understanding why content validity assumes primary importance in AI-generated assessment contexts. Central to their framework is the concept of "impact by design," which they argue should guide validity considerations from the earliest stages of test development (Chalhoub-Deville & O'Sullivan, 2020). This approach places consequences at the forefront of the evidence chain, requiring that the effects of testing on individuals, groups, and society be prioritized throughout the development process rather than evaluated post hoc.

Their theoretical framework emphasizes that validity must be understood through systematic connections between test content and intended constructs via explicit theoretical

frameworks. This perspective proves particularly relevant for AI-generated items, where the relationship between generation specifications and resulting content may be less transparent than in traditional human-authored assessments. As Chalhoub-Deville and O'Sullivan (2020) note, testing in reform-driven systems "is policy-driven and is intended to effect fundamental and encompassing changes at the individual, group, and societal levels" (p. 142), highlighting the critical importance of establishing clear theoretical foundations that can guide automated generation processes.

The "validity by design" framework demands that test developers articulate explicit theories about how assessment content will function within broader educational and social contexts before generation begins. For AI-generated items, this requirement becomes even more complex because automated systems must operationalize these theoretical relationships across potentially thousands of generated items while maintaining consistency with intended impact goals. The systematic specification of content domains and cognitive processes that must guide AI generation requires explicit theoretical frameworks that can be translated into algorithmic parameters without losing sight of the broader consequential goals that justify the assessment enterprise.

Chalhoub-Deville and O'Sullivan's (2020) emphasis on integrated arguments that synthesize multiple forms of evidence provides a methodological foundation for addressing the unique challenges posed by AI generation. Their framework requires that content validity evidence be systematically connected to other validity considerations through coherent theoretical arguments rather than treated as isolated technical requirements. For AI-generated items, this integrated approach ensures that content-related validation procedures maintain appropriate relationships to the broader validity argument while addressing the specific transparency and

consistency challenges introduced by automated generation processes. In this dissertation, this integration is operationalized through the DET assessment ecosystem's responsible AI framework (Burstein et al., 2022), which organizes validity evidence across four dimensions validity and reliability, fairness, privacy and security, and accountability and transparency enabling systematic mapping of each stage of the agentic generation and review pipeline to specific evidentiary claims. Chalhoub-Deville and O'Sullivan's framework thus provides the theoretical justification for why this dissertation requires a validation approach that is both integrated and operationally specific one that connects content-related evidence to fairness, transparency, and accountability within a single coherent structure. The DET responsible AI framework (Burstein et al., 2022) meets this requirement.

For this dissertation, content-related evidence serves as the primary evidentiary concern not because other sources of evidence are unimportant, but because content fidelity is a prerequisite for all downstream validation. If AI-generated items fail to represent the intended cardiovascular disease constructs, evidence based on internal structure, external relationships, or consequences becomes uninterpretable. This prioritization is inclusive of the broader validity frameworks discussed above. It reflects Messick's unified conception by treating content evidence as one essential strand within an integrated argument, operationalizes Kane's interpretive chain by establishing that the first inferential link from item content to construct must hold before subsequent links can be evaluated, and aligns with Lissitz and Samuelsen's position that content-representation deserves centrality without dismissing the relevance of external evidence.

Validity Frameworks in the Context of AI-Generated Items

The foundational theories reviewed above collectively establish that validating AI-generated assessment content requires a framework that integrates content-related evidence within a broader argument addressing fairness, transparency, and accountability concerns that traditional validity frameworks acknowledge in principle but do not operationalize for AI contexts. The DET Responsible AI Standards (Burstein et al., 2022) were selected for this dissertation because they uniquely satisfy this requirement: they preserve the integrated evidence-based argumentation advocated by Messick and Kane, foreground content representation as Lissitz and Samuelson recommend, embed design-centered specification consistent with Mislevy's ECD, and operationalize impact-by-design as Chalhoub-Deville advocates while adding the AI-specific governance dimensions (privacy, security, accountability, transparency) that none of these foundational frameworks address. This is not a departure from classical validity theory but an extension of it into a domain where automated generation introduces risks that require explicit governance structures.

Framework Overview and Theoretical Foundations

The DET Responsible AI framework addresses test equity and quality through four integrated standards that represent ethical principles aligned with contemporary validity theory while addressing AI-specific concerns (Burstein et al., 2022). The framework builds on classical validity theory while incorporating principles from responsible AI governance, creating a systematic approach to validation that addresses both technical quality and ethical considerations. The four standards are: Validity and Reliability, Fairness, Privacy and Security, and Accountability and Transparency. These standards collectively address the full scope of validity concerns while

providing specific guidance for AI-powered assessment contexts. Each standard includes explicit goals and subgoals that operationalize theoretical principles into practical validation procedures, offering concrete guidance for test developers working with AI-generated content.

1. Validity and Reliability Standard: Content-Focused Validation

The Validity and Reliability standard serves as the foundation of the DET framework, directly addressing content related validity concerns that are central to AI-generated assessment. This standard focuses on ensuring that tests are suitable for their intended purposes and that AI systems contribute to it rather than compromise validity evidence. The standard includes two primary goals: specifying processes required to build validity arguments and evaluating AI used in test item creation, item calibration, and scoring. The content validity components of this standard are operationalized through specific subgoals that address the relationship between AI generation processes and construct representation.

Subgoal 1.1.1 requires developing descriptions of test target domains to ensure that test items align with intended constructs. This requirement directly addresses the content specification challenges identified by Chalhoub-Deville (2020) and operationalizes ECD principles by requiring explicit articulation of construct domains before AI generation begins. The framework's approach to AI-generated content emphasizes systematic human expert involvement in domain specification and generation oversight. Subject matter experts articulate target constructs using established frameworks, such as the Common European Framework of Reference for language assessment. These expert specifications then guide AI generation processes, ensuring that automated systems maintain appropriate relationships to intended constructs. This systematic approach addresses the

content validity challenges posed by AI generation while maintaining the efficiency advantages of automated processes.

Subgoal 1.1.2 addresses AI scoring system accuracy and fairness by requiring systematic evaluation of automated scoring processes using human expertise. This requirement recognizes that content validity extends beyond item generation to encompass how generated items are scored and interpreted. The framework requires that AI scoring systems be evaluated against human expert judgments to ensure that automated processes maintain appropriate construct interpretation.

The emphasis on explainable scoring methods in Subgoal 1.1.3 directly addresses transparency concerns in AI-generated assessment. The framework requires that AI scoring features have clear alignment with domain constructs and that scoring decisions can be explained in terms of theoretically grounded construct interpretations. This requirement operationalizes the theoretical emphasis on construct clarity advocated by Lissitz and Samuelson (2007) while addressing the unique interpretability challenges posed by AI systems.

2. Fairness Standard: Representational and Procedural Equity

The Fairness standard addresses content related validity concerns by ensuring that AI-generated items represent diverse populations appropriately and avoid systematic biases that could affect construct measurement. This standard recognizes that content validity requires not only appropriate domain coverage but also equitable representation that does not introduce construct-irrelevant variance.

Subgoal 2.1.5 requires developing fairness and bias item review principles that eliminate construct-irrelevant barriers while ensuring cultural and linguistic accessibility. This requirement directly addresses content validity by ensuring that AI-generated items measure intended

constructs rather than irrelevant cultural knowledge or linguistic complexity. The systematic human review processes specified in this subgoal operationalize content validation procedures while addressing AI-specific bias risks.

The framework's approach to demographic representation in AI training data (Subgoal 2.2.1) addresses content validity by ensuring that AI systems are trained on data that represents the diversity of intended test populations. This requirement recognizes that biased training data can lead to systematic differences in how AI systems generate content for different demographic groups, potentially introducing construct-irrelevant variance that threatens content validity.

3. Privacy and Security Standard: Protecting Assessment Integrity

While primarily focused on data protection and test security, the Privacy and Security standard includes provisions that directly support content validity. The standard's emphasis on fair and reliable test security protocols (Goal 3.3) ensures that AI-powered security systems do not introduce construct-irrelevant factors that could affect score interpretation. The human-in-the-loop AI proctoring protocols specified in Subgoal 3.3.1 demonstrate how AI systems can enhance rather than compromise content validity by providing more detailed and consistent security monitoring while maintaining human oversight of critical decisions. This approach addresses concerns about AI transparency while leveraging technological capabilities to improve assessment conditions.

4. Accountability and Transparency Standard: Documentation and Stakeholder Communication

The Accountability and Transparency standard supports content related validity by requiring complete documentation of how AI systems are used throughout the assessment ecosystem. Goal 4.2 specifically requires documenting AI used for building validity arguments, test item creation,

test item calibration, and scoring. This documentation requirement operationalizes the transparency principles essential for content validity evaluation in AI-generated assessment contexts. The framework's emphasis on stakeholder communication (Goal 4.5) ensures that validity evidence is communicated clearly to diverse audiences, supporting the broader validity argument while maintaining public confidence in AI-powered assessments. This requirement recognizes that content validity depends not only on technical quality but also on stakeholder understanding and acceptance of assessment processes.

Integration with Traditional Validity Theory

The DET Responsible AI framework demonstrates how traditional validity principles can be adapted to address AI-specific challenges while maintaining theoretical coherence with established validity frameworks. The framework's emphasis on content validity aligns with the theoretical positions advanced by Chalhoub-Deville, Mislevy, and Lissitz and Samuelsen while providing practical guidance for implementation in AI-powered assessment contexts. The integrated nature of the framework reflects contemporary understanding that validity is not a property of tests but a relationship between test scores and their intended interpretations, supported by multiple sources of evidence. By addressing technical quality, fairness, security, and transparency within a unified framework, the responsible AI dimensions introduced in the DET ecosystem (Burstein et al., 2022) operationalize the validity arguments required for high-stakes assessment while addressing the unique challenges posed by AI technologies. This framework provides a practical model for how validity theory can be extended to address AI-generated assessment content while maintaining the rigor required for appropriate score interpretation. The explicit attention to content validity within each standard ensures that fundamental validity

requirements are addressed throughout the assessment development process, providing a foundation for validity arguments in AI-powered assessment contexts.

In adopting this framework, the present study takes the position that content-related evidence is the foundational but not exclusive concern for validating AI-generated items. The DET framework operationalizes this position by embedding content validation within each of its four standards ensuring that validity and reliability, fairness, security, and transparency all serve the overarching goal of producing items that faithfully represent intended constructs while meeting the ethical obligations that accompany AI use in high-stakes assessment. This inclusive approach honors the field's progression from fragmented validity types to integrated argumentation while responding to the new demands that AI technologies impose on the validation enterprise.

Beyond Duolingo, general AI ethics guidelines also inform these practices. The IEEE Global Initiative on Ethics (2017) and the U.S. NIST AI Risk Management Framework (2023) both stress principles like transparency, explainability, bias mitigation, and human oversight in AI. In the testing domain, these translate into concrete steps: documenting prompt engineering, continually monitoring item statistics for unexpected drift or bias, and engaging experts in reviewing AI outputs. Duolingo's DET RAI standards have been explicitly validated against the NIST framework's trustworthiness characteristics, showing alignment between test-specific and general AI best practices. Importantly, the DET RAI model is one of the most detailed test-specific frameworks proposed for AI item generation. It demonstrates how classic validity evidence (content alignment, statistical fit, consequence analysis) can be extended into an "AI-compatible" protocol that also addresses fairness, security, and accountability. This dissertation follows a similar multi-faceted strategy: using test blueprints and SME review to enforce content validity, train and calibrate human raters, conduct small and large peer reviews of AI items, and then pilot

them with students. All these steps mirror traditional test development but with added documentation and review steps to satisfy the RAI principles, thereby ensuring high-quality, valid AI-generated items for medical education.

The literature review done in this section establishes that validating AI-generated assessment content demands a multi-stage approach grounded in complementary theoretical traditions. No single validation event, whether automated screening, expert review, or psychometric analysis can satisfy the evidentiary requirements that Messick (1989), Kane (2006), and the Standards (AERA et al., 2014) articulate for high-stakes assessment. Content-related evidence must be established first, as Lissitz and Samuelson (2007) argue, because downstream evidence based on internal structure, external relationships, or consequences is uninterpretable if the generated items fail to represent intended constructs. Yet content based evidence alone is insufficient: Mislevy et al.'s (2003) evidence-centered design demonstrates that construct alignment must be specified before generation begins, Chalhoub-Deville and O'Sullivan (2020) insist that fairness and impact considerations be embedded throughout the development process, and the responsible AI literature (Burstein et al., 2022) identifies transparency and accountability as governance requirements that classical validity frameworks were not designed to address. Taken together, these perspectives converge on a validation model in which multiple stages each informed by a distinct theoretical rationale contribute complementary evidence to a unified validity argument. It is this convergence that motivates the multi-stage validation pipeline and the adoption of the DET responsible AI framework in the present study.

Cost and Efficiency Evidence

The economic rationale for AIG rests on the well-documented costs of traditional item development. Rudner (2010), drawing on operational data from the Graduate Management

Admission Test, estimated that producing a single high-stakes item costs \$1,500 to \$2,500 when accounting for the full pipeline of SME writing, editorial review, field testing, psychometric analysis, bias screening, and the substantial attrition of items that fail to survive this process. Kosh et al. (2019) corroborated a similar range of \$1,500 to \$2,000 per item for large-scale K–12 assessments. These figures reflect a development model in which each item is, as Drasgow et al. (2006, p. 473) observed, “individually hand-crafted written, reviewed, revised, edited, entered into a computer, and calibrated as if no other like it had ever been created before.” Drasgow et al. (2006) further argued that item construction represents the critical bottleneck in test development, as the anticipated cost savings from computerized adaptive testing have not materialized precisely because of continuous item production demands.

Template-based AIG fundamentally restructures this cost equation by shifting the unit of analysis from individual items to generative models (Gierl & Lai, 2016). Empirical throughput data are compelling: Gierl et al. (2012) generated 1,248 unique postoperative surgery items from a single item model in approximately six hours, while Falcão et al. (2023) reported that SMEs trained in AIG principles produced 9,496 items compared to only 25 items manually a roughly 380-fold increase in output capacity. The Medical Council of Canada’s IGOR system, a well-documented operational deployment, generated tens of thousands of items across more than 50 cognitive maps over five years of collaborative development with the University of Alberta (De Champlain, 2015; Gierl & Lai, 2016a). Kosh et al. (2019) conducted the only published cost–benefit analysis of AIG, comparing step-by-step costs against manual item writing for K–12 mathematics. Their results indicated that AIG becomes more cost-effective when producing a minimum of 173 to 247 items within a single fine-grained content area. Below this threshold, the

upfront investment in cognitive model development and technology deployment exceeds per-item savings. Critically, their analysis also revealed that subsequent batches of content areas require only 126 items to reach break-even, as initial infrastructure costs are not repeated (Kosh et al., 2019). Despite these gains, setup costs remain substantial: cognitive model development requires approximately three hours per model, item model creation an additional two hours, and the MCC invested over five years of research collaboration before achieving full operational deployment (De Champlain, 2015; Gierl et al., 2021).

The transition to LLM-based AIG represents a qualitative shift in the cost structure. Whereas template-based systems require extensive upfront model development before generation can begin, LLMs can generate items from natural-language prompts in seconds. Kıyak and Kononowicz (2025) provided the most direct comparison, showing that their hybrid AI-template approach completed item model development in only 10 minutes compared to approximately five hours for traditional template methods a roughly 30-fold reduction in template development time. Raw API costs are negligible: at current GPT-4o pricing, generating a single multiple-choice question (MCQ) costs approximately \$0.002 to \$0.006, compared to \$1,500–\$2,500 for traditional development. However, raw generation speed does not equate to operational efficiency. Review cycles constitute the most consequential hidden cost across all AIG approaches. Embretson and Kingston (2018) stated directly that “the efficiency of AIG will be mitigated if the generated items must be submitted to traditional, time-consuming review processes.” For template-based items, review can be conducted at the model level rather than the item level (Gierl & Lai, 2016), and blinded evaluations have shown that expert panelists cannot distinguish AIG items from human-authored ones at rates above chance (Gierl & Lai, 2013; Pugh et al., 2020). For LLM-generated

items, the evidence is more mixed. Linde et al. (2026) found no significant differences in difficulty or discrimination between GPT-4o and human-authored medical MCQs, but a recent Medical Teacher study reported that AI-generated items had a higher rate of requiring major revision (Wu et al., 2025). Gorgun and Bulut (2024) positioned LLM-based automated evaluation as a potential intermediate step between generation and field testing to reduce the cost and time of multiple revision rounds.

The multi-agent paradigm represents the most recent evolution in AIG efficiency, yet empirical cost data remain entirely absent. Lee et al. (2025) introduced the LM-AIG framework using specialized agents for item generation, content review, linguistic evaluation, bias assessment, and revision mirroring the roles of human test development teams. Firoozi et al. (2025) developed a related system for automated quality control of MCQ items. The theoretical efficiency case is that automated review agents could reduce human review bottlenecks by catching content errors, linguistic flaws, and bias before SME evaluation. However, neither study reports comparative efficiency metrics: no items-per-hour, no cost-per-item, and no comparison of first-pass acceptance rates between multi-agent and single-model generation. Circi et al. (2023) observed that “very little guidance exists in the current AIG literature for testing programs that want to decide if AIG is right for them,” and Tan et al. (2025), reviewing 60 studies using LLMs for AIG, found that most focused-on quality evaluation rather than cost analysis. Three critical measurement gaps thus limit the evidence base for evaluating agentic LLM frameworks: (1) no study has reported standardized efficiency metrics for LLM-based item generation; (2) no comparative data exist on first-pass acceptance rates for multi-agent versus single-model generation; and (3) No reported costs of prompt engineering, automated review infrastructure, and validation overhead

remain unquantified. These gaps represent significant opportunities for empirical contribution, particularly for research comparing agentic frameworks against both traditional and single-model baselines using standardized cost and time metrics.

CHAPTER III: METHODS

Building on earlier chapters, this chapter detailed the methodological design for developing and validating an agentic LLM based framework for AIG in medical education assessment.

In chapter one we introduced the background and rationale for this study, highlighting the limitations of traditional and template-based AIG systems and articulating the need for a new framework that integrates LLM within an agentic, human-in-the-loop system. The chapter also presented the overarching research purpose: to determine whether LLM-driven agentic systems can generate high-quality, valid, and psychometrically defensible assessment items aligned with established medical education standards.

Chapter 2 provided a thorough literature review tracing the evolution of AIG from its early prose-based and template-based foundations to modern neural and LLM-based paradigms. It examined how cognitive modeling, item modeling, and emerging multi-agent systems inform current practices, and emphasized content-related validity and content fidelity as the most critical validity considerations for AI-generated items. The review introduced the Duolingo English Test (DET) Responsible AI framework, whose four standards are: Validity & Reliability, Fairness, Privacy & Security, and Accountability & Transparency offer guiding principles for responsible and valid AI-based assessment practices.

Research Questions

Grounded in these foundations, the present chapter outlines the methodological framework designed to answer the following research questions:

- 1) How can large language models be integrated into a designed agentic framework to develop assessment items?
- 2) To what extent do we have alignment between the agentic framework items and SME reviewed items based on established test specifications and content standards?
 - a. How effective is the electronic review system developed to help SME review AI generated items for medical education assessment?
 - b. What validation criteria and checkpoints are necessary to ensure content related validity, fairness, and psychometric properties of AI-generated test items, and how do these compare to those used for human-authored items?
- 3) To what extent does the agentic LLM framework improve efficiency in terms of item development time, review cycles, and cost per item compared to conventional item development methods?
- 4) What are the psychometric properties (difficulty, discrimination, reliability, dimensionality) of test items generated by the agentic LLM approach, and how do these properties compare to those of traditionally developed items from previous and current test administrations?

Methodological Overview

To address these questions, this chapter details the methodological procedures used to design, validate, and evaluate the agentic AIG framework. The methodology employs a sequential two-phase approach with iterative refinement between phases based on expert feedback. The framework integrates dual large language model architecture (generative and evaluative functions) operating through a multi-agent system coupled with structured human expert review processes. Validation procedures align with the DET Responsible AI framework, and psychometric evaluation employ Classical Test Theory (CTT), Item Response Theory (IRT) models for comparative analysis against traditionally developed items. Resource utilization metrics including computational costs, processing time, and development efficiency are systematically tracked to evaluate the framework's scalability and cost-effectiveness relative to conventional item development approaches.

Research Design

This dissertation employs a mixed-methods concurrent embedded design to systematically evaluate an agentic AIG framework using LLMs for medical education assessment. The research approach integrates quantitative psychometric evaluation with qualitative validation processes to assess the quality, validity, and utility of AI-generated test items. The study addresses fundamental questions regarding development, validation, efficiency, and psychometric properties of items generated through a multi-agent system, while establishing benchmarks for comparison with traditionally developed assessment items.

The methodological framework transforms the established three-step AIG paradigm from *Create Cognitive Model* → *Create Item Model* → *Generate Items* (Gierl & Lai, 2013) to *LLM Cognitive Model* → *Agentic System* → *Item Generation*. This transformation represents a

significant departure from template-based approaches, incorporating contemporary artificial intelligence capabilities while maintaining rigorous psychometric standards required for high-stakes medical assessment.

Figure 5. Gierl et al. 2021 Workflow and data transformation required to generate items

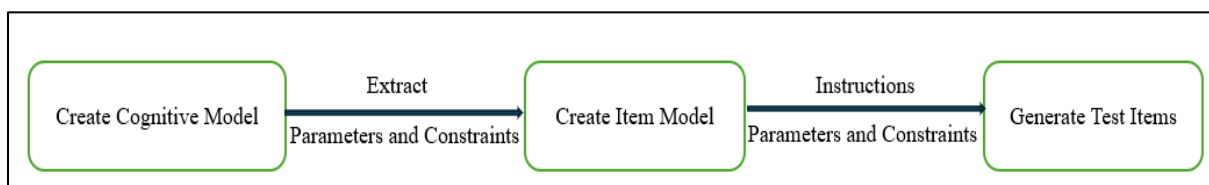
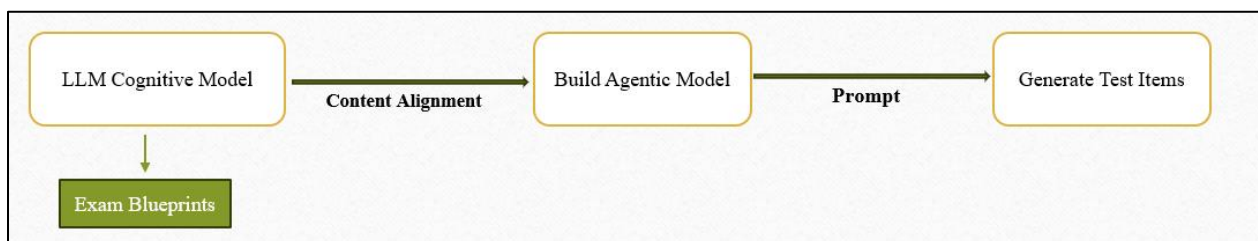


Figure 6. AIG Workflow for LLM Generated Assessment Items



Study Setting and Context

This research was conducted in partnership with the Physician Assistant Education Association (PAEA), which administers End-of-Rotation (EOR) examinations for physician assistant programs nationwide. The study focuses on cardiovascular content across five core clinical specialties required for PA education: Emergency Medicine (EMEOR), Family Medicine (FMEOR), Internal Medicine (IMEOR), Surgery (SUREOR), and Pediatrics (PEDEOR). This selection reflects both PAEA's examination structure and the prevalence of cardiovascular conditions across clinical settings, ensuring generated items address clinically relevant content with broad educational applicability. Cardiovascular topics provide sufficient complexity and

clinical diversity to evaluate the agentic framework's capabilities while maintaining practical relevance for PA education and practice.

Subject Matter Experts

The study engaged ten subject matter experts (SMEs) distributed equally across the five medical specialties (two reviewers per specialty) for the small peer review and four to eight reviewers for the large peer review. SME selection criteria included board certification in the relevant medical specialty, minimum five years of clinical experience, current faculty or staff affiliation with a PAEA member program, prior involvement in assessment development, and active participation in medical education. This composition ensures specialized expertise while providing sufficient reviewer independence for systemic quality evaluation.

Ethical Considerations

This study was determined to be exempt from IRB review per University of North Carolina at Greensboro Institutional Review Board guidelines because it utilizes de-identified secondary data provided by the Physician Assistant Education Association (PAEA). All data were de-identified prior to transmission to the researcher and contain no personal identifiable information. SME participation in item review processes was voluntary and professional in nature, involving standard test development activities. The LLM implementation via OpenAI Enterprise API ensures data privacy and security through enterprise-grade protections, with expert reviewer identities and individual ratings remaining confidential.

Methodological Scope

This study focused on cardiovascular exam items for PA End of Rotation™ examinations. The two-phase generation approach provided data on iterative improvement. Psychometric validation relied on pilot testing data from a single administration cycle conducted by PAEA.

Detailed discussion of methodological limitations and their implications for generalizability appears in more detail in Chapter 5.

Technical Implementation

The framework employed a multi-agent system where specialized AI agents perform discrete item development functions including cognitive planning, item generation, quality evaluation, and targeted revision. As detailed in Chapter 2, contemporary agentic systems leverage LLMs to autonomously interpret cognitive specifications, generate contextually appropriate items, and conduct systematic quality evaluations while maintaining human oversight through structured expert review processes. The framework consists of four hierarchically organized agents, each with specialized responsibilities within the item generation workflow.

Cognitive Planning Agent

The Cognitive Planning Agent transforms examination blueprints and content specifications into structured cognitive models that guide item generation. This agent analyzes PAEA test specifications, learning objectives, and content standards detailed in exam blueprints to create systematic representations of domain knowledge and clinical reasoning requirements. The agent establishes the foundational framework for all downstream item generation processes. The Cognitive Planning Agent operates at the cognitive modeling stage, transforming blueprint specifications and topic inputs into structured Item Briefs that constrain subsequent generation. Each Item Brief specifies:

- Track assignment: EOR specialty (EMEOR, FMEOR, IMEOR, SUREOR, PEDEOR)
- Blueprint alignment: Specific blueprint code and learning objective
- Topic specification: Cardiovascular topic from approved content list

- Vignette outline: Structured plan including clinical setting, patient summary, and key clinical features (3-6 elements per vignette)
- Clinical key: The correct answer concept or intervention
- Distractor map: Three misconception-driven distractor intents reflecting common clinical reasoning errors
- Task constraints: Task area (e.g., diagnosis, management) and required lead-in template

The agent receives inputs including extracted blueprint text, cardiovascular topic lists, optional user-specified constraints (forced track, topic, or task area), and a per-item nonce for scenario variation. Outputs conform to a validated JavaScript Object Notation (JSON) schema (ItemBrief) that ensures downstream agents receive complete, well-structured specifications.

Track-Specific Writer Agents

Five specialized Writer Agents generate complete multiple-choice items from Item Briefs, with each agent operating under track-specific constraints that ensure age-appropriate patient presentations, clinically authentic settings, and specialty-aligned clinical reasoning patterns:

- Emergency Medicine Writer (EMEOR): Acute presentations, time-critical decisions, emergency department settings, adult patients (ages 18-65)
- Family Medicine Writer (FMEOR): Broad age range (newborn-geriatric), outpatient/home settings, preventive care, chronic disease management
- Internal Medicine Writer (IMEOR): Adult medical presentations (≥ 18 years), inpatient and outpatient contexts, complex diagnostic reasoning
- Surgery Writer (SUREOR): Perioperative scenarios, adult patients, surgical indications, procedural complications

- Pediatrics Writer (PEDEOR): Pediatric populations (0-18 years), developmental considerations, pediatric-specific pathophysiology

All Writer Agents employed the same underlying LLM (GPT-5 Nano) with identical model parameters but received track-specific Item Briefs that define specialty-appropriate content constraints, clinical contexts, and reasoning patterns. Generated items include a clinical vignette stem, four to five response options, a keyed correct answer, and an evidence-based rationale with supporting citations.

Reviewer Agent

The Reviewer Agent, powered by GPT-5 Mini, conducted systematic evaluation of generated items using an eight-dimensional rubric aligned with the SME evaluation framework detailed in Evaluation Framework section. The agent scores each item on:

- Clinical Relevance (1-5 scale)
- Content Accuracy (Yes/No)
- Currency (1-5 scale)
- Distractor Plausibility (1-5 scale)
- Format Compliance (1-5 scale)
- Language Clarity (1-5 scale)
- Bias-Freedom (1-5 scale)
- Clarity of Stem (1-5 scale)

Acceptance Criteria: Items must score ≥ 3.0 on all seven scaled dimensions (with Clinical Accuracy assessed as Yes/No) to advance to human SME review. Any dimension scoring below 3.0 triggers automatic rejection. This threshold established a minimum quality baseline for initial filtering, ensuring only fundamentally sound items proceeded to expert evaluation and reducing

SME review burden while maintaining rigorous quality control. The Reviewer Agent provided structured feedback for rejected items, identifying specific dimensions failing to meet criteria and suggesting improvement areas. Accepted items proceeded to the small peer review.

SME-Driven Rewrite Agent

Following an initial large peer review, the SME-Driven Rewrite Agent was incorporated to support targeted revision of items requiring minor corrections rather than complete regeneration. The agent accepted two inputs: (1) original item JSON containing complete item structure and metadata, and (2) free-text SME feedback indicating required changes.

The agent applies targeted revisions while preserving:

- Original track assignment (unless feedback explicitly requires change)
- Topic and blueprint alignment (unless fundamental content issues identified)
- Task area and cognitive complexity level (unless misalignment documented)

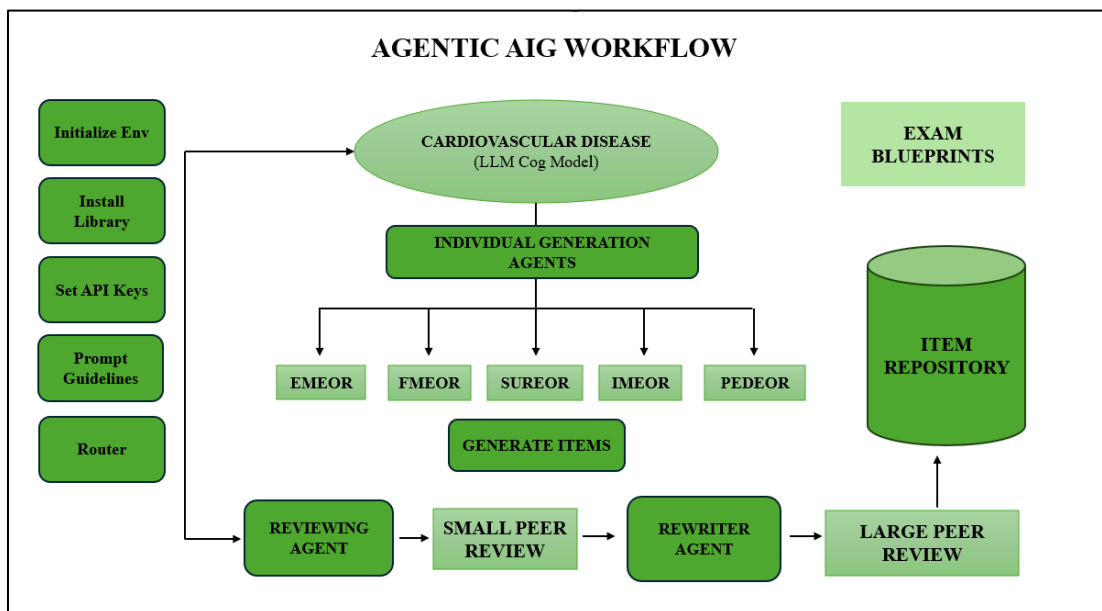
This design allows SME-guided refinement to be integrated efficiently without altering the core generation pipeline. Rewritten items are exported separately and retained alongside original versions for traceability. The Rewrite Agent maintains audit trails documenting original content, feedback received, and specific revisions applied, supporting transparency in the revision process. This agent operationalized standard test development practice where items with salvageable content are returned to item writers for correction rather than being discarded entirely, improving efficiency while maintaining quality standards.

System Infrastructure and Workflow Components

The agentic AIG workflow operates through several integrated technical infrastructure components that support the multi-agent development process:

- **Initialize Env:** Establishes the computational environment and system parameters required for the multi-agent system to operate, including setting up the runtime configuration and workspace.
- **Install Libraries:** Loads the necessary software dependencies and frameworks (LangChain, LangGraph, OpenAI SDK) that enable agent functionality and LLM communication.
- **Set API Keys:** Configures authentication credentials required to access the GPT-5 Enterprise API and other external services used by the system.
- **Prompt Guidelines:** Defines the standardized instructional templates and constraints that govern how agents formulate queries to the LLM and structure their outputs to ensure consistency with item writing standards.
- **Router:** Manages the flow of tasks and information between agents by directing items through appropriate review pathways and determining which agent processes each component based on specialty or function.
- **Item Repository:** Serves as the centralized database storage system that archives all generated and validated items for subsequent pilot testing and psychometric analysis.

Figure 7. Agentic AIG Workflow for LLM based assessment Items



Model Architecture

The framework employed dual LLM architecture optimized for distinct functional requirements. GPT-5 Nano powered the Cognitive Planning Agent and all five specialty-specific Writer Agents, handling creative scenario generation, clinical vignette development, and option formulation with cost-effectiveness for large-scale production. GPT-5 Mini also powered the Reviewer Agent, providing systematic quality evaluation across eight dimensions optimized for consistent multi-dimensional scoring. Both models were operated via OpenAI Enterprise API, ensuring enhanced data privacy protections and computational stability required for research applications involving sensitive medical content.

To ensure diverse item generation while maintaining structural constraints, the system employed a per-item nonce (random seed value) during cognitive planning that influenced task selection, patient demographics, clinical settings, and distractor frameworks. The nonce operated within track-specific constraints (age ranges, clinical settings) and blueprint specifications,

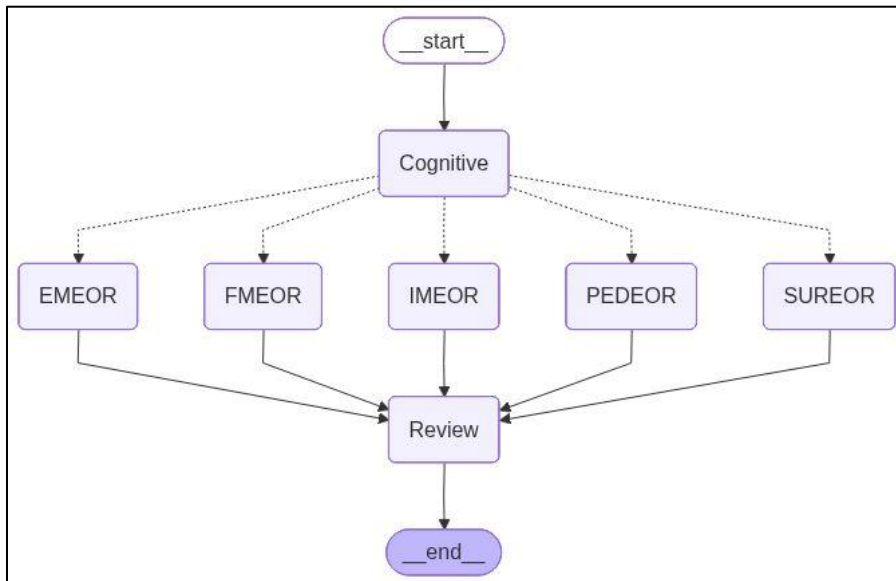
ensuring variation which enhanced item pool diversity rather than introduces construct-irrelevant variance.

The multi-agent system employed LangChain and LangGraph frameworks to coordinate workflow and communication between agents, with LangGraph structuring the system as a directed graph where nodes represent agents and edges define information flow. The framework maintained detailed logs of all agent interactions and decision points, providing complete audit trails supporting quality assurance and regulatory compliance requirements.

State-Based Information Flow and Agent Communication

The multi-agent workflow operates through a state-based processing model where a shared state object is passed sequentially between agents. This state encapsulates both content and metadata required at each stage of item development, ensuring complete information transfer and audit trail maintenance.

Figure 8. State-Based Information Flow



Items progressed through a fixed pipeline with routing decisions following explicit rules. When users designated a specific track, the system enforced this constraint throughout otherwise, routing defaults to the Cognitive Planning Agent's specification. Fallback mechanisms handle conflicts by defaulting to track assignments and logging discrepancies for manual review. Inter-agent interoperability relied on two standardized JSON schemas. The ItemBrief Schema from the Cognitive Planning Agent specifies track assignment, blueprint alignment, vignette structure, clinical key, and distractor framework. The Item Schema from the Writer Agents contains the complete item with vignette, response options, answer key, rationale, reviewer scores, and traceability metadata. Items failing validation at any handoff point are rejected immediately to prevent error propagation.

Rule-Based Compliance and Quality Assurance

Beyond LLM-based generation, the framework employs deterministic rule-based controls operating independently during post-processing. Automated compliance verification confirms correct lead-in placement, detects prohibited language patterns (negative phrasing, absolute qualifiers), validates required template elements, and ensures citation formatting compliance. Items violating critical rules are automatically rejected before SME review.

Automated bias detection flags stereotypical naming patterns, cultural assumptions, socioeconomic bias indicators, and monitors demographic representation across item pools, directing prioritized SME attention to flagged items. Response option randomization with programmatic answer key remapping eliminates systematic test-taking advantages and key-position bias. All items receive standardized metadata including structured identifiers, timestamps across processing stages, track and phase designations, and reviewer decisions, enabling complete

item-level traceability. This dual-layer approach (LLM-based generation plus rule-based verification) enhances system reliability and reduces systematic error propagation.

SME Review Platform Development

Custom R Shiny application was developed to facilitate systematic item review by subject matter experts. The application provides dual interfaces: an administrative interface enabling item assignment, progress monitoring, data export, and analytics generation, and a user interface presenting generated items in standardized formats with embedded evaluation criteria and streamlined data collection procedures. Security features include user authentication, data encryption, and detailed audit trail maintenance. The application, hosted on PAEA's Shiny server, supports both item quality assessment and in-platform item revision capabilities. The system captures both quantitative ratings across all evaluation dimensions and qualitative feedback through reviewer comments and recommendations.

Item Generation and Validation Procedures

The study employs a sequential two-phase generation approach designed to optimize item quality and pool composition based on iterative SME feedback.

Phase 1: Initial Large-Scale Generation

Following multi-agent system validation, the framework generated a substantial initial pool of multiple-choice test items distributed across the five medical specialties. This initial phase established a foundational item pool while allowing assessment of automated review performance and identification of systematic quality issues across tracks. All items underwent automated review using the Reviewer Agent's eight-dimensional rubric. Items scoring ≥ 3.0 on all dimensions advanced to the small peer review followed by a large peer review. Items that failed the automated review were rejected. The ≥ 3.0 advancement threshold was established in consultation with

PAEA's psychometric team based on the rubric's scale design, where 3.0 corresponds to "Satisfactory needs minor revision," representing the minimum acceptable quality for items to warrant further investment of review resources.

Inter-Phase Refinement Protocol

Following completion of Phase 1 reviews, generation prompts were systematically refined based on structured analysis of expert feedback from both small and large peer review sessions. The refinement process targeted recurring issues identified during human review, particularly format and style compliance, and distractor quality. Additionally, an SME-Driven Rewriter Agent was developed and integrated into the workflow to support targeted revision of items requiring minor corrections. The model architecture and configuration parameters remained unchanged, with improvements implemented through enhanced prompt engineering. This design allowed for isolation of prompt refinement effects on item quality while maintaining consistent model capabilities.

Refinement focused on enhanced PAEA formatting requirements and constraints, improved distractor development instructions with explicit misconception mapping, strengthened clinical authenticity criteria and guideline currency, explicit lead-in template enforcement, and simplified language requirements for clarity. The effectiveness of inter-phase refinement in improving item quality was assessed by comparing Phase 1 and Phase 2 outcomes, as reported in Chapter 4.

Phase 2: Targeted Supplementary Generation

Following prompt refinement and system enhancement, Phase 2 generated supplementary items to address gaps identified during the Phase 1 SME evaluation. The number of items and distribution across tracks were determined adaptively based on Phase 1 outcomes. Phase 2 targeted

specialties with higher rejection rates to maintain target pool sizes, underrepresented topics or task areas requiring strategic development to ensure exhaustive blueprint coverage, and specific content areas flagged for enhancement by subject matter experts.

Phase 2 volume is determined adaptively based on Phase 1 results, with unequal distribution across specialties reflecting differing approval rates and content needs. This targeted approach optimized resource allocation by concentrating generation effort where item pools required reinforcement while avoiding unnecessary item production in well-performing specialties. All Phase 2 items underwent the same multi-stage review process as Phase 1: automated review, small peer review, and large peer review. The SME-Driven Rewriter Agent is available during Phase 2 reviews to facilitate rapid revision of items requiring minor corrections.

Combined Generation Target

The two-phase strategy produced a combined item pool for the study, with total generation volume adapting to observed approval rates to ensure sufficient high-quality items advance to pilot testing while maintaining efficient resource utilization. Item generation across both phases followed established PAEA content blueprints specifying topic coverage, cognitive complexity, and clinical scenario requirements, integrating authentic clinical presentations with realistic patient demographics, symptom patterns, physical examination findings, and diagnostic information consistent with actual clinical practice. Distractor development emphasized plausible but incorrect options based on common clinical reasoning errors and alternative diagnoses. Specific generation volumes, approval rates, and comparative outcomes across phases are reported in Chapter 4.

Evaluation Framework

The review process employed an eight-dimensional evaluation framework that holistically assessed item quality across critical domains. The eight-dimensional evaluation framework used

in this study was developed during my internship at the National Commission on Certification of Physician Assistants (NCCPA), where it was designed to evaluate AI-generated medical assessment items based on the quality criteria applied by experienced NCCPA item writers in operational review (Makinde, 2025, NCME). The framework's dimensions were subsequently validated by PAEA item development specialists, who confirmed alignment with the quality criteria applied in their operational item review process.

Table 3. Eight-Dimensional Item Evaluation Framework

Dimension	Definition
Content Accuracy	Medical accuracy and currency of information presented in clinical scenarios, diagnostic procedures, and treatment options
Clinical Relevance	Real-world applicability and authenticity of clinical scenarios
Currency	Up-to-date medical information and current practice standards
Format Compliance	Adherence to standardized item writing guidelines
Clarity of Stem and Options	Precision and unambiguous nature of item language
Language Clarity	Overall linguistic precision and readability
Distractor Plausibility	Quality and defensibility of answer choices
Bias Freedom	Absence of cultural, gender, demographic, or socioeconomic bias

Reviewers evaluated each dimension using a five-point Likert scale ranging from 1.0 (Unacceptable) to 5.0 (Excellent), except for Content Accuracy which used a binary (Yes/No) assessment to determine whether medical information was factually correct and aligned with current evidence-based practices. The same acceptance thresholds applied at different review stages: the automated review required ≥ 3.0 average on all scaled dimensions, the small peer review also required ≥ 3.0 average across dimensions (averaging both reviewers' scores), and the large peer review addressed minor quality issues through collaborative revision during the review

session, with corrections applied to ensure items met PAEA standards before advancing to pilot testing. Items requiring excessive revision time (>15 minutes) were rejected. The custom R shiny system captured both quantitative ratings and qualitative feedback through dimension-specific comments and general notes at the conclusion of review.

Small Peer Review Procedure

Following automated review, items underwent small peer review conducted by two SMEs per examination specialty. This review was conducted asynchronously through the R Shiny electronic review system, enabling SMEs to evaluate items independently. This initial expert evaluation provided structured quality assessment and identified items requiring revision before advancing to large-scale collaborative review. Each item was independently evaluated by both reviewers using the eight-dimensional framework described above. Reviewers assigned scores on the 1-5 Likert scale for each dimension (with Content Accuracy assessed as Yes/No) and provided qualitative feedback identifying specific concerns or improvement recommendations. Items must have achieved an average score ≥ 3.0 across all eight dimensions (averaging both reviewers' scores) to advance to large peer review. Items scoring below this threshold were flagged for revision or rejection. The small peer review served as an intermediate quality gate, filtering items before the more resource-intensive large peer review process.

Large Peer Review Procedure

The large peer review constituted the final expert evaluation before items advanced to pilot testing. This process employed a structured collaborative review methodology that integrated multiple expert perspectives through role-based evaluation. Each large peer review session included 4-8 SMEs, typically 6 with complementary areas of expertise. Reviewers were drawn from the pool of SME's that participated in the previous small peer review process.

To ensure rigorous and efficient evaluation, each reviewer in the session was assigned a specific role:

1. Item Tagger: Assigned metadata categories and ensures appropriate item classification
2. Timer: Monitored review duration per item to maintain session efficiency
3. Question Reader: Read each item aloud to the group, ensuring all reviewers engage with identical content presentation
4. Medication Checker: Verified pharmaceutical information accuracy and currency with current formularies
5. Laboratory Values Checker: Validated diagnostic test reference ranges and clinical significance thresholds
6. Format and Style Checker: Ensured consistency with PAEA item writing standards and guidelines

This role-based approach distributed cognitive load and ensured systematic attention was paid to all quality dimensions. Each item underwent collaborative discussion following initial individual review. The target review time was five to ten minutes per item, with items requiring more than fifteen minutes flagged for rejection or revision based on feasibility considerations. Items requiring excessive review time typically reflected fundamental clarity or complexity issues that would impair operational use in time-constrained testing environments. Review sessions were scheduled for three hours, allowing for approximately fifteen to twenty-five items per session depending on complexity.

Following collaborative discussion, the review team reached consensus on item disposition either to approve for pilot testing, recommend for revision, or reject. Approved items demonstrated

acceptable quality across all eight evaluation dimensions, clinical appropriateness, and compliance with PAEA standards. Systematic content analysis of SME feedback, thematic coding of improvement suggestions, and comparative evaluation of review patterns across item types provide contextual understanding of quantitative findings while identifying specific areas for system enhancement and validation framework refinement. Six large peer review sessions were conducted, and specific session outcomes are reported in Chapter 4.

Reviewer Agreement and Item Advancement Criteria

This study operationalized reviewer agreement through a mean aggregate scoring approach applied consistently across all review stages. For each item, two assigned specialty-specific reviewers independently rated seven dimensions using a 5-point scale:

- 5.0: Excellent - Exceeds standards
- 4.0: Good - Meets standards
- 3.0: Satisfactory - Needs minor revision
- 2.0: Poor - Needs major revision
- 1.0: Unacceptable - Rejected.

Content Accuracy was assessed on a binary scale: Yes or No

Reviewer scores were averaged across both reviewers for each dimension, and items were required to achieve a mean score ≥ 3.0 across all scaled dimensions to advance to large peer review.

Two indicators support the consistency of this approach. First, dimension-level mean scores aggregated across all reviewed items indicated whether reviewers apply the evaluation criteria discriminately, with systematic differentiation across dimensions reflecting meaningful

criterion application. Second, examination-level mean scores across all five specialties indicated whether the scoring framework functions reliably across diverse clinical content areas.

The 3.0 threshold served as the consistent quality standard applied at both the automated review stage and the small peer review stage, ensuring methodological coherence across the human and AI components of the validation framework.

Psychometric Analysis Framework

AIG items approved through the SME review process were administered as unscored pretest items embedded within PAEA's operational End of RotationTM examination forms across five specialties. Each form contained approximately 120 items, with AIG items distributed across eight forms. PAEA's standardized computer-based testing platform maintained detailed response tracking for subsequent psychometric analysis.

Classical Test Theory Analysis

Item-level CTT statistics were computed from full-form item analyses for each test form. Item difficulty was measured as the proportion of examinees answering correctly (p-value). Item discrimination was measured using the external point-biserial correlation, computed as the Pearson correlation between the item score (0/1) and the total score derived exclusively from operational items on the same form, following standard pretest item evaluation procedures (Haladyna & Rodriguez, 2013; Lane et al., 2016). Distractor functioning was evaluated at the option level; distractors selected by fewer than 5% of examinees were classified as non-functioning (Haladyna & Downing, 1993).

Item Response Theory Analysis

One-parameter logistic (1PL) models were calibrated on each complete test form using the TAM package in R (Robitzsch et al., 2022), which implements marginal maximum likelihood estimation. The probability of a correct response under the Rasch model is expressed as:

$$P(X_{ni} = 1 | \theta_n, b_i) = \frac{\exp(\theta_n - b_i)}{[1 + \exp(\theta_n - b_i)]}$$

Where θ_n is the ability of person n and b_i is the difficulty of item i . Full-form calibration was employed so that operational items anchored the ability scale and AIG items borrowed estimation strength from well-estimated ability parameters.

Person ability was estimated using the Expected A Posteriori (EAP) method (Bock & Mislevy, 1982). EAP reliability was computed for each form to confirm adequate measurement precision, defined as:

$$EAP \text{ Reliability} = \frac{Var(\theta_{EAP})}{[Var(\theta_{EAP}) + SE^2]}$$

where $Var(\theta_{EAP})$ is the variance of the ability estimates across examinees and SE^2 is the mean squared standard error of the estimates. Values above .70 are generally considered adequate for group-level measurement decisions (Bock & Mislevy, 1982).

Model-data fit was evaluated using infit and outfit mean-square residual (MNSQ) statistics (Wright & Masters, 1982). Infit (information-weighted mean square) weights residuals by their statistical information, giving greater influence to responses from examinees whose ability is near the item's difficulty:

$$Infit \text{ MNSQ}_i = \Sigma(x_{ni} - E_{ni})^2 / \Sigma W_{ni}$$

$$\text{Outfit MNSQ}_i = (1/N) \sum [(x_{ni} - E_{ni})^2 / W_{ni}]$$

where x_{ni} is the observed response of person n to item i , $E_{ni} = P_{ni}(\theta_n)$ is the expected response under the model, and $W_{ni} = P_{ni}(1 - P_{ni})$ is the response variance. Values below 1.0 indicated overfit (responses more predictable than expected), while values above 1.0 indicated underfit (responses less predictable than expected). Items with infit or outfit MNSQ values outside the 0.7–1.3 range were flagged for review (Wright et al., 1994).

Comparative Psychometric Analysis

Two complementary comparisons evaluated whether AI-generated items exhibit psychometric properties comparable to traditionally developed items. The within-form comparison leveraged the embedded pilot design, in which AIG and operational items were administered to the same examinees on the same test forms, controlling for examinee ability and testing conditions. Item-level CTT statistics and IRT difficulty parameters from the same calibration were compared between AIG and operational items using independent samples t-tests, and Cohen's d effect sizes.

Given the substantial sample size imbalance between AIG items and operational items, distributional assumptions underlying the independent-samples t-test were evaluated prior to analysis. Shapiro-Wilk tests were conducted to assess normality for both difficulty (proportion correct) and discrimination (point-biserial correlation) distributions within each group. Levene's test was used to evaluate homogeneity of variance across groups. Where normality assumptions were violated, Mann-Whitney U tests were conducted as non-parametric alternatives to supplement the parametric results. Results from both parametric and non-parametric tests are reported to demonstrate robustness of findings regardless of distributional assumptions.

The cross-administration comparison evaluated AIG items against historically administered cardiovascular disease items from previous PAEA examinations across the same five specialties, comparing difficulty, discrimination, infit MNSQ, and outfit MNSQ. Rasch difficulty was excluded from the cross-administration comparison because the two groups were calibrated in separate analyses with different examinee samples, producing parameters on different measurement scales. Effect size benchmarks followed conventional thresholds: $|d| < 0.20$ = negligible, $0.20-0.49$ = small, $0.50-0.79$ = medium (Cohen, 1988).

Content Related Validity Evidence Collection

Content validation procedures encompassed both pre-generation and post-generation phases to ensure construct representation, aligned with the Duolingo English Test (DET) Responsible AI framework's Validity and Reliability standard.

Before item generation began, the system underwent three critical validation checks. First, cognitive models were verified against PAEA examination blueprints to ensure content specifications, learning objectives, and competency frameworks align precisely with accreditation standards, addressing the DET framework's requirement for cognitive alignment (Goal 1a: Identify AI item methods). Second, domain coverage analysis to confirm that cognitive models adequately represent the breadth and depth of each medical specialty's knowledge base. Third, SMEs reviewed the theoretical frameworks underlying the cognitive models to validate their clinical accuracy and pedagogical soundness.

Post-generation validation employed the eight-dimensional SME review process with ten expert reviewers conducting systematic evaluation of content accuracy, clinical relevance, and format compliance (DET Goal 1b: Build content validity). Mean aggregate scoring across the eight evaluation dimensions with a consistent ≥ 3.0 advancement threshold fulfills accuracy and

reliability requirements (DET Goal 1c: Ensure accuracy & reliability). Clinical authenticity assessment and quantitative analysis of content distribution patterns across specialties provided additional validation evidence. Psychometric evaluation using Item Response Theory and comparative benchmarking against traditionally developed items, detailed in the Psychometric Analysis Framework section, completed the framework's validity standards. This multi-layered validation approach ensured generated items met established standards for responsible AI use in high-stakes educational assessment.

Summary

This chapter detailed the methodological framework for developing and evaluating an agentic large language model system for automatic item generation in medical education assessment. Conducted in partnership with PAEA, the study employed a two-phase generation approach with iterative refinement based on expert feedback. The framework integrated dual large language model architecture (GPT-5 Nano for generation, GPT-5 Mini for evaluation) through a multi-agent system comprising Cognitive Planning, Writer, Reviewer, and Rewrite Agents. Multi-layered validation procedures aligned with the DET Responsible AI framework included pre-generation cognitive model verification and post-generation eight-dimensional expert review. Ten subject matter experts across five medical specialties conducted systematic small peer review using a custom-developed R Shiny application, and large peer review was done with PAEA internal review App with reviewer agreement operationalized through mean aggregate scoring and consistent advancement thresholds. Rule-based compliance verification provided additional quality assurance. Psychometric validation employed IRT models for difficulty and comparative benchmarking against traditionally developed items. The methodology tracked computational resources, processing time, and costs to evaluate efficiency gains relative to conventional

approaches. This framework integrates AI capabilities with rigorous psychometric standards and structured expert validation, establishing foundations for responsible implementation of multi-agent systems in high-stakes educational assessment.

CHAPTER IV: RESULTS

This chapter presents the empirical results of the study examining automatic item generation using a multi-agentic LLM framework for medical assessment. The study examined how LLMs can be integrated into structured multi-agent systems to generate, review, and refine cardiovascular assessment items for PAEA End of Rotation™ examinations across five clinical specialties.

In Chapter 1, I introduced the background and rationale for this study, highlighting the limitations of traditional and template-based AIG systems and articulating the need for a new framework that integrates LLMs within an agentic, human-in-the-loop system. The chapter presented the overarching research purpose to determine whether LLM-driven agentic systems can generate assessment items aligned with established medical education standards.

Chapter 2 provided a comprehensive literature review tracing the evolution of AIG from early prose-based and template-based foundations to modern neural and LLM-based paradigms. It examined how cognitive modeling, item modeling, and emerging multi-agent systems inform current practices, emphasizing content-related validity and content fidelity as critical validity considerations for AI-generated items. The review introduced the Duolingo English Test (DET) Responsible AI framework, whose four standards Validity & Reliability, Fairness, Privacy & Security, and Accountability & Transparency offer guiding principles for responsible AI-based assessment practices.

Chapter 3 detailed the methodological design implementing this framework, describing the multi-agent architecture comprising Cognitive Planning, Item Generation, Review, and Rewriter agents. It outlined the two-phase generation approach targeting cardiovascular content across five End of Rotation™ examinations, the electronic review system developed for structured SME

evaluation, and the multi-stage validation process including small peer review and large peer review sessions.

Results Organization

Grounded in these foundations, this chapter reports the findings organized according to the four research questions that guided this study:

- 1) How can large language models be integrated into a designed agentic framework to develop assessment items?
- 2) To what extent is there alignment between the agentic framework items and SME reviewed items based on established test specifications and content standards?
 - a. How effective is the electronic review system developed to help SME review AI generated items for medical education assessment?
 - b. What validation criteria and checkpoints are necessary to ensure content related validity, fairness, and psychometric properties of AI-generated test items, and how do these compare to those used for human-authored items?
- 3) To what extent does the agentic LLM framework improve efficiency in terms of item development time, review cycles, and cost per item compared to conventional item development methods?
- 4) What are the psychometric properties (difficulty, discrimination, reliability, dimensionality) of test items generated by the agentic LLM approach, and how do these properties compare to those of traditionally developed items from previous and current test administrations?

Research Question 1: Integration of LLMs into Agentic Framework

RQ1: How can large language models be integrated into a designed agentic framework to develop assessment items?

Responding to this research question, we report how LLMs were operationalized within the multi-stage agentic framework to generate cardiovascular assessment items for End of Rotation™ examinations. Results focus on the integration approach, generation volumes, and item progression through the development workflow.

LLM Integration Architecture

The framework integrated two distinct large language models into specialized agent roles. GPT-5 Nano was employed for cognitive planning and item writing across five track-specific Writer Agents (EMEOR, FMEOR, IMEOR, SUREOR, PEDEOR). GPT-5 Mini was employed for automated review evaluation across the eight quality dimensions discussed in Chapter III. This dual-model architecture leveraged GPT-5 Nano's generative capabilities for creative scenario development and GPT-5 Mini's analytical precision for systematic quality evaluation. Items progressed through a staged pipeline: Cognitive Planning Agent → Track-Specific Writer Agent → Automated Reviewer Agent → [optional] SME-Driven Rewrite Agent. Rule-based post-processing and metadata stamping occurred prior to export for human review. Across both generation phases and re-writing, the framework utilized 7,741,445 total tokens through 1,189 model API calls, with a combined cost of \$14.20. Detailed efficiency metrics including per-item costs and throughput analysis are reported in Research Question 3.

Sample Characteristics

The study generated 333 cardiovascular multiple-choice items across two sequential phases. Phase 1 produced 250 items distributed across the five specialty tracks, while Phase 2

generated 83 items with strategic distribution based on Phase 1 outcomes. Items underwent multi-stage quality review starting with the automated review that passed 210 items, then a small peer review conducted by 10 subject matter experts evaluating the 210 items with 2 reviewers per specialty yielded 151 items, and large peer review comprising 6 collaborative sessions with 6-8 SMEs per session. Following comprehensive large peer review processes, 76 items advanced to pilot testing administered by PAEA to PA students in batches.

Phase 1: Item Generation and Progression

In Phase 1, 250 generated items were distributed across the five specialty tracks.

Table 4. Phase 1 Item Generation and Automated Review

Track	Generated (N)	Auto Review Pass (N)	Pass Rate (%)
EMEOR	50	25	50.0
FMEOR	50	27	54.0
IMEOR	50	25	50.0
SUREOR	50	25	50.0
PEDEOR	50	25	50.0
Total	250	127	50.8

In table 4, we showed the Phase 1 items generated and how the automated review went. The automated Reviewer Agent approved 127 items (50.8%) that met the ≥ 3.0 threshold on all eight dimensions. These 127 approved items advanced to the human review pipeline which consisted of two subject matter experts reviewing the items. Of the 127 items evaluated in small peer review, 90 items (70.9%) achieved average scores ≥ 3.0 across the eight quality dimensions

by averaging both reviewers' scores. The remaining 37 items (29.1%) scored below the 3.0 threshold and were rejected.

Three large peer review sessions involving 4-8 SMEs per session evaluated the 90 items that passed small peer review. In the first review session, SME's identified systematic format/style violations and significant enemy item concerns approved only 8 items. Subsequent enemy item analysis revealed 50 items of the 90-item pool under large peer review exhibited content overlap that could provide inadvertent answer cues or assess identical constructs with superficial presentation variations. These items were systematically eliminated to maintain test validity.

The remaining 32 items were processed through the SME-Driven Rewriter Agent which was developed following the initial large peer review session to salvage items with correctable issues. Of these, 26 items (81.3%) were approved after targeted revision, while 6 were rejected in the large peer review session for requiring excessive review time (> 15 minutes per item). A total of 34 (8 FMEOR, 14 EMEOR, 7 IMEOR, 3 PEDEOR, and 2 SUREOR items) Phase 1 items advanced to pilot testing, representing 26.8% of the 127 items that entered the human review pipeline.

Inter-Phase Refinement

Systematic analysis of SME feedback revealed recurring issues that informed prompt refinement:

- Format and style non-compliance with PAEA standards (most frequent issue)
- Distractor implausibility or insufficient discrimination
- Lead-in placement errors

Following this, the generation prompts were systematically refined. The underlying LLM architecture and configuration parameters remained unchanged, with improvements implemented through enhanced prompt engineering targeting these specific issues. Additionally, the SME-Driven Rewriter Agent was integrated into the workflow for a second item generation sequence, Phase 2 to facilitate rapid revision of items requiring minor corrections.

Phase 2: Item Generation and Progression

Following prompt refinement and system enhancement, Phase 2 generated 83 items strategically distributed across tracks based on Phase 1 pretesting needs, with items requiring excessive review time (> 15 minutes) excluded from advancement.

Table 5. Phase 2 Item Generation and Review Progression

Track	Generated (N)	Auto Review Pass (N)	Auto Pass Rate (%)
EMEOR	12	12	100.0
FMEOR	10	10	100.0
IMEOR	10	10	100.0
PEDEOR	26	26	100.0
SUREOR	25	25	100.0
Total	83	83	100.0

All 83 Phase 2 items passed automated review, demonstrating improved generation quality following prompt refinement. These 83 items proceeded to small peer review, 58 of which (69.9%) achieved average scores ≥ 3.0 across the eight quality dimensions and advanced to large peer review. The remaining 25 items (30.1%) were rejected representing comparable performance to Phase 1's 70.9% pass rate.

Following 3 large peer reviews, 42 Phase 2 items were approved for pilot testing (72.4% of the 58 items reaching large peer review), with 16 rejected (4 EMEOR, 1 FMEOR, 3 IMEOR, 4 PEDEOR, and 4 SUREOR) for requiring excessive review time (> 15 minutes per item). No enemy item concerns emerged in Phase 2, suggesting that prompt refinements successfully addressed the content repetition issues identified in Phase 1. The 42 Phase 2 items advanced to pretesting, representing 50.6% of the 83 Phase 2 generated items, a 272% improvement over Phase 1's 13.6% success rate.

Combined Generation Outcomes

Table 6 summarizes item progression from generation through pretesting advancement across both phases. Across both phases, the Review Agent approved 210 items (63.1% of 333 generated) that entered the human review pipeline. Through multi-stage validation, 76 items advanced to pilot testing. Phase 2 demonstrated substantial improvement in human review pipeline success, with 50.6% items reaching pretesting compared to 26.8% in Phase 1 an 89% relative improvement reflecting effective inter-phase refinement and elimination of enemy item concerns.

Table 6. Combined Item Progression Across Both Phases

Stage	Phase 1		Phase 2		Total
	N	%	N	%	N
# of Items Generated	250	100.0	83	100.0	333
Approved by Review Agent	127	50.8	83	100.0	210
Approved by Small Peer Review	90	70.9	58	69.9	148
Approved by Large Peer Review	34	26.8	42	50.6	76

Track Distribution to Pretesting

Table 7 shows the distribution across track of the items that advanced to pretesting. All 76 items advancing to pretesting conformed to single-best-answer multiple-choice format with clinical vignette stems, 4-5 response options, keyed correct answer, and evidence-based rationale. Track-specific items adhered to required age ranges and clinical setting constraints.

Table 7. Track Distribution of Items Advancing to Pretesting

Track	Phase 1 (N)	Phase 2 (N)	Total (N)	% of Pretesting Pool
EMEOR	14	8	22	28.9
FMEOR	8	4	12	15.8
IMEOR	7	3	10	13.2
SUREOR	3	11	14	18.4
PEDEOR	2	16	18	23.7
Total	34	42	76	100.0

To finalize this, Research Question 1 examined how LLMs can be integrated into a designed agentic framework to develop assessment items. The implemented framework successfully operationalized dual LLM architecture producing 333 cardiovascular assessment items across two sequential phases. The Review Agent approved 210 items (63.1% of generated), with great improvement from Phase 1 (50.8%) to Phase 2 (100%), demonstrating effective prompt refinement. Through multi-stage human review (small peer: 148 of 210 passed, 70.5%; large peer:

6 sessions across both phases), 76 items (36.2%) approved advanced to pretesting. Human review pipeline success improved substantially from Phase 1 (26.8%) to Phase 2 (50.6%), an 89% relative improvement reflecting combined effects of enhanced prompt engineering and elimination of enemy item concerns. Total computational resource utilization was 7,741,445 tokens across 1,189 API calls at \$14.20. These findings demonstrate that large language models can be effectively integrated into structured agentic frameworks for assessment item generation, with iterative refinement capabilities that substantially enhance output quality while maintaining production efficiency. Critically, the Phase 2 configuration represents the finalized framework architecture; future implementations can begin directly from this optimized configuration, bypassing the iterative calibration costs incurred during Phase 1 development and achieving production-quality outputs from the outset.

Research Question 2: Alignment Between Agentic and SME-Reviewed Items

RQ2: To what extent do we have alignment between the agentic framework items and SME-reviewed items based on established test specifications and content standards?

Alignment in this context refers to the degree to which items produced by the agentic LLM framework meet the quality standards, content specifications, and professional expectations that SMEs apply when evaluating assessment content. Guided by the Duolingo English Test (DET) Responsible AI Framework (Burstein et al., 2022) which operationalizes validity theory into four actionable standards for AI-powered assessment this study examined alignment evidence across four complementary dimensions: (i) automated review performance, which assesses whether the framework's internal quality criteria produce items consistent with expert expectations and whether SME feedback improves that consistency over time; (ii) SME quality ratings, which provide direct expert judgment on item quality across defined dimensions; (iii) peer review

outcomes, which measure the proportion of items that survive progressively rigorous expert scrutiny against operational standards; and (iv) electronic review system effectiveness, which evaluates the reliability of the instrument through which alignment evidence was collected. Additionally, RQ2b examined the validation criteria and checkpoints necessary to ensure content-related validity, fairness, and psychometric properties, operationalized through the DET Responsible AI Framework. Together, these analyses addressed whether the agentic framework produced items that subject matter experts judged as suitable for operational assessment use.

Automated Review Performance

The automated Review Agent functioned as the first quality gate in the validation pipeline, evaluating all 333 generated items across the eight quality dimensions using a rating ≥ 3.0 as an acceptance threshold. Quality issues identified during Phase 1's expert review informed systematic prompt refinements that shaped Phase 2 generation reflecting the effectiveness of the SME-driven feedback loop in improving item quality prior to expert review. Table 8 presents comparative performance across phases.

Table 8. Automated Review Agent Performance by Phase

Phase	Items Evaluated (N)	Items Approved (N)	Approval Rate (%)	Mean Score (sd)
Phase 1	250	127	50.8	3.4 (0.6)
Phase 2	83	83	100.0	4.1 (0.3)
Combined	333	210	63.1	3.6 (0.6)

Note. Numbers in parentheses are standard deviations.

Phase 2 achieved perfect Review Agent approval (100%) compared to Phase 1's 50.8%, with higher mean scores (4.1 vs. 3.4) and reduced variability (0.3 vs. 0.6). This improvement did

not result from changes to the underlying model or the Review Agent’s evaluation criteria; rather, it was driven entirely by prompt refinements targeting systematic quality issues that SMEs identified during Phase 1 review, including format/style compliance, distractor plausibility, lead-in placement, and clinical currency.

Small Peer Review Outcomes

Small peer review provided the proportion of items that independent SMEs judged as meeting professional quality thresholds which represented a concrete measure of how closely AI-generated content aligns with expert standards and expectations. It involved two independent SMEs per specialty evaluating all 210 Review Agent-approved items using the eight-dimensional framework. Table 9 presents outcomes across phases.

Table 9. Small Peer Review Performance by Phase

Phase	Items Reviewed (N)	Items Passed (N)	Pass Rate (%)	Items Rejected (N)
Phase 1	127	90	70.9	37
Phase 2	83	58	69.9	25
Combined	210	148	70.5	62

Of the 210 Review Agent-approved items entering small peer review, 148 items (70.5%) achieved average scores ≥ 3.0 from reviewers across dimensions and advanced to the large peer review, while 62 items (29.5%) were rejected. Phase 2 displayed comparable performance with acceptance rate of 69.9% as compared to 70.9% in Phase 1, indicating that while prompt

refinements substantially improved automated review performance, small peer review maintained rigorous human expert standards with consistent evaluation criteria.

Small Peer Review Quality Evaluation

While the overall pass rate provided a global measure of alignment, examining performance across individual quality dimensions revealed where alignment was strongest and where systematic gaps remained. Independent SME reviewers evaluated items across eight quality dimensions using a structured rubric. Each item received ratings from two specialty-specific reviewers, with advancement to large peer review requiring an average score ≥ 3.0 across all criteria. Content Accuracy was assessed as binary (accurate/inaccurate), while the other seven dimensions used 5-point scales.

Table 10. Quality Dimension Ratings Across All Reviewed Items (N = 210)

Criterion	Mean	SD
Bias Freedom	3.79	0.85
Currency	3.44	0.84
Clinical Relevance	3.32	0.85
Language Clarity	3.22	0.85
Distractor Plausibility	2.80	0.96
Clarity of Stem	2.80	0.99
Format Compliance	2.63	0.85
Content Accuracy (proportion)	0.86	0.35

In table 10, items demonstrated strongest performance on Bias Freedom (Mean = 3.79), Currency (Mean = 3.44), and Clinical Relevance (Mean = 3.32), all exceeding the advancement threshold. Lower scores on Format Compliance (Mean = 2.63), Clarity of Stem (Mean = 2.80), and Distractor Plausibility (Mean = 2.80) indicate areas requiring continued prompt refinement. Content Accuracy reached 85.9%, indicating the Review Agent's clinical verification provided effective screening. Note that the Content Accuracy proportion here reflects the rate of 'Yes' (Accurate) ratings across all reviews. These mean scores across both phases align with the format non-compliance, lead-in placement errors, and distractor implausibility issues identified during Phase 1 feedback analysis, confirming these dimensions remain priority areas for continued refinement.

Table 11. Small Peer Review Performance by Examination

Examination	# of Items	Mean Score	Review Time (min)
EMEOR	37	3.35	13
FMEOR	37	3.23	15
IMEOR	35	3.19	16
SUREOR	50	3.06	17
PEDEOR	51	3.04	16

Note: Review time is the average time it will take to fix the item and get it ready

Table 11 breaks down the performance by examination. All five examinations achieved mean scores exceeding the 3.0 advancement threshold, with Emergency Medicine demonstrating highest quality ratings (Mean = 3.35) and shortest time to review readiness (13 minutes), while

Surgery and Pediatrics exhibited lower but acceptable mean scores (Mean =3.06 and Mean =3.04, respectively). Review times ranged from 13-17 minutes across specialties, with a general trend suggesting higher-quality items required less SME deliberation. Of the 210 items entering small peer review, 148 items (70.5%) achieved the ≥ 3.0 threshold and advanced to large peer review, while 62 items (29.5%) required revision or rejection.

Large Peer Review Outcomes

Large peer review represents the most stringent test of alignment, requiring items to survive collaborative committee-level scrutiny against operational deployment standards. While small peer review evaluated individual item quality, large peer review assessed whether items were suitable for inclusion on live PAEA examinations a higher bar that incorporates considerations of content overlap, blueprint coverage, and operational feasibility. Sessions employed structured collaborative evaluation with 6-8 SMEs per session using assigned roles (Item Tagger, Timer, Question Reader, Medication Checker, Laboratory Values Checker, Format/Style Checker).

Table 12. Large Peer Review Session Characteristics

Metric	Phase 1	Phase 2	Combined
Sessions conducted	3	3	6
Items reviewed	90	58	148
Average items per session	30	20	25
Average time per item	9 min	7 min	8.5 min
Items >15 min review time	6	16	22
Items approved for pretesting	34 (37.8%)	42 (72.4%)	76 (51.4%)

Table 12 shows the review session characteristics across the two phases. Of the 148 items entering large peer review, 76 items (51.4%) were approved for pretesting on operational PAEA examinations. Phase 2 demonstrated substantially higher approval (72.4%) compared to Phase 1 (37.8%), a 34.6 percentage point improvement primarily attributable to the elimination of enemy item concerns in Phase 2. The convergence of higher Phase 2 approval rates with shorter average review times (7 vs. 9 minutes) provides corroborating evidence that Phase 2 items achieved stronger alignment with expert standards.

Enemy Item Identification and Elimination

During Phase 1 large peer review, subject matter experts identified 50 enemy items (55.6% of the 90-item pool) items creating content overlap, providing inadvertent answer cues, or assessing identical constructs with superficial variations. The primary concerns included identical clinical scenarios with superficial presentation variations (40%), shared diagnostic/management pathways creating answer dependencies (35%), and content overlap providing cueing information (25%). When enemy item pairs or clusters were identified, reviewers systematically retained items with superior quality scores, better blueprint alignment, greater clinical authenticity, and optimal difficulty levels, eliminating the remainder to prevent construct contamination and answer cueing. Notably, no enemy item concerns emerged during Phase 2 large peer reviews, again demonstrating that prompt refinements successfully addressed content repetition tendencies and largely explaining the improvement in large peer review approval rates from 37.8% (Phase 1) to 72.4% (Phase 2).

SME-Driven Rewriter Agent Effectiveness

The Rewriter Agent served as an additional alignment mechanism within the pipeline, addressing the gap between items that met individual quality thresholds in small peer review but

fell short of operational readiness during committee-level scrutiny. Following enemy item elimination in Phase 1, 35 items with correctable issues were processed through the SME-Driven Rewriter Agent. The agent successfully revised 29 items (82.9% success rate) with an average processing time of 1.84 minutes per item (64.4 minutes total), demonstrating efficient AI-assisted revision capabilities. The 6 rejected items (17.1%) required excessive review time even after automated revision, indicating fundamental issues beyond targeted correction. It is worth noting that in subsequent small peer review sessions, items requiring further review were routed through the Rewriter Agent for targeted correction before re-evaluation, creating an iterative refinement process that improved item quality while maintaining reviewer efficiency.

As an alternative to automated revision, 10 items in Phase 2 underwent manual pre-editing by an SME prior to large peer review. This strategic human intervention required approximately 3-5 minutes per item and substantially accelerated subsequent large peer review sessions. Items that received manual pre-editing moved through large peer review more efficiently than both automated-revised items and unrevised items, suggesting that targeted human expertise applied at critical revision points optimizes downstream workflow efficiency.

Electronic Review System Evaluation

RQ2a: How effective is the electronic review system developed to help SME review AI-generated items for medical education assessment?

To facilitate the review process, I created a custom R Shiny electronic review platform supported the small peer review. It provided a structured evaluation interface, automated scoring calculations, and comprehensive data collection capabilities. The application, hosted on PAEA's Shiny server, enabled asynchronous systematic quality assessment and facilitated both quantitative scoring and qualitative feedback collection from SMEs. Following completion of review sessions,

10 SMEs completed a feedback survey evaluating system effectiveness across usability, functionality, and technical performance dimensions.

Usability and Satisfaction

SME satisfaction ratings demonstrated generally positive reception of the review system.

Table 13. SME Review System Usability Ratings

Dimension	Rating	n	%
Ease of Use	Very Easy	1	10.0
	Easy	6	60.0
	Neutral	1	10.0
	Difficult	2	20.0
Interface Clarity	Agree (clear/intuitive)	5	55.6
	Neutral	2	22.2
	Disagree	2	22.2
System Efficiency	Agree (efficient review)	6	60.0
	Neutral	2	20.0
	Disagree	2	20.0
Overall User Experience	Good	2	30.0
	Acceptable	5	60.0
	Poor	1	10.0

The results in Table 13 show that most reviewers (70.0%) rated the system as easy or very easy to use, and 90% reported acceptable or good overall experience. However, 20.0% reported difficulty with ease of use or interface clarity, indicating areas for improvement. The review time data aligned with operational feasibility targets. Most SMEs (63.6%, n = 7) reported spent 6-10 minutes per item on average, with 9.1% (n = 1) completing reviews in 2-5 minutes and 27.3% (n = 3) requiring more than 10 minutes. This distribution demonstrates that the electronic system enabled most reviewers to complete evaluations within the target 5–10-minute operational range while accommodating variability in reviewer thoroughness and item complexity.

Technical challenges emerged during implementation despite successful facilitation of review processes. Data persistence issues were the most critical limitation, with multiple reviewers reporting that previously reviewed items were not saved, requiring re-review and manual progress tracking outside the system. Navigation challenges included difficulty determining which items had been completed, unclear saving indicators, and time-tracking resetting when navigating backwards. Interface layout presented additional usability challenges, with reviewers reporting that the side-by-side item and review view, while conceptually helpful, made item text too small for comfortable reading, and excessive scrolling between item content and evaluation criteria reduced review efficiency.

The electronic review platform demonstrated moderate effectiveness in supporting SME review of AI-generated items. Strengths included structured evaluation framework (8-criterion rubric), automated scoring calculations, efficient review time performance (72.7% within target range), comment and feedback functionality, and comprehensive data capture. Despite technical challenges, 81.8% of SMEs ($n = 9$) rated their overall experience as acceptable or good. The system successfully enabled completion of a comprehensive multi-stage review for 210 items across the small peer review process.

Validation Criteria and Checkpoints

RQ2b: What validation criteria and checkpoints are necessary to ensure content-related validity, fairness, and psychometric properties of AI-generated test items?

The validation framework presented in Table 14 operationalizes the theoretical foundations established in Chapter 2 by mapping each DET standard to the validity traditions that informs it. The Validity and Reliability standard addresses content-related evidence as prioritized by Lissitz and Samuelsen (2007), corresponding to the scoring and generalization inferences in Kane's (2006)

interpretive argument. The Fairness standard enacts Chalhoub-Deville and O'Sullivan's (2020) impact-by-design principle by embedding bias and equity checks throughout the development process. The Privacy and Security standard extends beyond classical frameworks to address AI-specific governance concerns. The Accountability and Transparency standard operationalizes Mislevy et al.'s (2003) emphasis on explicit evidentiary reasoning through complete workflow documentation and audit trails. Each validation checkpoint in Table 14 serves a specific evidentiary function within the broader validity argument, ensuring that the theoretical commitments articulated in Chapter 2 are realized in practice.

The validation framework addressed all four DET Responsible AI standards through systematic checkpoints. Validity and reliability were ensured through cognitive alignment with PAEA test specifications; multi-stage SME review using structured 8-dimensional rubrics and mean aggregate scoring with consistent advancement thresholds. Fairness considerations included explicit bias review criteria included in the 8-D framework and clinical diversity in item scenarios stipulated in the item generation prompts. Privacy and security were maintained through enterprise-grade LLM deployment (GPT-5 mini and nano with no institutional data training) and authenticated review systems. Accountability and transparency were established through comprehensive documentation, LangGraph audit trails, defined expert oversight criteria, and planned dissemination.

Table 14. Validation Framework: Responsible AI (RAI) Implementation

DET Standard & Standard	Key Implementations
1. Validity & Reliability	
a. Identify AI item methods	Cognitive Alignment: Pre-generation validity using PAEA specifications
b. Build content validity	SME Review: 8-D rubric, 10 reviewers (content, relevance, format)
c. Ensure accuracy & reliability	Psychometrics: 1PL IRT for difficulty Agreement: Mean aggregate scoring with ≥ 3.0 threshold Benchmarking: Compare to human-authored items Model Selection: Document LLM choice (GPT-5 Nano/Mini)
2. Fairness	
a. Representation	Bias Criterion: Explicit fairness dimension in 8-D review rubric
b. Bias mitigation	Clinical Diversity: Balanced patient demographics across items
c. Accessible language	Human-in-Loop: SME checks for culture/language neutrality (10 SMEs)
3. Privacy & Security	
a. Ensure data & test security	GPT-5 Enterprise: Private environment, no training on institutional data
b. Maintain privacy & scalable item pools	Shiny App: Authenticated, encrypted, logged access for SME review Scalability: Framework supports rapid, scalable item generation. Secure Servers: PAEA-hosted infrastructure with institutional controls Multi-Stage Review: Quality gates before deployment (Review Agent → Small → Large peer review)
4. Accountability & Transparency	
a. Document AI workflow	Documentation: Full framework description, LangGraph workflow, audit trails
b. Define human oversight	Expert Criteria: Board-certified SMEs, ≥ 5 yrs clinical experience, 10 reviewers total (2 per specialty)
c. Share research & results	Transparency: Complete audit trails via LangGraph (nonces, timestamps, versions) Dissemination: Dissertation and planned publications addressing 4 research questions Stakeholder Access: SME review system with open evaluation interface and feedback mechanisms

Research Question 3: Efficiency and Cost-Effectiveness

RQ3: What are the efficiency metrics (time, cost, development cycles) of the agentic LLM approach compared to traditional item development methods?

Computational Resource Usage

The agentic framework utilized two GPT-5 models via PAEA's enterprise OpenAI subscription where GPT-5 nano was used for the generation (input: \$0.05/1M tokens, cached input: \$0.005/1M tokens, output: \$0.40/1M tokens) and GPT-5 mini for automated review (input: \$0.250/1M tokens, cached input: \$0.025/1M tokens, output: \$2.000/1M tokens). Table 15 reveals the computational resources used.

Table 15. Computational Resource Utilization Across Phases

Metric	Phase 1	Phase 2	Combined
Items generated	250	83	333
Total tokens used	4,042,329	2,221,065	7,741,445
Total API calls	756	288	1,044
Total cost (USD)	\$8.00	\$4.00	\$14.20
Cost per generated item	\$0.032	\$0.048	\$0.043
Cost per pretesting item	\$0.24	\$0.07	\$0.16
Tokens per item	16,169	26,759	23,250

The total tokens used include 1,478,051 tokens for the Rewriter Agent automated revision processing and 145 Rewriter Agent API calls. The total study cost including development was \$24.20. The complete generation of 333 items required 7,741,445 tokens across 1,044 API calls at a production cost of \$14.20. The development costs were \$10.00, averaging \$0.043 per item

generated. This total includes 6,263,394 tokens for item generation for Phases 1 and 2, and 1,478,051 tokens for Rewriter Agent automated revision processing. For the 76 items advancing to pretesting, the cost per item was \$0.16, demonstrating highly cost-effective resource utilization. Enterprise licensing through PAEA provided access to GPT-5 models with institutional data privacy protections and prompt caching capabilities, ensuring compliance with educational data security requirements while optimizing computational costs.

Time Efficiency Analysis

Table 16 presents time requirements across each stage of the agentic development pipeline. This breakdown isolates the time contribution of each stage from automated generation through multi-stage expert review enabling direct comparison with published benchmarks for traditional item development.

Table 16. Time Requirements by Development Stage

Stage	Average Time	Range	Total Time
Item generation (per item)	2.4 min	1.8-3.2 min	13.3 hours (333 items)
Small peer review (per item)	4.5 min	3-7 min	15.8 hours (210 items)
Author/Rewrite processing (per item)	1.84 min	1.2-2.8 min	1.1 hours (35 items)
Large peer review (per item)	10 min	4-15 min	17 hours (98 items)

The agentic system generated items at approximately 2.4 minutes per item, maintaining consistent throughput across both phases. Combined SME review time (small + large peer review)

averaged approximately 16-19 minutes per item, substantially less than estimated 94-97 minutes per item for traditional development with expert review cycles.

Item Development Cycle Comparison

Table 17 compares the per-item development timeline between the agentic framework and traditional methods across equivalent stages.

Table 17. Development Timeline Comparison: Agentic vs. Traditional Methods

Item Development Stage	Agentic Framework	Traditional Approach
Item generation (per item)	2.4 min	45 min
Individual/Initial Review (per item)	< 1 min (automated)	15 min
Small peer review (per item)	4 min	17 min
Author/Rewrite processing (per item)	1.84 min	10 min
Large Peer Review (per item)	7-10 min	7-10 min
Total per item	16-19 min	94-97 min
For 76 pretesting items	20-24 hours	119-123 hours
Time savings	83-84%	Baseline

Traditional estimates are based on standard assessment development at PAEA. The agentic approach reduced total development time by approximately 80-83% compared to traditional methods, from an estimated 119-123 hours to 20-24 hours for the 76 items advancing to pretesting.

Cost-Effectiveness Analysis

Comparing item development costs across methods presents inherent challenges. Not all organizational expenses can be cleanly allocated per item (e.g., summit logistics, overhead, multi-purpose editorial work, item revisions, etc.), and "approved items" in various stages of the pipeline may not represent equivalent quality or operational readiness. The cost comparisons presented below therefore represent order-of-magnitude estimates intended to provide contextual benchmarking rather than precise financial accounting.

Two reference points informed cost comparisons. First, PAEA's Item development budget data indicates that \$484,654.45 is allocated across activities and yields 999 approved items, implying an approximate cost of \$485 per approved item. This figure, however, represents fixed costs for estimable elements only and likely underestimates true development costs given organizational overhead, shared resources, administrative support, and indirect expenses not directly allocated to item development. Second, published literature estimates traditional high-stakes item development costs at \$1,500–\$2,500 per operational item (Rudner, 2010), approximately 3-5 times higher than PAEA's internal estimate. This discrepancy likely reflects that PAEA's estimate captures direct allocable costs while literature estimates attempt comprehensive cost accounting including indirect expenses. This suggests that the actual per-item costs fall somewhere between these benchmarks.

The agentic framework incurred \$174.20 in total direct costs comprising \$24.20 in LLM API costs (\$10.00 development, \$14.20 production) and \$150.00 for OpenAI Enterprise subscription (\$50/month $\times$ 3 months allocated to this automatic item generation project). This yielded 333 generated items with 76 advancing to pretesting readiness, representing \$0.52 per generated item and \$2.29 per pretesting-ready item in direct computational costs.

Table 18. Cost Comparison: Agentic vs. Traditional Development

Cost Component	Agentic Framework	Traditional Approach
Direct Costs		
LLM API costs	\$24.20 total	—
Open AI Enterprise Cost(3months)	\$150	—
Total direct costs	\$174.20	—
Cost per generated item	\$0.52 (333 items)	—
Cost per pretesting item	\$2.29 (76 items)	\$485–\$2,500
Time Investment		
Expert review time	20-24 hours	119-123 hours
SME time savings	80-83%	Baseline

The Range reflects PAEA internal estimate (lower bound) and literature estimate (upper bound). The agentic approach introduced modest direct computational costs (\$2.29 per pretesting item) while reducing SME time requirements by approximately 80% (from estimated 150 hours to 30 hours for 76 pretesting-ready items). This resource reallocation enables subject matter experts to focus on higher-value validation and refinement rather than initial drafting. Total direct computational costs (\$174.20) represent a fraction of traditional development costs, whether compared to PAEA's internal benchmark ($\$485/\text{item} \times 76 \text{ items} = \$36,860$) or literature estimates ($\$1,500/\text{item} \times 76 \text{ items} = \$114,000$).

Direct computational costs were negligible compared to traditional approaches, representing less than 0.1% of estimated traditional development costs. The primary value proposition lies in SME time optimization rather than direct cost reduction, enabling experts to allocate effort toward quality assurance rather than content generation while maintaining comparable output quality through multi-stage validation. The enterprise subscription cost of \$50 monthly provides infrastructure for sustained item generation at scale, with per-item costs decreasing substantially as generation volume increases beyond this proof-of-concept study.

Scalability Considerations

The framework demonstrated consistent performance characteristics supporting scalability: stable generation time (~2.4 min/item) across both phases, constant automated review processing (<30 sec/item), linear cost scaling with volume (variable API costs only), and quality improvement through iterative refinement suggesting sustainable efficiency.

At current efficiency rates, generating 1,000 items would require approximately \$72.60 in computational costs (40 hours generation time, 8.3 hours automated review time), with human review time scaling based on Review Agent approval rates. The framework's modular architecture and stateless LLM processing enable parallel execution, potentially reducing generation time further with concurrent API calls while maintaining per-item cost consistency.

In summary, the agentic framework substantially improved operational efficiency while maintaining rigorous quality standards and traceable validation/validity checkpoints. The primary value lies in SME time optimization and workflow acceleration, enabling organizations to increase assessment development capacity without proportional increases in expert labor requirements.

Research Question 4: Psychometric Properties

RQ4: What are the psychometric properties (difficulty, discrimination, reliability, dimensionality) of test items generated by the agentic LLM approach, and how do these properties compare to traditionally developed items?

Addressing this research question, 34 AI-generated items advanced to the pilot testing stage and were administered as unscored pretest items embedded within PAEA's operational End of Rotation™ examination forms across five specialties. Each form contained 120 items, of which 100 items were operational (scored), and 20 items were pretest (unscored). A small number of pretest items were AIG pilot items.

Psychometric analysis proceeded in four approaches. First, the summary statistics with EAP reliability, then distractor analysis to evaluate the functioning of incorrect response options. Second, classical test theory (CTT) statistics were computed from the full-form item analysis for all 34 AIG items (one item was excluded due to all examinees answering correctly, yielding an analytic sample of 33 items). Third, one-parameter logistic (1PL) model was calibrated on each complete test form, and item parameter estimates were extracted for the AIG items. Fourth, two complementary comparative analyses were conducted: a within-form comparison of AIG items against operational items administered to the same examinees on the same test forms, and a cross-administration comparison of AIG items against historically administered cardiovascular disease items from previous PAEA examinations. Together, these comparisons evaluate psychometric parity from two independent perspectives.

Pilot Testing Sample

We first present the distribution of AIG items and examinees across the 8 test forms spanning five examinations. Table 19 presents the distribution of AIG items and examinees across

the eight test forms spanning five examinations. Average EAP reliability was computed twice for each form: once using all 120 items and once excluding only the AIG items. Full-form reliability ranged from .689 to .803. Note that neither set of estimates should be interpreted as the reliability of the scored operational exam; both include unscored field test items.

Table 19. Full-Form Calibration Summary

Exam	Forms	Total Items	AIG Items	Examinees	EAP Rel.	EAP Rel. (excl. AIG)
EMEOR	Form1	120	6	357	.766	.761
	Form2	120	8	350	.767	.749
FMEOR	Form1	120	4	76	.766	.760
	Form2	120	2	72	.803	.805
	Form3	120	2	97	.762	.761
IMEOR	Form1	120	7	436	.749	.741
PEDEOR	Form1	120	3	127	.689	.696
SUREOR	Form1	120	2	400	.737	.734

Across six of the eight forms, inclusion of AIG items increased EAP reliability by .001 to .018, with the largest gains on forms containing more AIG items (EMEOR Form 2: +.018, 8 items; IMEOR Form 1: +.008, 7 items). Two forms showed negligible decreases (FMEOR Form 2: −.002; PEDEOR Form 1: −.007), consistent with sampling variability given their small AIG counts ($k =$

2–3). No form exhibited a substantive reliability decline, indicating that AIG items functioned as productive, or at minimum neutral, contributors to measurement precision.

Distractor Analysis

Option-level analysis examined the proportion of examinees selecting each response alternative for the 33 AIG items. Distractors selected by fewer than 5% of examinees were classified as non-functioning distractors (NFD; Haladyna & Rodriguez, 2013).

Table 20. Distractor Functioning Summary for AIG Items by Examination

Exam	k	Items with NFD	% with NFD
EMEOR	14	11	78.6
FMEOR	8	8	100.0
IMEOR	7	6	85.7
PEDEOR	2	0	0.0
SUREOR	2	2	100.0
Overall AIG	33	27	81.8
Overall Operational	727	633	87.1

Note: k = number of items

Table 20 shows how the distractors functioned across the exams and we also reported the overall operational distractor analysis too. Twenty-seven of 33 AIG items (81.8%) contained at least one non-functioning distractor. Notably, operational items on these same forms exhibited a comparable rate (87.1%), indicating that non-functioning distractors are common not unique to just AI-generated items.

This finding corroborates the expert review results reported in Table 10 in which the Distractor Plausibility dimension received the lowest mean quality rating ($M = 2.80$) among the five dimensions evaluated. The convergence of expert judgment and empirical field test evidence

identifies distractor quality as the primary target for refinement of the agentic framework and in general.

Classical Test Theory Results

CTT item analysis was conducted on each complete 120-item form, and statistics were extracted for the AIG items and operational items. Two discrimination indices are reported: the corrected point-biserial correlation (r_{pbis}) computed from the full-form total score, and the operational-only total score as an independent criterion.

Table 21 presents the CTT statistics for AIG field test items, non-AIG field test items, and operational items across the five examinations. Mean difficulty for the 33 AIG items was .789 (.200), comparable to non-AIG field test items (.806, .201) and operational items (.830, .161), indicating that AIG items produced similar difficulty distributions to traditionally authored items at both the field test and operational stages. PEDEOR was a notable exception: the two AIG items averaged .500, substantially harder than the non-AIG field test (.760) and operational (.841) items on that exam, though interpretation is limited by the small number of AIG items.

Mean point-biserial discrimination for AIG items was .132 (.102), comparable to non-AIG field test items .118 (.106) but below operational items .155 (.107). All three values fell in the low-to-moderate range. This is expected for two reasons. First, field test items regardless of authorship have not undergone the iterative statistical screening that selectively retains higher-discriminating items in operational pools. Second, as noted by the principal psychometrician overseeing at PAEA (R. Furter, personal communication), medical education examinations span multiple clinical content areas within a single form treated as essentially unidimensional, which attenuates item-total correlations relative to narrowly focused assessments. The similarity between AIG and non-

AIG field test discrimination suggests that the AI generation process did not systematically degrade item quality on this metric. Complete item-level psychometric statistics for all 33 AI-generated items are presented in Appendix B.

Table 21. CTT Statistics for AIG Items by Examination

Exam	Items	k	Mean p	SD p	Mean r_{pbis}	SD r_{pbis}
EMEOR	AIG	14	.732	.224	.131	.098
	Non-AIG	26	.850	.121	.122	.070
	Operational	200	.813	.177	.151	.086
FMEOR	AIG	8	.863	.172	.145	.111
	Non-AIG	52	.781	.226	.132	.145
	Operational	300	.852	.147	.169	.135
IMEOR	AIG	7	.862	.113	.129	.092
	Non-AIG	13	0.827	.167	.117	.081
	Operational	100	.834	.153	.146	.081
PEDEOR	AIG	2	.500	.082	.136	.178
	Non-AIG	17	.760	.263	.101	.101
	Operational	100	.841	.167	.148	.119
SUREOR	AIG	2	.944	.041	.152	.090
	Non-AIG	18	.841	.195	.090	.066
	Operational	100	.780	.174	.141	.082
Overall	AIG	34	.796	.177	.132	.102
	Non-AIG	126	.806	.201	.118	.106
	Operational	800	.830	.161	.155	.107

Notes. k = number of items, p = proportion correct score, SD = standard deviation, r_{pbs} = point biserial correlation.

Item Response Theory Results

One-parameter logistic (1PL) models were calibrated on each of the 8 complete test forms using the TAM package in R (Robitzsch et al., 2022). Each calibration estimated item difficulty parameters (b) via marginal maximum likelihood (MML) and examinee ability parameters (θ) via expected a posteriori (EAP) estimation, fitting all 120 items and all examinees simultaneously. Item parameter estimates for the AIG items were then extracted from these full-form calibrations. Table 22 presents Rasch difficulty and fit statistics for AIG items alongside operational items.

Table 22. Rasch Difficulty and Fit Statistics for AIG Items by Examination

Exam	Exam Type	K	Mean b	SD b	Mean Infit	Mean Outfit
EMEOR	AIG	14	-1.441	1.477	1.012	.984
	Non-AIG	26	-1.407	1.489	.967	.934
	Operational	200	-1.98	1.360	.998	.972
FMEOR	AIG	8	-2.576	1.525	1.001	.957
	Non-AIG	52	-1.864	1.672	1.013	1.014
	Operational	300	-1.42	2.54	.977	.979
IMEOR	AIG	7	-2.311	1.229	1.002	.997
	Non-AIG	13	-2.002	1.154	1.015	0.974
	Operational	100	-1.440	1.711	.988	.958
PEDEOR	AIG	2	-0.016	.346	1.017	1.018
	Non-AIG	17	-1.513	1.725	0.994	0.967
	Operational	100	-1.92	1.000	.978	.935
SUREOR	AIG	2	-3.052	.849	.987	.909
	Non-AIG	18	-2.389	1.572	1.006	0.992
	Operational	100	-1.700	1.33	1.000	.973
Overall	AIG	33	-1.912	1.501	1.006	.978
	Non-AIG	126	-1.964	1.748	0.987	0.971
	Operational	800	-1.721	1.270	0.998	0.993

Note. b = Rasch difficulty in logits. k= number of items

Table 22 presents Rasch difficulty and fit statistics for AIG, non-AIG field test, and operational items across the five examinations. Mean Rasch difficulty for the 33 AIG items was -1.912 logits ($SD = 1.501$), compared to -1.964 ($SD = 1.748$) for non-AIG field test items and -1.721 ($SD = 1.270$) for operational items. Overall, all three groups occupied similar positions on the difficulty continuum. At the exam level, however, non-trivial discrepancies emerged. PEDEOR AIG items averaged -0.016 logits compared to -1.920 for operational items, reflecting the small AIG count ($k = 2$) on that exam. EMEOR and IMEOR showed closer alignment between AIG and operational difficulty estimates. These exam-level differences underscore the importance of interpreting Rasch parameters at the form level, particularly when AIG item counts are small.

Mean infit MNSQ was 1.006 and mean outfit MNSQ was .978 for AIG items, both close to the expected value of 1.0 (Wright & Masters, 1982), indicating adequate model-data fit. Non-AIG field test items showed comparable fit (infit = .987, outfit = .971), as did operational items (infit = .998, outfit = .993). All three item groups fell within the acceptable 0.7–1.3 range (Wright & Linacre, 1994). Complete item-level statistics are presented in Appendix B.

Figure 9. Infit and Outfit MNSQ for AI-Generated Items

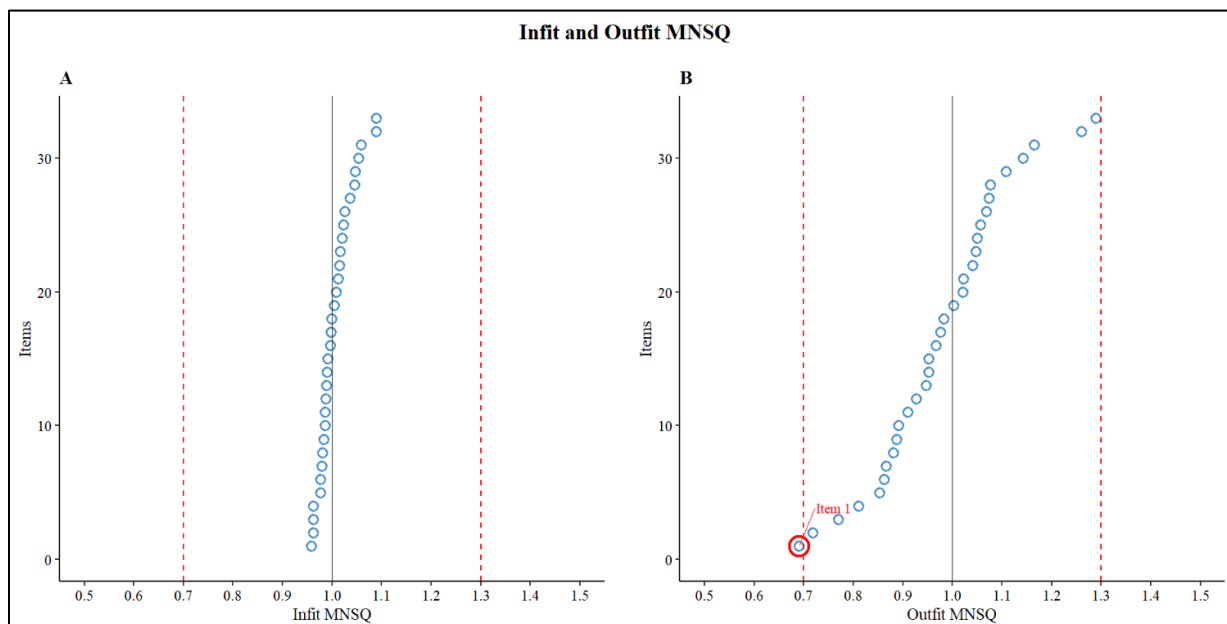


Figure 10 displays infit and outfit MNSQ statistics for the 33 AIG items. All items fell within the acceptable 0.7–1.3 range for infit (Panel A). One item (Panel B) exhibited outfit MNSQ below 0.7, indicating overfit. Overfit occurs when examinee responses are more predictable than the Rasch model expects that is, the item discriminates too consistently, producing a Guttman-like response pattern in which nearly all examinees above the item's difficulty answer correctly and nearly all below answer incorrectly. While this departs from the model's probabilistic expectations, overfit does not indicate construct-irrelevant variance or ambiguity. In practice, overfitting items function adequately for measurement purposes and are generally retained in item banks (Wright & Linacre, 1994). The more consequential form of misfit is underfit ($\text{MNSQ} > 1.3$), which signals unpredictable response patterns potentially caused by item ambiguity, multidimensionality, or construct-irrelevant difficulty. No AIG items exhibited underfit on either statistic.

Comparative Psychometric Analysis

Two complementary comparisons were conducted to evaluate the psychometric quality of AI-generated items. The first compared AIG items against operational items administered to the same examinees on the same test forms (within-form comparison). The second compared AIG items against historically administered cardiovascular disease pretest items from previous PAEA examinations (cross-administration comparison). Together, these comparisons provide convergent evidence from two independent perspectives.

Assumption Testing Results

Prior to conducting group comparisons, distributional assumptions were evaluated. Table 16 presents the results of Shapiro-Wilk normality tests and Levene's test for homogeneity of variance. Difficulty distributions departed significantly from normality for both AIG items ($W = .864, p < .001$) and operational items ($W = .843, p < .001$), indicating negatively skewed distributions consistent with the generally easy item pools characteristic of medical education examinations. Discrimination distributions did not violate normality for either group (AIG: $W = .965, p = .356$; Operational: $W = .999, p = .613$). Levene's test indicated no significant differences in variance between groups for either difficulty ($F = 2.84, p = .093$) or discrimination ($F = 1.12, p = .291$). Given the normality violations in difficulty distributions and the substantial sample size imbalance ($k = 34$ vs. $k = 926$), Mann-Whitney U tests were conducted alongside independent t-tests to ensure robustness of findings. Rasch difficulty distributions were normally distributed for AIG items ($W = .957, p = .210$) but departed from normality for operational items ($W = .994, p = .005$), likely due to sensitivity of the Shapiro-Wilk test at large sample sizes. Levene's test indicated no significant difference in variance ($F = 3.72, p = .054$). Mann-Whitney U tests confirmed the parametric finding ($p = .688$).

Table 23. Distributional Assumption Tests for Within-Form Comparison

Metric	Group	k	W	p	F	p (Levene)
Difficulty (p-value)	AIG	34	.864	< .001	2.84	.093
	Operational	926	.843	< .001		
Discrimination (rpbis)	AIG	34	.965	.356	1.12	.291
	Operational	926	.999	.613		
Rasch difficulty	AIG	33	.957	.210	3.72	.054
	Operational	775	.994	.005		

Note. *W* = Shapiro-Wilk test statistic. *F* = Levene's test statistic (center = median). *k* = number of items.

Within-Form Comparison: AIG vs. Operational Items

The pilot design enabled direct comparison controlling for examinee ability and testing conditions. CTT statistics for AIG items (*k* = 33) were compared against operational items pooled across all 8 forms (*k* = 926). Table 24 presents the within form comparison between AIG and Operational Items.

Table 24. Within-Form Comparison of AI-Generated and Operational Item Statistics

Metric	AIG	Operational	t	p	Cohen's d	MW p
Difficulty (p)	.789 (.200)	.827 (.169)	-0.89	.380	-0.17	.649
Discrimination (rpbis)	.132 (.098)	.151 (.110)	-0.92	.365	-0.15	.603
Rasch difficulty (b)	-1.912 (1.501)	-2.042 (1.287)	0.49	.628	0.09	.688

Notes. *t* = independent two-sample t-test statistic, *p* = two-tailed significance value, Cohen's *d* = standardized effect size

No significant differences emerged between AI-generated and operational items in difficulty ($t = -0.89$, $p = .380$, $d = -0.17$), discrimination ($t = -0.92$, $p = .365$, $d = -0.15$), or Rasch difficulty ($t = 0.49$, $p = .628$, $d = 0.09$). Effect sizes were negligible for all three metrics, falling below the conventional small-effect threshold of $|d| = 0.20$ (Cohen, 1988). Non-parametric Mann-Whitney U tests confirmed these findings (difficulty: $p = .649$; discrimination: $p = .603$; Rasch difficulty: $p = .688$), demonstrating that the results are robust to violations of normality assumptions. The convergence of parametric and non-parametric results strengthens the conclusion that AIG items are statistically indistinguishable from operational items on both CTT and IRT metrics.

Cross-Administration Comparison: AIG vs. Historical Items

Table 25 reveals the AIG items compared against 36 traditionally authored cardiovascular disease items from previous PAEA administrations across the same five specialty examinations.

Table 25. Historical Cardiovascular Item Comparison

Metric	AIG	Historical	t	p	Cohen's d
Difficulty (p)	.789 (.200)	.727 (.236)	1.192	.238	0.29
Discrimination (rpbis)	.132 (.098)	.179 (.080)	-2.158	.035*	-0.52
Rasch Difficulty	-1.912 (1.501)	0.261 (1.624)	-5.777	0.00	-1.39
Infit MNSQ	1.006 (.035)	1.022 (.115)	-0.795	.431	-0.19
Outfit MNSQ	.978 (.138)	.981 (.168)	-0.099	.921	-0.02

Note. t = independent two-sample t-test statistic, p = two-tailed significance value, Cohen's d = standardized effect size

The cross-administration comparison revealed no significant differences for difficulty ($p = .238$), infit ($p = .431$), or outfit ($p = .921$), with negligible effect sizes. AIG items were significantly

easier in Rasch difficulty than historical items (-1.912 vs. 0.261 logits, $d = -1.39$, $p < .001$), a large effect attributable to the different examinee samples and separate calibrations placing b values on non-equivalent logit scales; this difference should be interpreted cautiously as it conflates true item difficulty with sample ability differences. AIG items exhibited lower discrimination than historical items ($d = -0.52$, $p = .035$). This modest discrimination difference is expected given that the historical items have undergone multiple cycles of field testing and revision, whereas the AIG items are first-generation pilot items that have not yet been refined.

Figure 10. CTT Item Statistics Across Three Item Groups

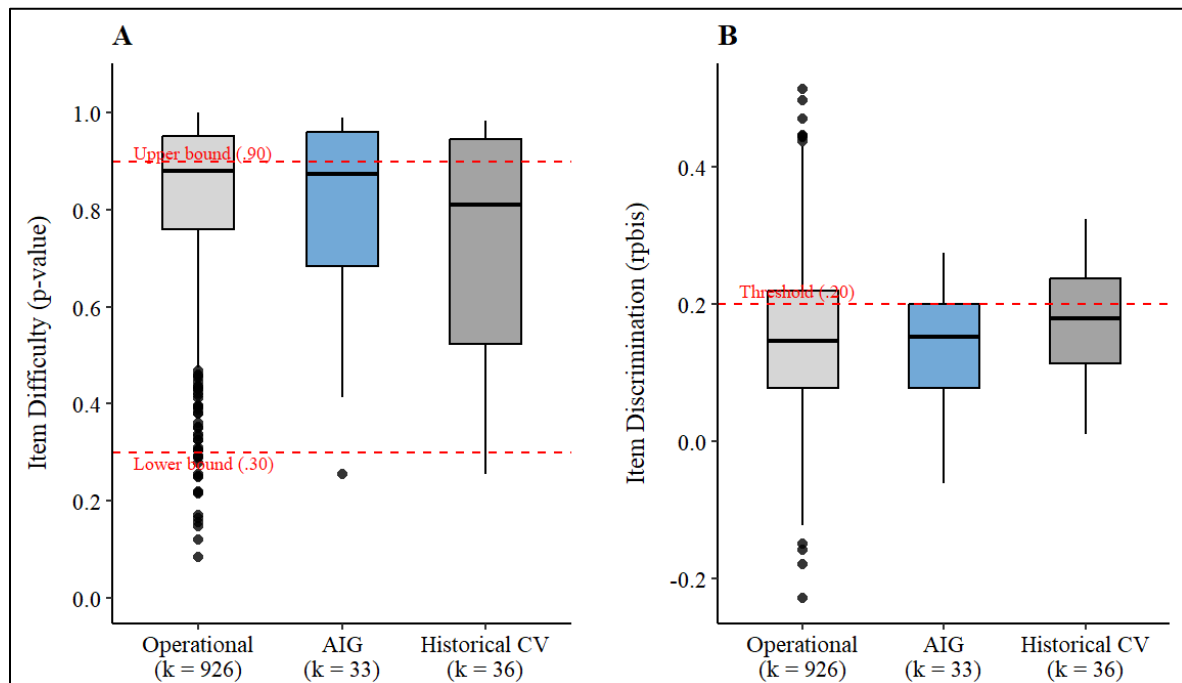


Figure 11 presents the distributions of item difficulty and discrimination across all three item groups. The substantial overlap in Panel A confirms that AIG items fall within the difficulty range of both operational and historical benchmarks. Panel B shows all three groups clustering below the .20 discrimination threshold, with comparable medians and spread. The within-form comparison is the methodologically stronger design controlling for examinee ability and testing

conditions showed no significant differences and negligible effect sizes. Taken together, the evidence supports psychometric parity between AI-generated and traditionally developed items.

To summarize research question four, full form 1PL Rasch calibration across 8 test forms produced stable difficulty estimates for all 33 AIG items, with 32 (97%) demonstrating acceptable model fit. Classical item statistics indicated moderate-to-easy difficulty ($M = .789$) and discrimination comparable to operational items on the same forms. Within-form comparison revealed no significant differences in difficulty or discrimination. Cross-administration comparison against historical cardiovascular items confirmed comparable difficulty and model fit, with a modest discrimination difference attributable to the AIG items' status as unrevised first-generation pilot items. Twenty-seven items (81.8%) contained non-functioning distractors, corroborating expert review findings and identifying distractor quality as the primary refinement target. These findings demonstrate that the agentic framework produces items whose core psychometric properties - difficulty and model fit - are statistically indistinguishable from traditionally developed items, while highlighting distractor generation as the area requiring further development.

CHAPTER V: DISCUSSION AND CONCLUSION

The overarching purpose of this study was to address critical gaps in AIG literature by investigating how LLMs could be effectively integrated into structured frameworks for medical assessment development. Building on Gierl et al.'s (2021) AIG framework, this study proposed a new framework for AIG and implemented an adapted multi-agentic approach specifically designed for LLM-based generation as described in Chapter III. This approach advanced beyond basic prompt engineering techniques (Susnjak, 2022) by operationalizing structured multi-agent systems with rigorous human oversight. Four research questions guided the investigation: 1) design and determine how LLMs can be integrated into multi-agent assessment frameworks; 2) examine alignment between AI-generated items and expert quality standards; 3) assess operational efficiency gains compared to traditional methods; and 4) evaluate psychometric properties of generated items. Each of these questions is addressed based on findings from the two-phase generation study in the previous section. This chapter concludes with practical implementation guidance, study limitations, and future research directions.

Summary of Findings

The agentic framework generated 333 cardiovascular assessment items across two phases, with 76 items (22.8%) advancing through multi-stage validation to pretesting readiness. The total computational cost of \$174.20 for item generation yielded \$2.29 per pretesting-ready item, representing less than 1% of traditional development costs estimated between \$500 and \$2,500 per item.

The phase-specific performance illustrated the critical importance of iterative refinement. Phase 1 generated 250 items with 34 advancing to pre-testing (13.6% success rate), while after the interphase refinement Phase 2 generated 83 items with 42 advancing to pre-testing (50.6% success

rate). This represented a 272% improvement. This quality enhancement resulted entirely from systematic prompt engineering based on expert feedback, with no model changes or architectural modifications, demonstrating how responsive to corrections the agentic approach can be. The Review Agent's approval rate improved from 50.8% to 100%, while human pipeline success increased from 13.6% to 50.6%, demonstrating that initial AI outputs require systematic enhancement rather than representing production-ready products.

Critical quality improvements through iterative refinement included complete elimination of enemy items, which represented the most significant validity threat. In the Phase 1 large peer review 50 enemy items were identified (55.6% of pool) creating construct contamination through content overlap or answer dependencies. Phase 2 achieved zero enemy items through targeted prompt modifications that addressed content redundancy. Format and style compliance improved substantially through correct and incorrect examples, step-by-step checklists, and format-specific validation criteria.

Quality dimension ratings across all 210 reviewed items demonstrated consistent performance, with items showing strongest results on Bias Freedom (M=3.79), Currency (M=3.44), and Clinical Relevance (M=3.32), all exceeding the 3.0 advancement threshold. Items for all five specialty examinations achieved acceptable mean scores: Emergency Medicine (M=3.35), Family Medicine (M=3.23), Internal Medicine (M=3.19), Surgery (M=3.06), and Pediatrics (M=3.04). Lower scores on Format Compliance (M=2.63), Clarity of Stem (M=2.80), and Distractor Plausibility (M=2.80) indicate areas for continued refinement. Content Accuracy reached 85.9%, indicating the Review Agent's clinical verification provided effective screening.

The framework demonstrated substantial operational efficiency: 20-24 hours total development time versus estimated 119-123 hours for traditional methods. This represented an

80% reduction in SME time requirements, and consistent performance characteristics including 2.4 minutes per generated item and linear cost scaling regardless of batch size. Of the 210 items that advanced to the small peer review sessions, 148 (70.5%) exceeded the 3.0 threshold and advanced to large peer review, with 76 ultimately reaching pretesting readiness.

Field testing was done for the first approved 34 items with one of the items excluded due to all examinees answering correctly ($p = 1.00$), yielding an analytic sample of 33 items. The psychometric analysis of the 33 field-tested AIG items confirmed that AI-generated items performed comparably to traditionally developed items across all core metrics. Within-form comparison against 926 operational items on the same test forms revealed no significant differences in difficulty ($d = -0.17$), discrimination ($d = -0.15$), or Rasch difficulty ($d = 0.09$), with all effect sizes below the negligible threshold. Cross-administration comparison against 36 historical cardiovascular items confirmed comparable difficulty, infit, and outfit, with a modest discrimination difference attributable to the AIG items' status as unrevised first-generation pilot items. Full-form Rasch calibration yielded adequate model fit, with 32 of 33 items (97%) falling within the acceptable 0.7–1.3 range. Distractor functioning was comparable to operational items (81.8% vs. 87.1% NFD rates), with distractor quality corroborated by expert review as the primary area for framework refinement. These results establish that the agentic framework produces items whose psychometric properties are statistically indistinguishable from traditionally developed items, completing the evidence chain from generation through expert review to empirical field performance.

Contributions to the Field

This study offers several distinct contributions to the AIG and educational measurement literature. First, it is the first study to design, implement, and empirically evaluate a multi-agent LLM-based agentic framework for high-stakes medical education assessment, moving beyond the single-prompt and template-based approaches that have dominated AIG research. The framework's architecture comprising a Cognitive Planning Agent, five specialty-specific Writer Agents, an automated Reviewer Agent, and an SME-Driven Rewrite Agent orchestrated through LangGraph represents a structural contribution that can be adapted by other testing organizations to their own content domains.

Second, the study developed the R Shiny electronic review system, a purpose-built application enabling asynchronous, authenticated SME review of AI-generated items using an eight-dimensional quality rubric. The system's design with role-based access, encrypted data handling, and integrated feedback mechanisms is itself a replicable contribution independent of the agentic framework.

Third, the study provides the first empirical cost and efficiency benchmarks for a multi-agent LLM-based AIG system, addressing a gap noted by Circi et al. (2023) and extending beyond the only prior cost-benefit analysis, which examined template-based systems exclusively (Kosh et al., 2019). The documentation of computational costs, token usage, API call volumes, and SME time allocation across all pipeline stages provides testing organizations with concrete reference points for planning their own implementations.

Fourth, it is the first AI-based AIG study to carry items through the complete pipeline from multi-agent LLM generation through multi-stage expert review to operational field testing with

psychometric analysis on live high-stakes examinations. Prior LLM-based AIG studies have typically terminated at expert quality evaluation (e.g., Sayin & Gierl, 2024; Kiyak & Kononowicz, 2025). The field-testing evidence closes this gap by demonstrating that AI-generated items are psychometrically indistinguishable from traditionally developed items under operational conditions.

Fifth, the study demonstrates how the DET Responsible AI framework (Burstein et al., 2022) can be adapted from its original language assessment context to organize validity evidence for AI-generated medical education items, mapping each pipeline stage to specific evidentiary claims across validity and reliability, fairness, privacy and security, and accountability and transparency. This mapping provides a replicable template for other testing organizations seeking to integrate responsible AI governance into their validation practices.

Lessons Learned: What Works and What Doesn't

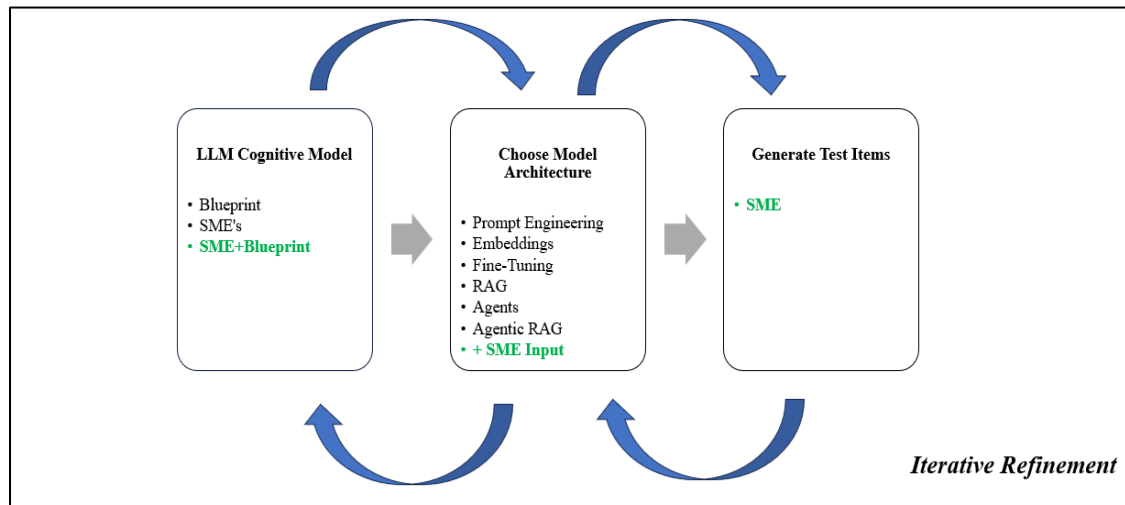
The implementation and evaluation of the agentic framework across two phases yielded several practical insights with implications for future AIG research and operational deployment.

Blueprint-Driven Cognitive Planning with Early SME Integration

The cognitive planning agent's blueprint-driven approach to scenario development before item generation proved essential for maintaining construct alignment. Rather than generating items directly from specifications, the framework first created detailed clinical scenarios establishing patient context, diagnostic considerations, and management pathways. This staged approach operationalized evidence-centered design principles while preventing construct drift through focused attention on distinct authoring subtasks.

However, the critical discovery concerned timing and extent of SME integration. Phase 1 built cognitive models from blueprint specifications alone, with SME involvement limited to downstream validation. This produced items with strong technical performance but systematic failures in format compliance, distractor quality, and clinical authenticity requiring professional judgment. Introducing structured SME feedback at multiple process stages dramatically improved as evidence by the performance of items generated in Phase 2. Future implementation should integrate subject matter expertise throughout the entire process rather than treating it as terminal validation. SME input should inform initial cognitive model design, shape generation prompt development, establish quality evaluation criteria, and provide ongoing validation. This integration will likely prevent quality issues requiring extensive downstream correction through revision.

Figure 11. Proposed framework for Automatic Item Generation Using LLM



The Critical Role of Pilot Testing of the Framework

A related lesson emerged regarding the critical importance of pilot testing before operational deployment. The need for two sequential phases with extensive iterative refinement between them could have been substantially reduced or eliminated through rigorous pilot testing

with SMEs prior to the first small peer review session. Had the framework undergone piloting where SMEs evaluated initial outputs and provided structured feedback before committing to large-scale generation, many of the systematic quality issues identified in Phase 1 could have been addressed proactively. This pilot-first approach would have enabled prompt refinement and system calibration in a controlled setting with smaller item volumes, potentially achieving in a single operational phase what this study required two phases to accomplish. Organizations implementing similar framework for item generation should prioritize pilot testing as a critical preparatory step rather than viewing it as optional, as the upfront investment in piloting substantially reduces downstream iteration requirements and accelerates achievement of production-quality outputs while managing stakeholder expectations about normal learning curves.

Model Architecture Complexity

The agentic architecture emerged as a significant strength requiring measured progression through increasingly sophisticated approaches. Organizations face multiple architectural options: prompt engineering, fine-tuning on specialized datasets, Retrieval-Augmented Generation with dynamic guideline access, and multi-agent systems. The multi-agent approach effectively captured medical assessment development complexity through specialized agents mirroring natural expert processes: cognitive planning replicated deliberate scenario construction, generation agents operationalize creative authoring, review agents systematically apply quality criteria, and rewriter agents enable targeted revision. This structural alignment proved valuable for generation quality and system troubleshooting, as agent-specific architecture made identifying required adjustments straightforward.

However, immediately implementing sophisticated multi-agent architecture without exploring simpler approaches proved challenging. Distributed components created debugging

challenges and maintenance overhead that diverted attention from optimizing prompt engineering. Organizations lacking substantial technical expertise would struggle with this complexity during initial implementation. A conservative progression beginning with simple prompt engineering, potentially advancing to RAG if clinical currency challenges persist, and progressing to agentic architectures only after demonstrating simpler approaches prove insufficient, will likely enable organizations to achieve adequate quality without full architectural complexity.

The recommendation is nuanced: organizations with sufficient technical capabilities pursuing high-volume, high-stakes assessment should seriously consider agentic approaches because that complexity appropriately reflects the sophistication required to ensure item development validity. The natural evolution points toward Agentic RAG as particularly promising, combining multi-agent advantages with dynamic knowledge retrieval to address clinical currency challenges. However, complexity should be justified through demonstrated need and empirical performance improvements at each progression stage.

Specialty-Specific Implementation and Multi-Agent Adaptability

The quality evaluation across the five specialties revealed consistent performance which supports the generalizability of the framework. All examinations achieved mean average scores exceeding the 3.0 advancement threshold: Emergency Medicine (M=3.35), Family Medicine (M=3.23), Internal Medicine (M=3.19), Surgery (M=3.06), and Pediatrics (M=3.04). This consistent performance of AI generated cardiovascular items across diverse clinical domains demonstrates the framework's adaptability without requiring specialty-specific architectural modifications.

This success aligns naturally with how most testing organization's structure item development workflows. Organizations typically maintain specialty-specific editorial boards,

separate boards with dedicated SMEs developing and validating items for their respective examinations. The framework's consistent performance across specialties suggests a single adaptable multi-agent system with specialty-specific prompt libraries and validation criteria may prove sufficient for most licensure/credentialing organizations.

Critically, the multi-agent architecture demonstrated remarkable responsiveness to targeted corrections. The 272% quality improvement from Phase 1 to Phase 2 through systematic prompt refinement, achieved without model changes or architectural modifications, illustrates rapid adaptation to quality issues as they emerge. This responsiveness suggests organizations can deploy a unified framework and calibrate different specialty requirements through targeted prompt refinement rather than building entirely separate specialty-specific systems. The optimal approach involves initial deployment focused on a single specialty to establish baseline quality, followed by systematic expansion through prompt adaptation, balancing specialty-appropriate content with efficiency and maintainability advantages of unified technical infrastructure.

System Iterative Refinement Enhancement

The current Review Agent implementation revealed both limitation and opportunity. Items underwent single-pass evaluation with binary approve-reject decisions, preventing the iterative refinement that characterizes effective human expert authoring. This terminal output architecture wasted generative effort on potentially salvageable items and failed to capitalize on iterative refinement capability representing AI systems' key advantages.

A more sophisticated architecture would implement closed-loop refinement where the Review Agent identifies specific deficiencies and returns items to the Cognitive Planning Agent with targeted feedback. The Cognitive Planning Agent would adjust scenario specifications, the Item Generation Agent would regenerate items incorporating corrections, and pass them to the

Review Agent to re-evaluate iteratively until approval or reaching maximum attempt limits. This would mirror human expert revision processes and likely improve overall efficiency, as the Rewriter Agent's 82.9% success rate demonstrates most quality issues involve correctable deficiencies rather than fundamental construct misalignment. The technical complexity of implementing such closed-loop refinement protocol would require careful design that includes the setting of maximum cycle limits among other considerations. The potential efficiency and quality gains, however, are likely to justify investment for organizations pursuing scaled implementations.

Human Oversight as a Non-Negotiable Foundation

Perhaps the most important lesson concerns the non-negotiable role of human expert oversight throughout generation and validation. While AI-enhanced generation promises substantial efficiency gains, the aspiration should not be eliminating human involvement but optimizing expert time allocation. The framework's primary value proposition lies in enabling organizations to maintain larger item banks through increased production capacity rather than replacing human judgment in quality assurance.

The multi-stage validation results underscore why human oversight cannot be bypassed: despite sophisticated automated screening, 63.8% of Review Agent-approved items were rejected during human validation, demonstrating that current AI capabilities cannot yet replicate the nuanced professional judgment that is essential for high-stakes assessment validity. In short, relying solely on automated screening would have released numerous flawed items. Each validation layer served distinct purposes that automated systems cannot yet fulfill. The small peer review evaluated specialty-specific appropriateness which requires domain expertise, while the large collaborative review identified construct contamination which requires multi-expert

consensus. This layered defense-in-depth approach proves essential for high-stakes contexts despite efficiency costs.

An important discovery emerged regarding optimal human-AI collaboration points within the workflow. As an alternative to automated revision, 10 items in Phase 2 underwent manual pre-editing by an SME prior to large peer review. This strategic human intervention required approximately 5-10 minutes per item but substantially accelerated subsequent large peer review sessions. Items receiving manual pre-editing moved through large peer review more efficiently than both automated-revised items and unrevised items, suggesting that targeted human expertise applied at critical revision points can optimize downstream workflow efficiency. This hybrid approach combining automated revision for high-volume processing with strategic manual editing for complex cases demonstrates that a flexible adaptation of the framework to balance automation efficiency with human expert judgment is beneficial. Organizations intending to implement multi-agentic system should consider identifying workflow bottlenecks where small investments of expert time yield disproportionate downstream efficiency gains, rather than rigidly applying either full automation or full manual processes throughout.

The study's efficiency gains of 80% reduction in SME time requirements and 83-90% reduction in total development time represent the achievable optimization model. Organizations can generate substantially more items and reduce expert burden from initial drafting, but established standards for content validation, construct alignment verification, and psychometric quality assurance remain essential and cannot be outsourced to AI. Organizations must budget for ongoing expert engagement, though involvement shifts from primarily authoring to primarily guiding and validating.

The largest win lies not in cost reduction but capacity expansion, enabling maintenance of rigorous quality standards while producing larger item banks necessary for secure, frequently refreshed assessment programs. AI enhances rather than replaces the human expertise that assessment validity fundamentally depends upon.

Format Compliance Requires Extensive Exemplification

Format and style compliance proved unexpectedly challenging despite explicit specifications. Simply stating requirements like "Follow PAEA format standards" failed to ensure compliance, as LLMs interpreted instructions through diverse training data patterns rather than organization-specific preferences. Models trained on diverse medical literature defaulted to commonly encountered formats rather than organizational conventions.

The breakthrough came through exemplification. Showing the agentic system specific correct and incorrect examples for every guideline element and demonstrating application through contrasting cases rather than merely stating rules greatly improved format and style compliance. Phase 2 achieved near-perfect format compliance through explicit style examples, step-by-step format checklists embedded in generation prompts, format-specific validation instructions in Review Agent prompts, and post-processing rule-based corrections. Organizations should expect similar challenges and invest substantial effort in creating detailed format exemplars rather than relying on abstract rule statements.

How to Operationalize AI for Item Development

AI fundamentally transformed the item development process and economics by generating thousands of items in hours rather than weeks or months. This capacity will enable organizations to maintain the substantially larger item banks that are essential for Computer Adaptive Testing that require extensive pools across difficulty levels and high-stakes examinations that demand

large banks for form construction, security through reduced reuse, and frequent content refreshment. The challenge in item generation shifts from production scarcity to quality curation through systematic validation processes.

1. **Identify Optimal Model Architecture:** Select the appropriate architectural approach based on technical capabilities. Prompt engineering for organizations with limited expertise, agentic or Agentic RAG systems for those with substantial capabilities pursuing high-volume, high-stakes assessment.
2. **Select the Most Capable Foundation Model:** Choose models with strong medical reasoning benchmarks, clinical knowledge accuracy, and reliable API infrastructure. Current state-of-the-art models (e.g., recent GPT and Claude variants) offer superior clinical authenticity.
3. **Establish Data Security and Privacy Protections:** Implement secure API connections with encrypted transmission, vendor policies ensuring confidentiality, access controls, audit trails, and secure storage meeting regulatory requirements.
4. **Integrate Subject Matter Expertise Throughout the Entire Process:** Structure SME engagement across all stages. AI cannot replace professional judgment for validity assurance. Human oversight remains non-negotiable for high-stakes assessment.
5. **Conduct Rigorous Pilot Testing Before Operational Deployment:** Generate items in a single specialty with intensive SME evaluation before large-scale implementation. Upfront piloting investment substantially reduces downstream iteration requirements.
6. **Establish Multi-Stage Validation as Non-Negotiable Quality Assurance:** Implement multiple validation layers editorial reviews, small peer reviews, and large peer reviews aligned with the organization's existing quality assurance policies and validation workflows.

7. Plan for Continuous Maintenance and Quality Monitoring: Maintain ongoing updates reflecting evolving guidelines, psychometric monitoring tracking performance, feedback collection from deployments, and periodic system reviews. Establish governance frameworks defining AI uses, oversight requirements, and incident protocols.

Implications for Assessment Practice

AI-enhanced generation shifts the central challenge in assessment item development from producing a sufficient volume of items to managing and validating the large volume of items that can now be produced efficiently. Traditional item development maintains rigorous quality standards but operates under capacity constraints that limit production volume, with high per-item development costs requiring careful resource allocation. This framework's dramatically reduced costs enabled generating large candidate pools and sustaining larger operational banks essential for Computer Adaptive Testing, high-stakes examinations that require multiple parallel forms, and frequent content refreshment maintaining security and currency.

AI integration transforms rather than eliminates expert roles. Traditional teams must evolve to include SMEs focused on validation and guidance, psychometricians designing validation frameworks, data scientists developing prompts, and software engineers maintaining systems. Essential emerging competencies include prompt engineering expertise, validation framework design, system performance analysis, and AI ethics and governance understanding. Graduate training and professional development implications extend through academic programs incorporating AI literacy, specialized training providers, and certification bodies considering AI oversight competencies.

Abundant item availability enables evolution from absolute to relative quality standards. Abundance conditions enable selecting top performers from large candidate pools based on optimal difficulty at critical cut-scores, minimal bias as evaluated through expert review, and parameter stability. This continuous improvement approach could potentially gradually raise quality standards as generation capabilities mature.

AIG represents one component of AI's transformation of educational measurement. AI technologies are revolutionizing automated scoring systems, automated feedback systems, enemy item detection, and many other labor-intensive quality assurance tasks. Organizations must develop strategic approaches to AI integration across these interconnected functions rather than treating automatic item generation as isolated innovation. This will require careful implementation and safety protocols as assessment practice undergoes this fundamental paradigm shift.

This transformation demands robust governance frameworks ensuring ethical, transparent, and accountable AI deployment. Organizations must establish documented policies defining appropriate AI uses and mandatory human oversight, quality monitoring procedures, incident response protocols, and explicit ethical principles. This study aligned with the DET Responsible AI Standards, which informed the framework's emphasis on transparency, human oversight, and systematic validation. However, the assessment field critically requires updated validity frameworks addressing AI integration. The current Standards for Educational and Psychological Testing (2014, AERA, APA, NCME) predate widespread AI deployment and inadequately address validity considerations specific to AI-generated items and algorithmic decision-making. Professional organizations including AERA, APA, and NCME should prioritize developing updated standards explicitly addressing AI usage, validation requirements for AI-generated content, human oversight specifications, and fairness considerations. Emerging guidance from

organizations like the Association of Test Publishers and frameworks like DET Responsible AI Standards provide interim direction, but professional standards revision proves essential for responsible AI adoption at scale. Organizations implementing AI-enhanced assessment should align governance with existing standards and emerging frameworks while anticipating updated standards addressing AI-specific validity and fairness considerations.

Limitations

Although this proof-of-concept study demonstrated AI-enhanced generation feasibility across multiple quality dimensions and validation stages, findings reflect highly controlled conditions which may differ from operational practice. The investigation focused exclusively on cardiovascular content within PAEA's specific context, with generalization to other specialties, healthcare professions, or testing organizations requiring empirical verification. Cardiovascular medicine's well-established diagnostic algorithms and extensive evidence base may make it particularly amenable to AI generation compared to specialties with greater diagnostic ambiguity.

Several methodological constraints reflect the study's proof-of-concept nature and practical realities. First, item volumes remained limited to accommodate dissertation timeline constraints and SME availability. While AI systems can generate thousands of items rapidly, this study generated 333 items across two phases to enable intensive expert review within reasonable timeframes and prevent SME burnout in this voluntary research project. Larger-scale implementations with dedicated resources would enable generating substantially larger pools supporting more robust validation. Second, content coverage focused on cardiovascular assessment across five End-of-Rotation examinations rather than coverage across all PAEA specialties. Extending the framework to additional content domains would strengthen evidence of broad applicability.

Third, cost comparisons relied on published literature estimates (Rudner, 2010) and PAEA historical budget data rather than a concurrent controlled comparison with traditional item development. Prospective studies with randomized or matched-pair designs would strengthen causal claims about cost-effectiveness.

Finally, framework performance depended on specific GPT-5 Mini and GPT-5 Nano capabilities current at the time of implementation. Model capabilities evolve rapidly thus framework performance using different models or future versions remain uncertain. Architectural principles including dual-model design, multi-stage validation, and iterative refinement likely transfer across models, but this generalization requires empirical verification.

In summary, these findings are limited in generalizability to the controlled proof-of-concept conditions, single content domain focus, limited item volumes, and specific organizational context. The framework demonstrated feasibility and identified critical implementation requirements, but operational deployment at scale requires additional research examining performance across diverse domains and organizational contexts.

Future Research Directions

This study outlined the feasibility and potential impact of AI-enhanced AIG for the development of a medical assessment under controlled conditions. Substantial empirical research remains needed to support frameworks for operational implementation across other assessment contexts.

First, exploring Agentic Retrieval-Augmented Generation architecture represents a critical next step addressing clinical currency challenges. An agentic RAG system combining specialized agents with real-time access to current clinical guidelines could maintain content currency without constant manual prompt updating (Lewis et al., 2020).

Second, comparative studies evaluating performance across different foundation models would clarify model selection guidance. This study utilized GPT-5 Mini and GPT-5 Nano, but rapidly evolving models including GPT-5.2, Claude Opus, and emerging reasoning-optimized variants may offer superior performance.

Third, exploring multimodal AI capabilities for image generation in clinical scenarios represents a promising extension given advance in text-to-image models and medical imaging AI (Moor et al., 2023). Multimodal systems generating integrated vignettes combining clinical narratives with diagnostic imaging, clinical photographs, and laboratory results would substantially expand applicability across specialties that require visual diagnosis such as dermatology, radiology, and pathology.

Fourth, systematic comparative studies examining framework performance across medical specialties with adequate sample sizes would produce a taxonomy of content characteristics facilitating or impeding generation success, clarifying generalization boundaries and specialty-specific requirements (Gierl et al., 2021).

Fifth, while this study conducted IPL IRT analysis and comparative benchmarking against traditionally developed PAEA items, psychometric evidence was derived from a single pilot administration cycle with a limited sample size. Large-scale replication across multiple administration cycles with larger examinee samples would strengthen the generalizability of psychometric findings and support more robust conclusions about the operational equivalence of AI-generated items.

Finally, implementation of science research examining organizational adoption across diverse contexts would identify facilitators, barriers, change management strategies, and workforce training approaches supporting successful implementation (Greenhalgh et al., 2017).

Thus, several promising directions exist for future research, working toward guidelines and quality control frameworks supporting responsible AI-enhanced automatic item generation implementation while maintaining rigorous validity and fairness standards.

This study demonstrated that LLMs can be effectively integrated into structured multi-agent frameworks for high-stakes medical assessment item development. the agentic approach reduced development time by 80–83% and computational costs to under \$2.30 per pretesting-ready item while maintaining quality standards through multi-stage agent and human validation. these findings establish both the feasibility and the boundaries of llm-based aig, confirming that ai enhances rather than replaces the human expertise upon which assessment validity fundamentally depends.

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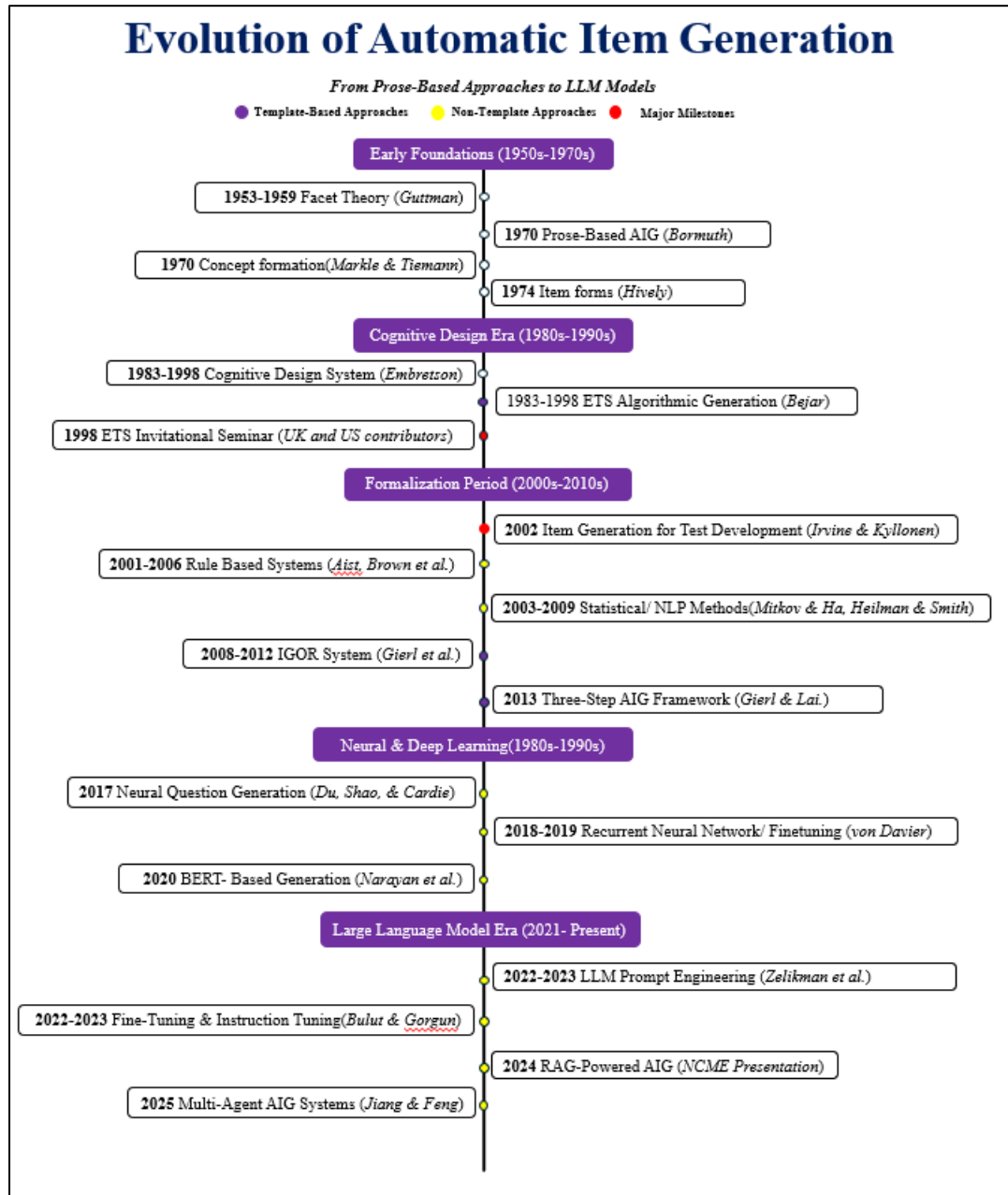
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APPENDIX A: EVOLUTION OF AIG

Figure A12. Evolution of AIG



APPENDIX B: COMPLETE ITEM-LEVEL & DATA STRUCTURE SPECIFICATION

Exam	Item_ID	N	PValue	rpbis_corrected	rbis	alphalfDeleted	rpbis_external
EMEOR	1	357	0.414566	0.110425	0.139575	0.776355	0.108155
EMEOR	2	357	0.683473	0.193833	0.253268	0.773932	0.20078
EMEOR	3	357	0.957983	0.168546	0.377058	0.775055	0.172596
EMEOR	4	357	0.254902	-0.0506	-0.06868	0.780124	-0.0618
EMEOR	5	357	0.7507	0.146874	0.200246	0.775131	0.15174
EMEOR	6	357	0.963585	0.150052	0.352135	0.775325	0.142158
EMEOR	7	350	0.694286	0.08279	0.108778	0.764955	0.082942
EMEOR	8	350	0.971429	0.128381	0.327335	0.763628	0.121082
EMEOR	9	350	0.934286	0.029779	0.057703	0.764947	0.022741
EMEOR	10	350	0.494286	0.196679	0.24651	0.761623	0.186972
EMEOR	11	350	0.845714	0.26606	0.404555	0.760194	0.262053
EMEOR	12	350	0.611429	-0.03315	-0.04216	0.768684	-0.04276
EMEOR	13	350	0.905714	0.166793	0.289986	0.762636	0.171949
EMEOR	14	350	0.765714	0.277329	0.382884	0.759462	0.275498
FMEOR	15	76	0.960526	0.244699	0.55891	0.755361	0.230697
FMEOR	16	76	0.934211	0.080502	0.155933	0.757627	0.081963
FMEOR	17	76	0.960526	0.282479	0.645203	0.754848	0.269337
FMEOR	18	76	0.513158	0.240924	0.302013	0.753386	0.231591
FMEOR	19	72	0.875	0.067269	0.108073	0.81571	0.072577
FMEOR	20	72	0.680556	0.193108	0.251962	0.813848	0.196985
FMEOR	21	97	0.989691	0.091452	0.337509	0.77958	0.077746
FMEOR	22	97	0.969072	-0.03652	-0.0906	0.780852	-0.04452
IMEOR	23	436	0.977064	-0.04379	-0.12059	0.753294	-0.04721
IMEOR	24	436	0.701835	0.068758	0.090716	0.753264	0.067943
IMEOR	25	436	0.977064	0.166571	0.458693	0.751105	0.172842
IMEOR	26	436	0.740826	0.121159	0.163941	0.751516	0.126563
IMEOR	27	436	0.917431	0.162657	0.294036	0.750464	0.160916
IMEOR	28	436	0.800459	0.231926	0.33154	0.748278	0.224079
IMEOR	29	436	0.919725	0.193383	0.352532	0.749892	0.197646
PEDEOR	30	129	0.44186	0.262007	0.329657	0.716364	0.250527
PEDEOR	31	129	0.55814	0.010375	0.013054	0.726915	-0.00158
SUREOR	32	400	0.9725	0.088473	0.228595	0.740611	0.089093
SUREOR	33	400	0.915	0.215962	0.387046	0.738074	0.216343

exam	itemID	b_difficulty	Infit_MNSQ	Infit_ZSTD	Outfit_MNSQ	Outfit_ZSTD
EMEOR	1	0.419792	1.042128	1.453182	1.04366	1.503891
EMEOR	2	-0.82291	0.998306	-0.0277	0.987043	-0.28953
EMEOR	3	-3.38304	0.991997	0.046437	0.896665	-0.37199
EMEOR	4	1.209024	1.077625	1.302481	1.140601	2.298473
EMEOR	5	-1.12406	1.007746	0.154652	1.011521	0.219913
EMEOR	6	-3.38304	0.986818	0.024792	0.84544	-0.60173
EMEOR	7	-0.88112	1.031495	0.710043	1.042648	0.952075
EMEOR	8	-3.63253	0.996352	0.077296	0.890073	-0.33227
EMEOR	9	-2.69653	1.021416	0.184437	1.083961	0.552899
EMEOR	10	0.09226	1.001672	0.077769	0.998933	-0.04144
EMEOR	11	-1.76991	0.966701	-0.34299	0.894804	-1.1786
EMEOR	12	-0.49772	1.08143	2.572579	1.1006	3.155976
EMEOR	13	-2.45774	0.998725	0.037231	0.974313	-0.14011
EMEOR	14	-1.28325	0.972873	-0.41439	0.922913	-1.23595
FMEOR	15	-2.97626	0.956931	0.023425	0.737525	-0.59574
FMEOR	16	-2.97626	1.010426	0.158061	1.085149	0.322201
FMEOR	17	-3.2142	0.93946	0.016575	0.662663	-0.69588
FMEOR	18	-0.16536	1.03747	0.708555	1.046844	0.877626
FMEOR	19	-1.73271	1.003774	0.079046	1.030522	0.214757
FMEOR	20	-0.82654	1.01419	0.177739	1.009283	0.123273
FMEOR	21	-4.2071	0.977306	0.193325	0.565093	-0.5479
FMEOR	22	-3.78972	1.022789	0.223105	1.341435	0.726549
IMEOR	23	-4.04827	1.014944	0.150179	1.315383	1.037502
IMEOR	24	-0.90474	1.033439	0.815544	1.04623	1.116574
IMEOR	25	-3.95047	0.991316	0.066211	0.813682	-0.59734
IMEOR	26	-1.08412	1.030152	0.632694	1.052753	1.086256
IMEOR	27	-2.45514	0.990062	-0.03759	0.970581	-0.1982
IMEOR	28	-1.43224	0.985892	-0.20141	0.96413	-0.54886
IMEOR	29	-2.61377	0.981943	-0.08515	0.880137	-0.85716
PEDEOR	30	0.097504	1.000334	0.013715	0.99984	-0.00104
PEDEOR	31	-0.21528	1.0389	1.009645	1.045656	1.178971
SUREOR	32	-3.70661	0.995773	0.075997	0.904487	-0.27221
SUREOR	33	-2.31873	0.981271	-0.10855	0.924559	-0.57752

The framework employs two primary data structures (JSON schemas) that govern information transfer between agents. These schemas ensure structural consistency, enable automated validation, and support systematic audit trail maintenance.

ItemBrief Schema (Cognitive Planning Output)

```
{
  "track": "EMEOR|FMEOR|IMEOR|SUREOR|PEDEOR",
  "blueprint": {
    "code": "string (e.g., CV-2.1.4)",
    "objective": "string (learning objective text)"
  },
  "topic": "string (cardiovascular topic)",
  "vignette_outline": {
    "setting": "string (clinical context)",
    "patient_summary": "string (demographics + chief complaint)",
    "key_features": ["string", "string", "string"]
  },
  "clinical_key": "string (correct answer concept)",
  "distractor_map": [
    "string (misconception intent 1)",
    "string (misconception intent 2)",
    "string (misconception intent 3)"
  ],
  "constraints": {
    "task_area": "string (diagnosis|management|interpretation|etc.)",
    "lead_in": "string (required lead-in template)"
  }
}
```

Item Schema (Writer Agent Output / Reviewer Agent Input)

```
{
  "item_id": "string (TRACK-TOPIC-YYYYMMDD-NNN)",
  "track": "EMEOR|FMEOR|IMEOR|SUREOR|PEDEOR",
  "specialty": "string (human-readable specialty name)",
  "topic": "string",
  "blueprint_code": "string",
}
```

```

"stem": "string (complete clinical vignette with lead-in)",
"options": {
  "A": "string",
  "B": "string",
  "C": "string",
  "D": "string",
  "E": "string (optional)"
},
"key": "A|B|C|D|E",
"rationale": "string (explanation with citations)",

"reviewer_scores": {
  "currency": float (1.0-5.0),
  "clinical_accuracy": float (1.0-5.0),
  "blueprint_alignment": float (1.0-5.0),
  "distractor_plausibility": float (1.0-5.0),
  "format_compliance": float (1.0-5.0),
  "clarity": float (1.0-5.0),
  "bias_free": float (1.0-5.0),
  "appropriate_difficulty": float (1.0-5.0)
},
"acceptance_status": "approved|rejected",
"rejection_reasons": ["string"] (if rejected),

"metadata": {
  "phase": "Phase 1|Phase 2",
  "timestamp_brief": "ISO 8601 datetime",
  "timestamp_generated": "ISO 8601 datetime",
  "timestamp_reviewed": "ISO 8601 datetime",
  "generation_duration_seconds": float,
  "nonce": "string (scenario variation seed)"
}
}

```

These schemas are enforced through automated validation at agent handoff points, ensuring data integrity throughout the generation pipeline. All field requirements, data types, and enumerated values are verified before processing proceeds, with validation failures triggering detailed error logging for system debugging and improvement.